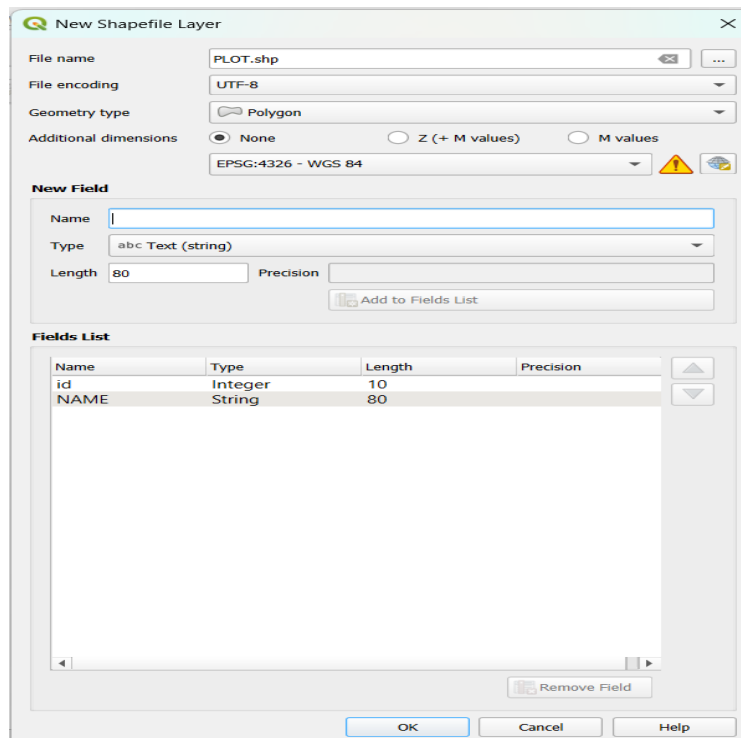
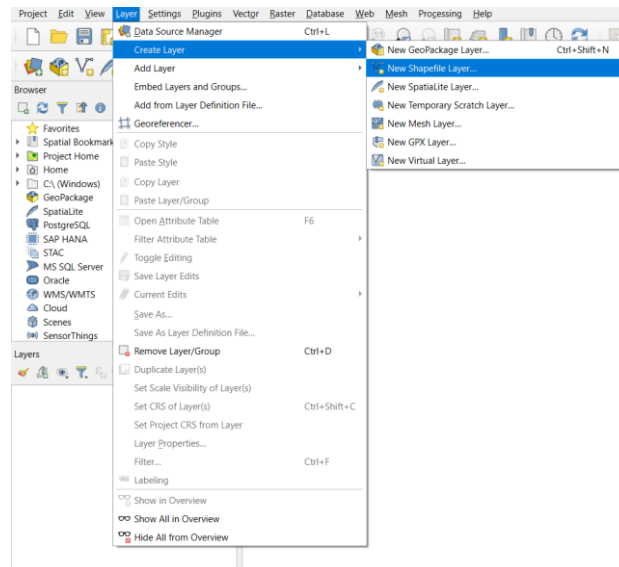


PRACTICAL NO.1

AIM : Creating and Managing Vector Data

- a) Adding vector layer
- b) Setting properties
- c) Vector Layer Formatting.

Steps:



New Shapefile Layer

File name: Garden.shp

File encoding: UTF-8

Geometry type: Polygon

Additional dimensions: ☒ None ☐ Z (+ M values) ☐ M values

EPSG:4326 - WGS 84

New Field

Name: name

Type: abc Text (string)

Length: 80 Precision:

Add to Fields List

Fields List

Name	Type	Length	Precision
id	Integer	10	

Remove Field

OK Cancel Help

New Shapefile Layer

File name: Road.shp

File encoding: UTF-8

Geometry type: LineString

Additional dimensions: ☒ None ☐ Z (+ M values) ☐ M values

EPSG:4326 - WGS 84

New Field

Name:

Type: abc Text (string)

Length: 80 Precision:

Add to Fields List

Fields List

Name	Type	Length	Precision
id	Integer	10	
name	String	80	

Remove Field

OK Cancel Help

New Shapefile Layer

File name: College.shp

File encoding: UTF-8

Geometry type: Polygon

Additional dimensions: ☒ None ☐ Z (+ M values) ☐ M values

EPSG:4326 - WGS 84

New Field

Name:

Type: abc Text (string)

Length: 80 Precision:

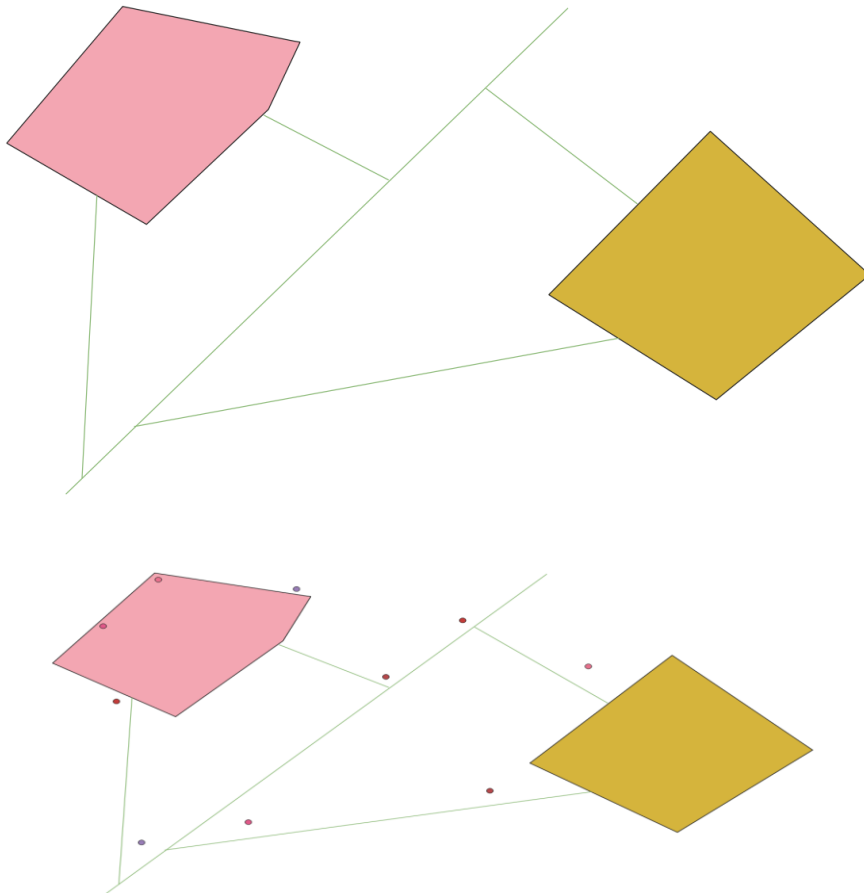
Add to Fields List

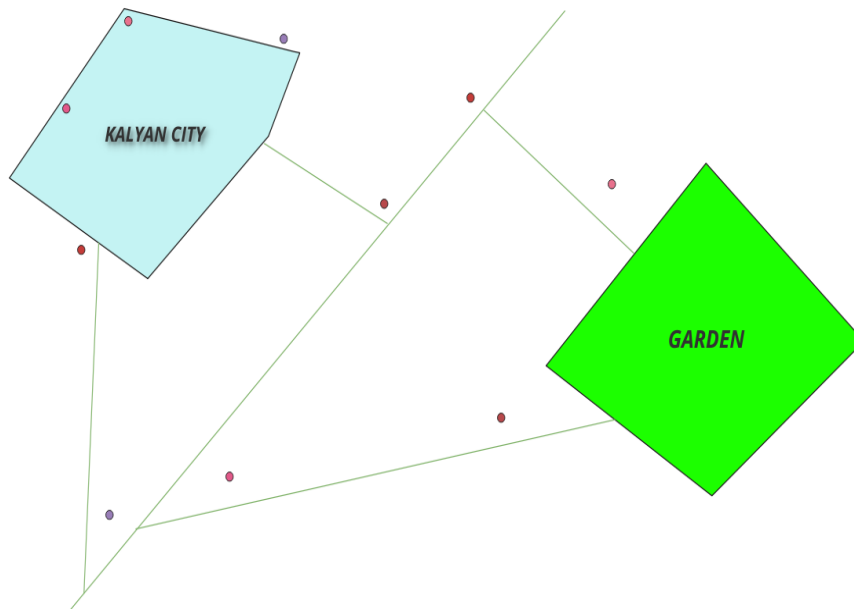
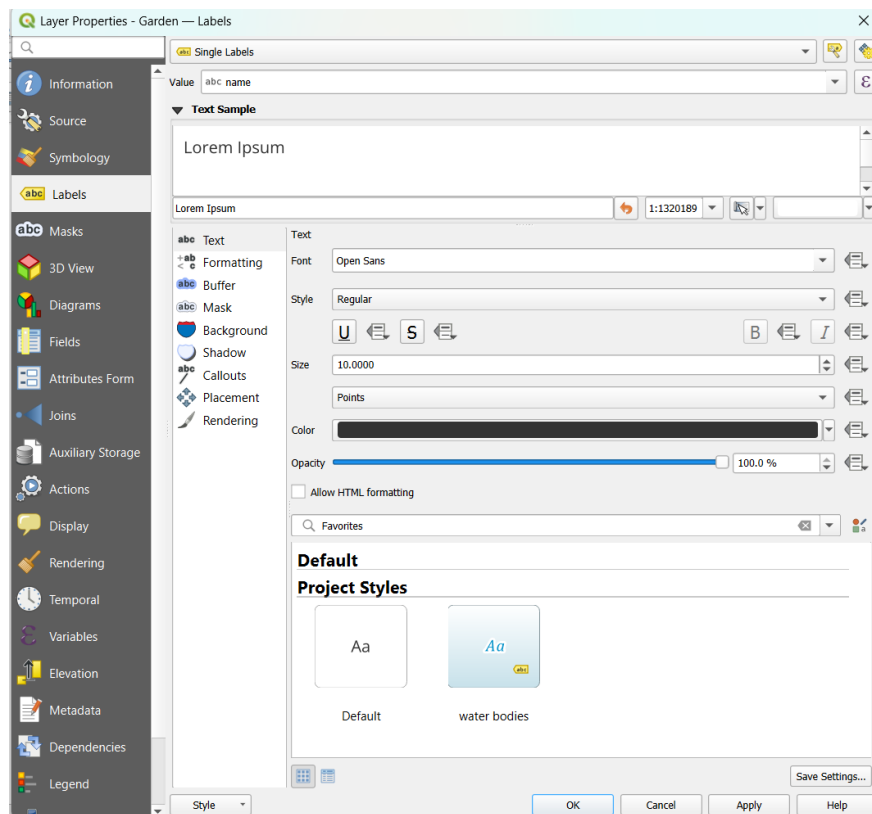
Fields List

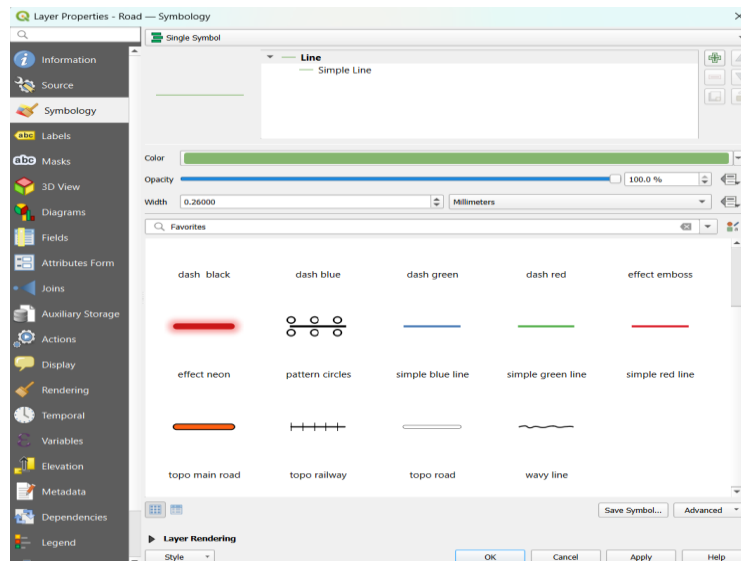
Name	Type	Length	Precision
id	Integer	10	
name	String	80	

Remove Field

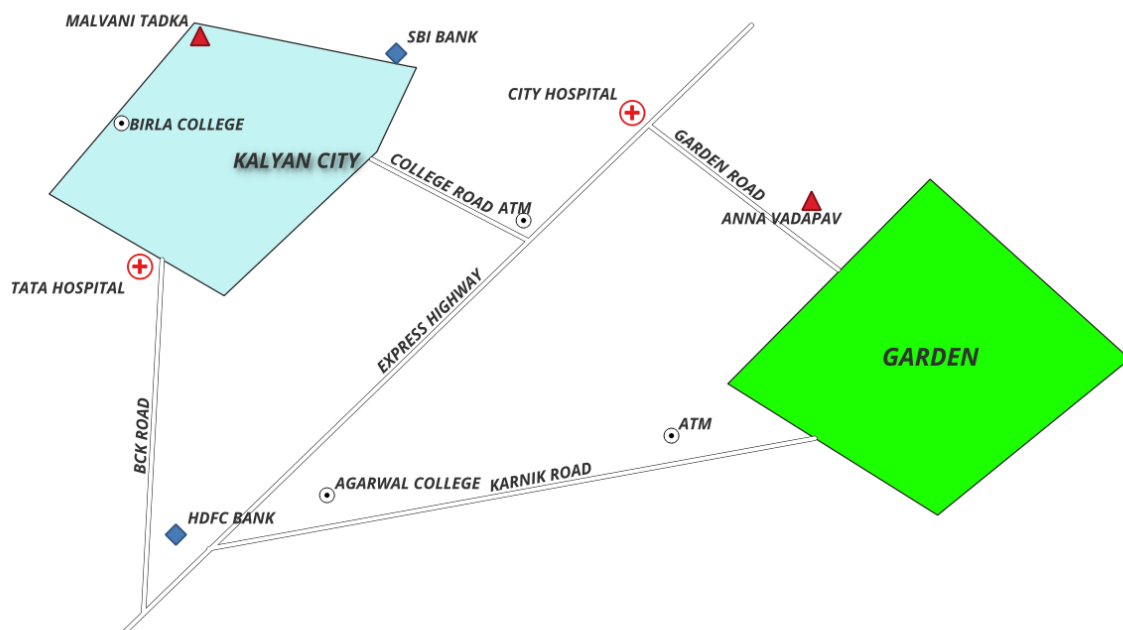
OK Cancel Help







FINAL OUTPUT :

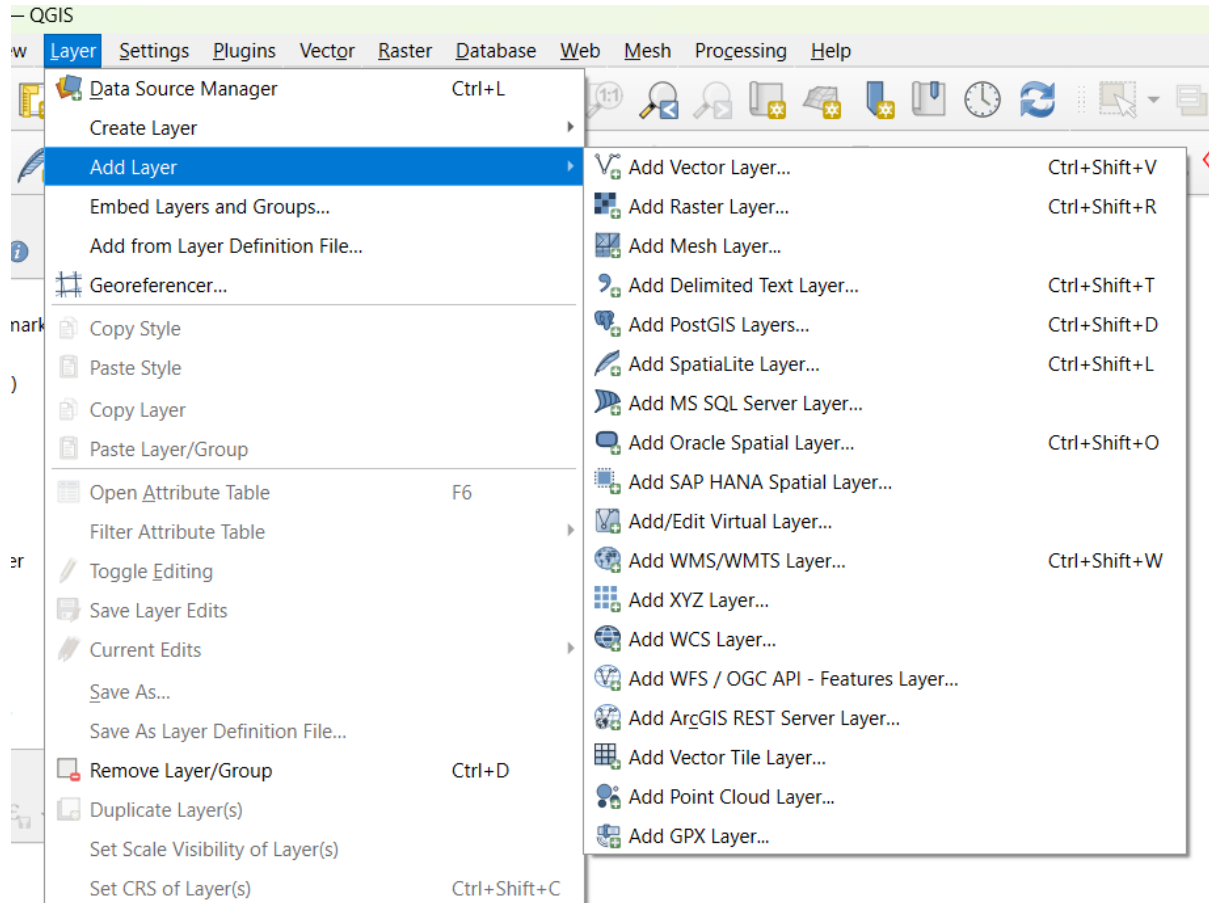


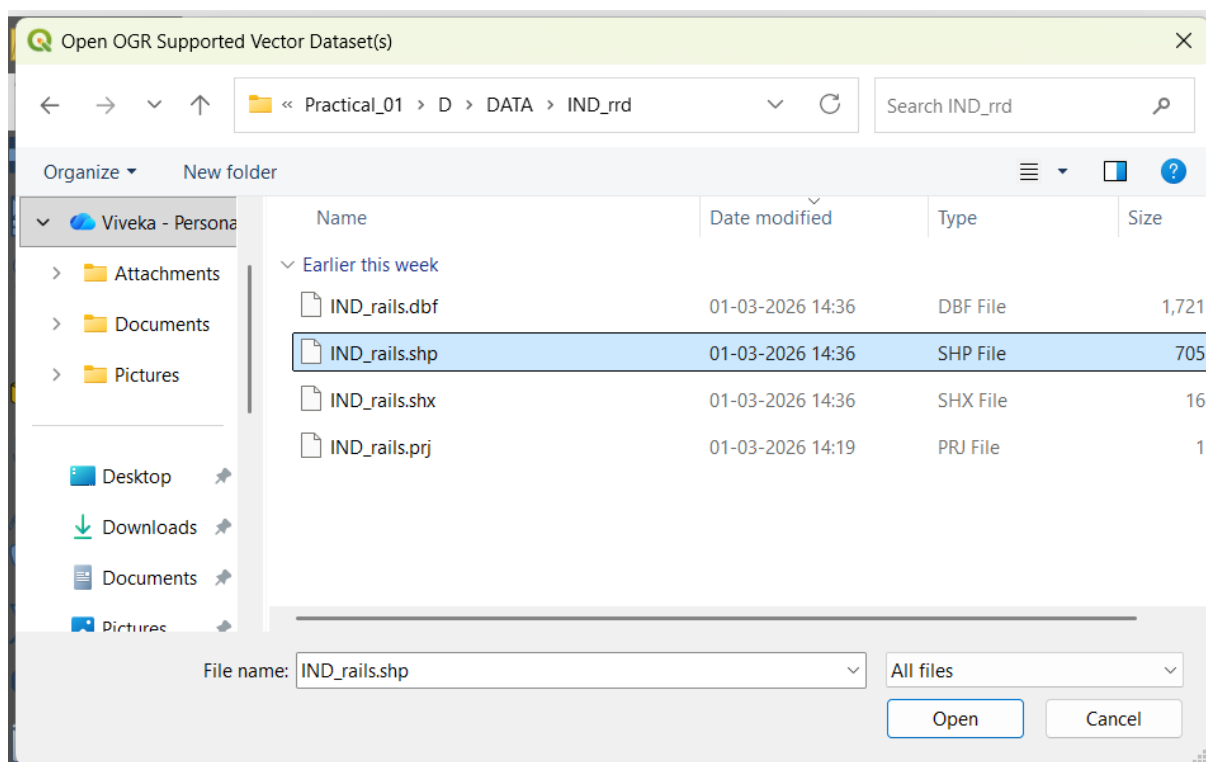
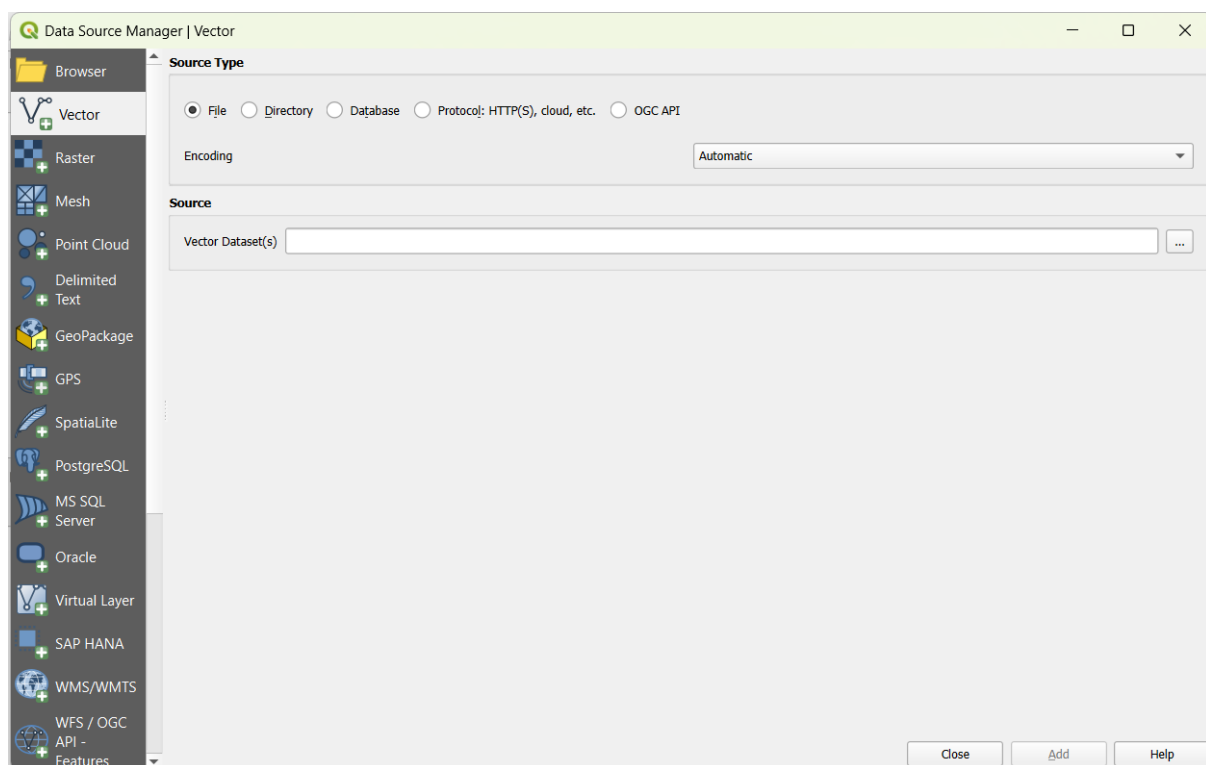
CONCLUSION : Vector data is created and managed successfully .

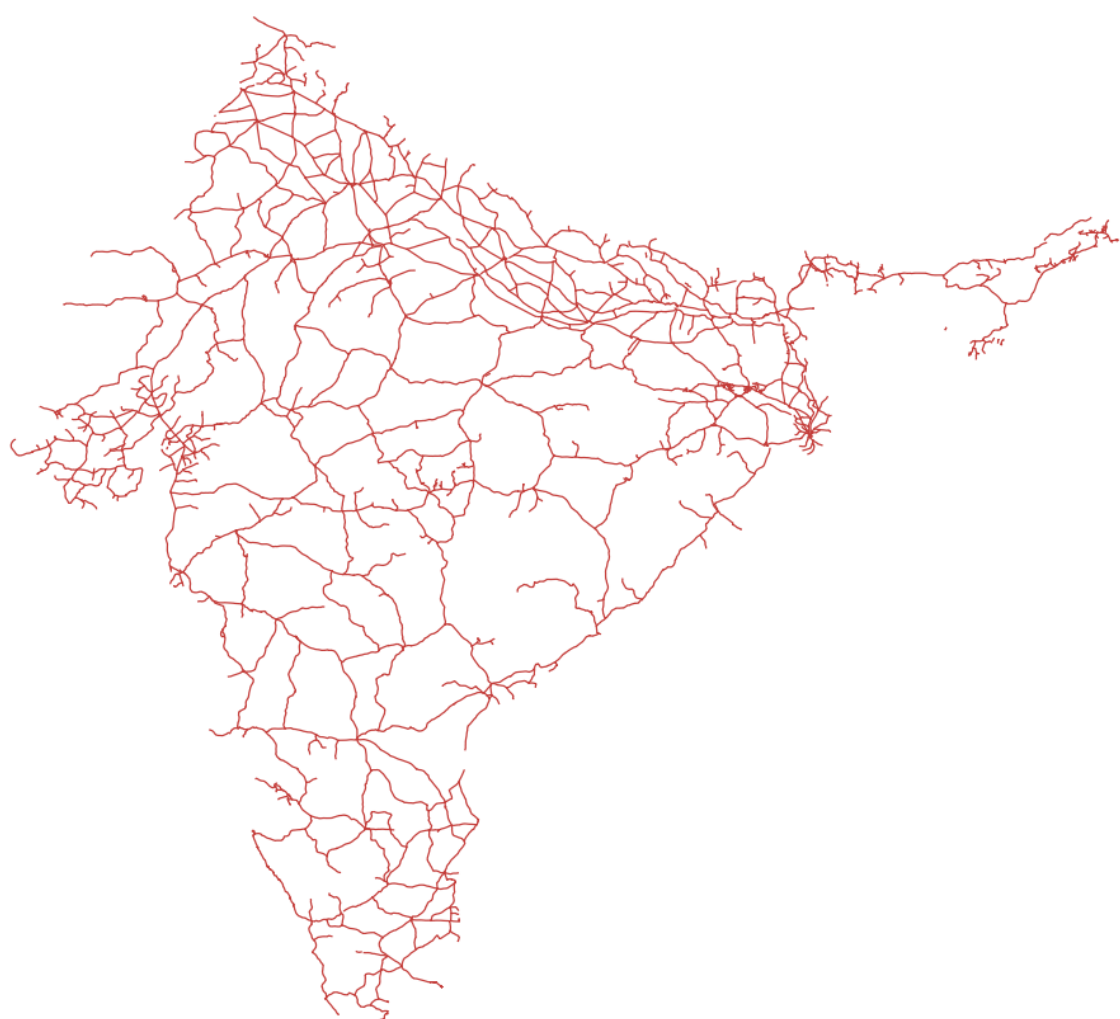
PRACTICAL NO.2

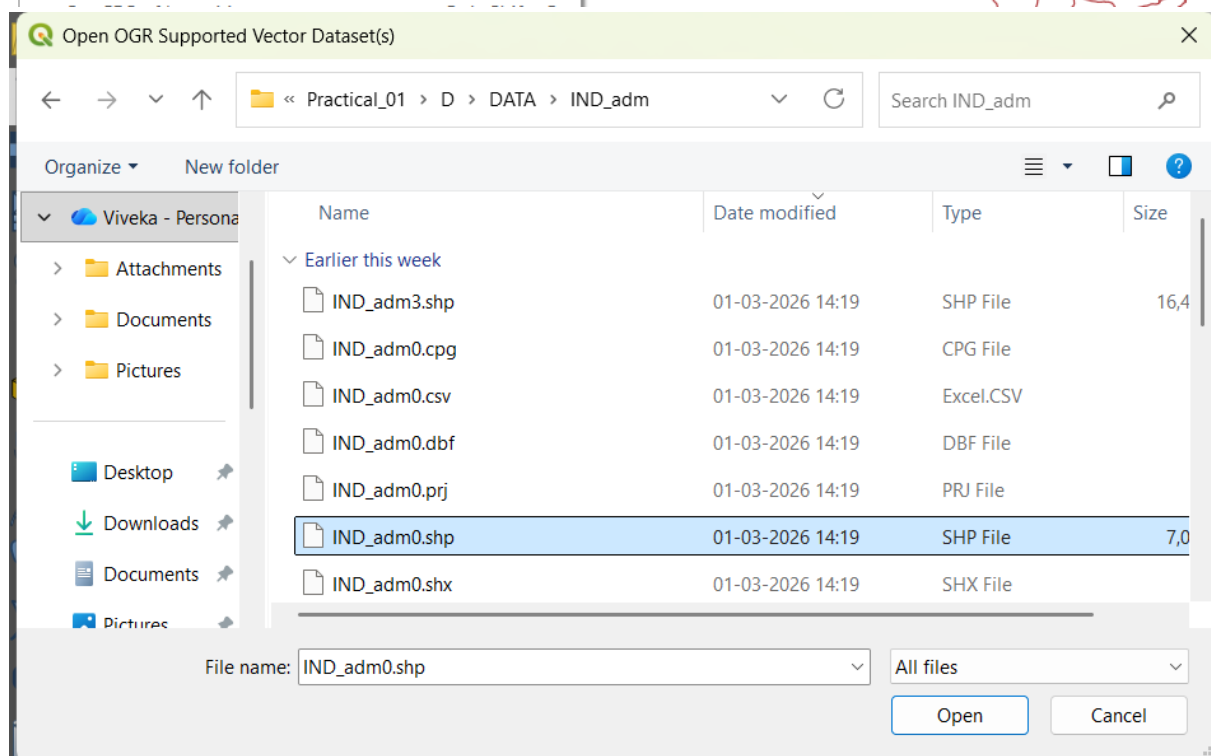
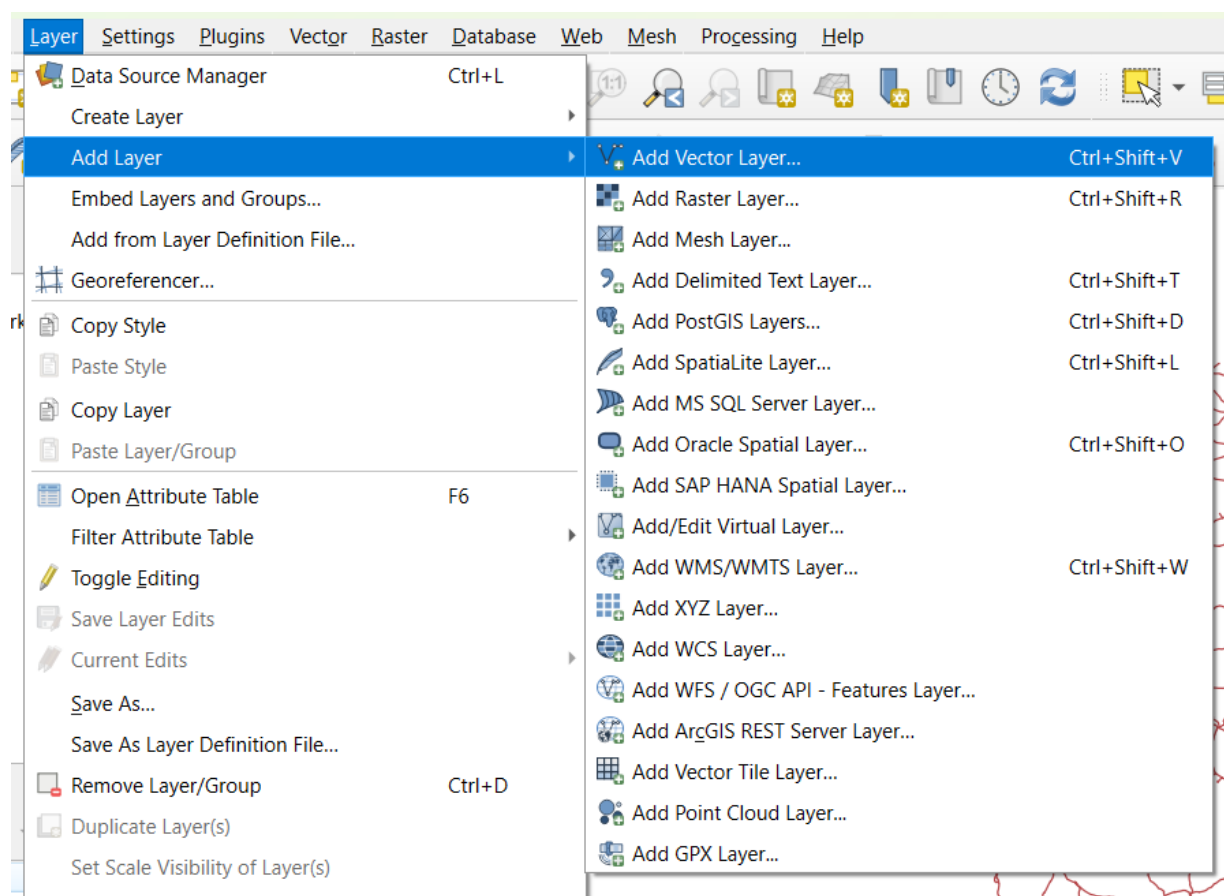
AIM : Calculating line lengths and statistics.

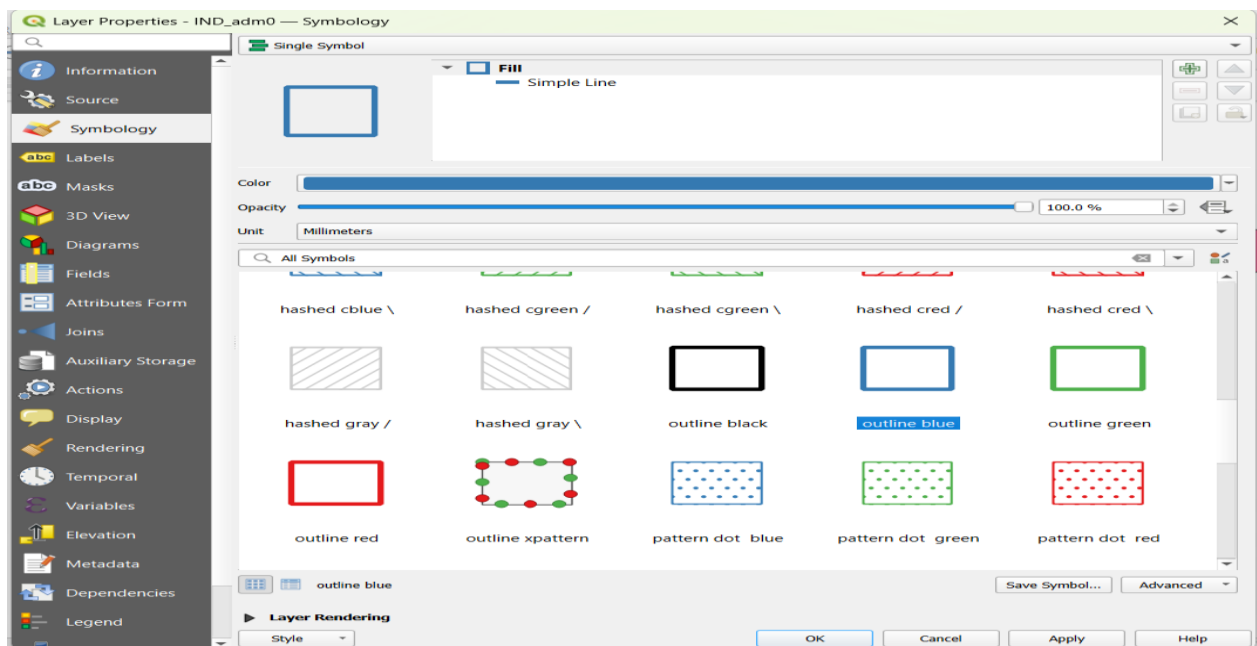
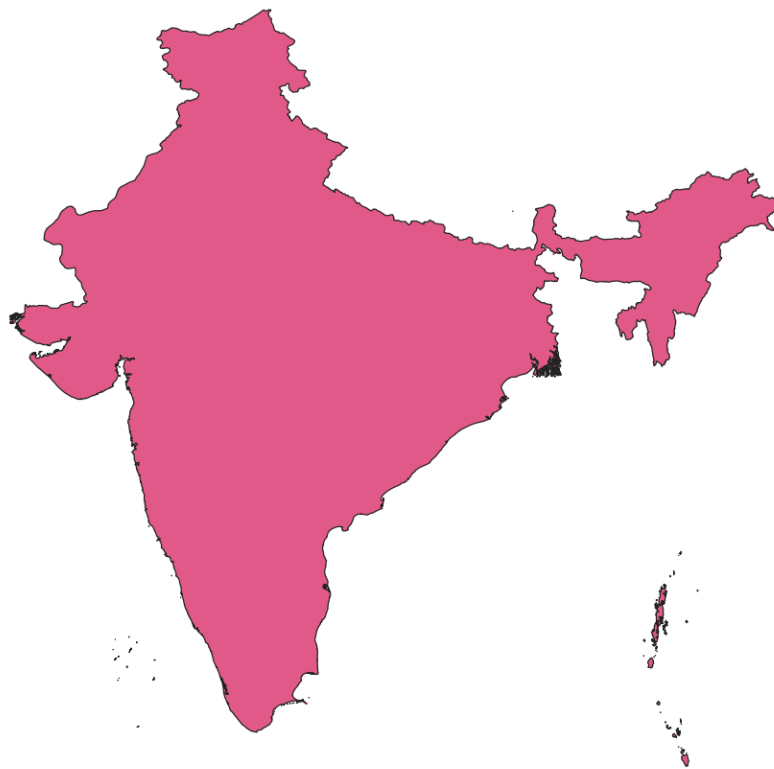
Steps:

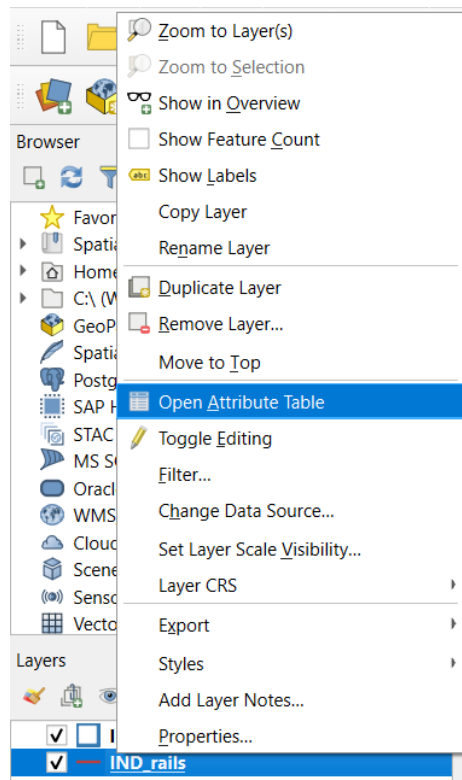












IND_rails — Features Total: 2013, Filtered: 2013, Selected: 0

	FID_rail_d	F_CODE_DES	EXS_DESCRI	FCO_DESCRI	FID_countr	ISO
1	144645	Railroad	Operational	Single	102	IND
2	145991	Railroad	Operational	Single	102	IND
3	146001	Railroad	Operational	Single	102	IND
4	146008	Railroad	Operational	Single	102	IND
5	146096	Railroad	Operational	Single	102	IND
6	146394	Railroad	Operational	Single	102	IND
7	146464	Railroad	Operational	Single	102	IND
8	146693	Railroad	Operational	Single	102	IND

Show All Features

IND_rails — Field Calculator

☐ Only update 0 selected feature(s)

☒ Create a new field ☐ Update existing field

☐ Create virtual field

Output field name:

Output field type: 123 Integer (32 bit)

Output field length: 10 Precision: 3

Expression Function Editor

Feature: Operational

Preview:

feature
geometry
id
row_number
Aggregates
Arrays
Color
Conditionals
Conversions
Date and Time
Fields and Values
Files and Paths
Fuzzy Matching
General
Geometry
Map Layers
Maps
Math
Operators
Rasters

OK Cancel Apply Help

IND_rails — Field Calculator

☐ Only update 0 selected feature(s)

☒ Create a new field ☐ Update existing field

☐ Create virtual field

Output field name: ROUTE_LEN

Output field type: 1.2 Decimal number (real)

Output field length: 10 Precision: 2

Expression: $\$length/1000$

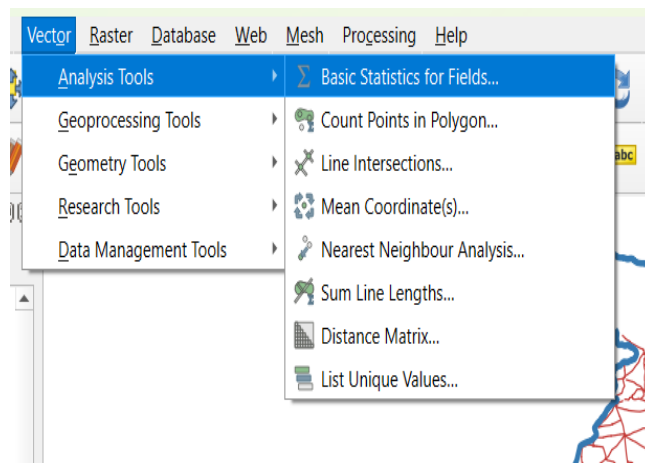
Function Editor

Feature: Operational

Preview: 29.005154962444088

OK Cancel Apply Help

FID_rail_d	F_CODE_DES	EXS_DESCR	FCO_DESCR	FID_country	ISO	ISOCOUNTRY	route_len	Route_Len_1	route	ROUTE_LEN
1	144645	Railroad	Operational	Single	102	IND	INDIA	29.01	29.01	29.01
2	145991	Railroad	Operational	Single	102	IND	INDIA	66.13	66.13	66.13
3	146001	Railroad	Operational	Single	102	IND	INDIA	2.33	2.33	2.33
4	146008	Railroad	Operational	Single	102	IND	INDIA	63.81	63.81	63.81
5	146096	Railroad	Operational	Single	102	IND	INDIA	92.71	92.71	92.71
6	146394	Railroad	Operational	Single	102	IND	INDIA	22.24	22.24	22.24
7	146464	Railroad	Operational	Single	102	IND	INDIA	77.24	77.24	77.24
8	146693	Railroad	Operational	Single	102	IND	INDIA	48.99	48.99	48.99
9	146708	Railroad	Operational	Single	102	IND	INDIA	32.27	32.27	32.27
10	147055	Railroad	Operational	Single	102	IND	INDIA	26.83	26.83	26.83
11	147427	Railroad	Operational	Single	102	IND	INDIA	17.26	17.26	17.26
12	147450	Railroad	Operational	Single	102	IND	INDIA	4.41	4.41	4.41
13	147522	Railroad	Unexamined/U...	Unknown	102	IND	INDIA	4.48	4.48	4.48
14	147524	Railroad	Unexamined/U...	Unknown	102	IND	INDIA	26.35	26.35	26.35
15	147558	Railroad	Operational	Single	102	IND	INDIA	74.95	74.95	74.95
16	147598	Railroad	Operational	Single	102	IND	INDIA	63.03	63.03	63.03
17	148187	Railroad	Unexamined/U...	Unknown	102	IND	INDIA	2.86	2.86	2.86
18	148188	Railroad	Unexamined/U...	Unknown	102	IND	INDIA	49.84	49.84	49.84
19	148191	Railroad	Unexamined/U...	Unknown	102	IND	INDIA	79.31	79.31	79.31
20	148201	Railroad	Unexamined/U...	Unknown	102	IND	INDIA	29.28	29.28	29.28
21	148486	Railroad	Operational	Single	102	IND	INDIA	29.12	29.12	29.12
22	148508	Railroad	Operational	Single	102	IND	INDIA	34.71	34.71	34.71
23	148509	Railroad	Operational	Single	102	IND	INDIA	51.17	51.17	51.17



Vector Analysis - Basic Statistics for Fields

ParametersLog

Input layer

IND_rails [EPSG:4326]

☐ Selected features only

Field to calculate statistics on

123 FID_rail_d

Statistics [optional]

[Create temporary layer]

☒ Open output file after running algorithm

Statistics report [optional]

[Save to temporary file]

Basic statistics for fields

Generates basic statistics from the analysis of a values in a field in the attribute table of a vector layer. Numeric, date, time and string fields are supported. The statistics returned will depend on the field type.

0%

Cancel

AdvancedRun as Batch Process...RunCloseHelp

123 FID_rail_d

abc F_CODE_DES

abc EXS_DESCRI

abc FCO_DESCRI

123 FID_countr

abc ISO

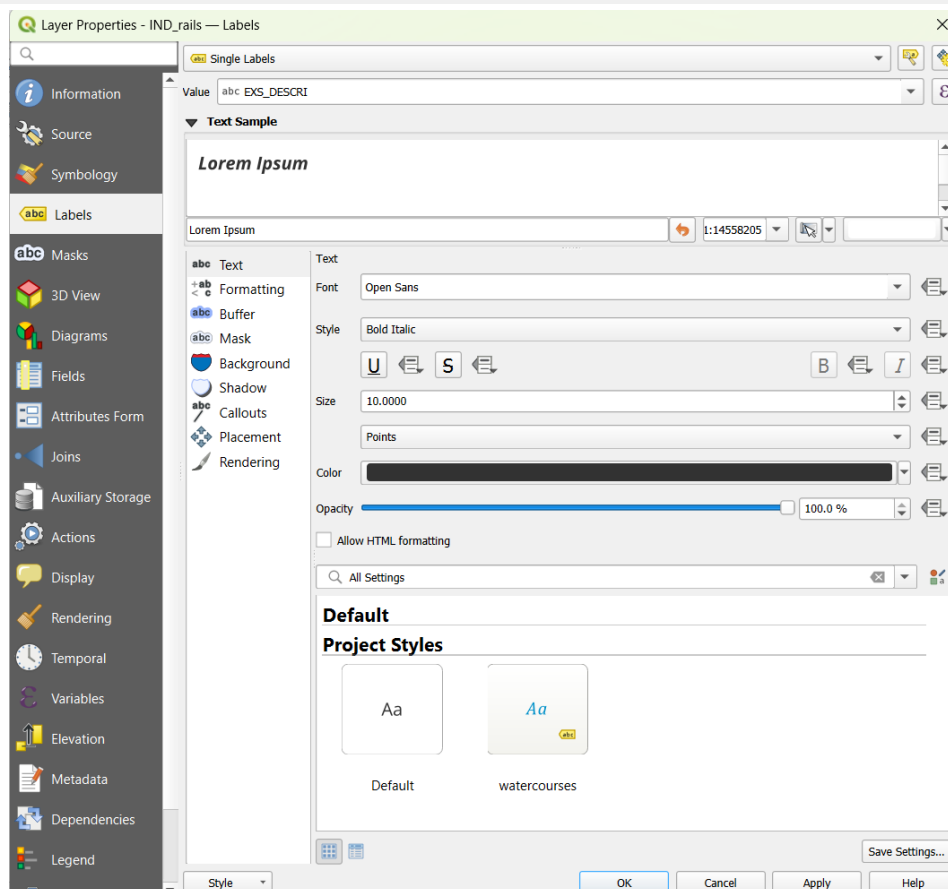
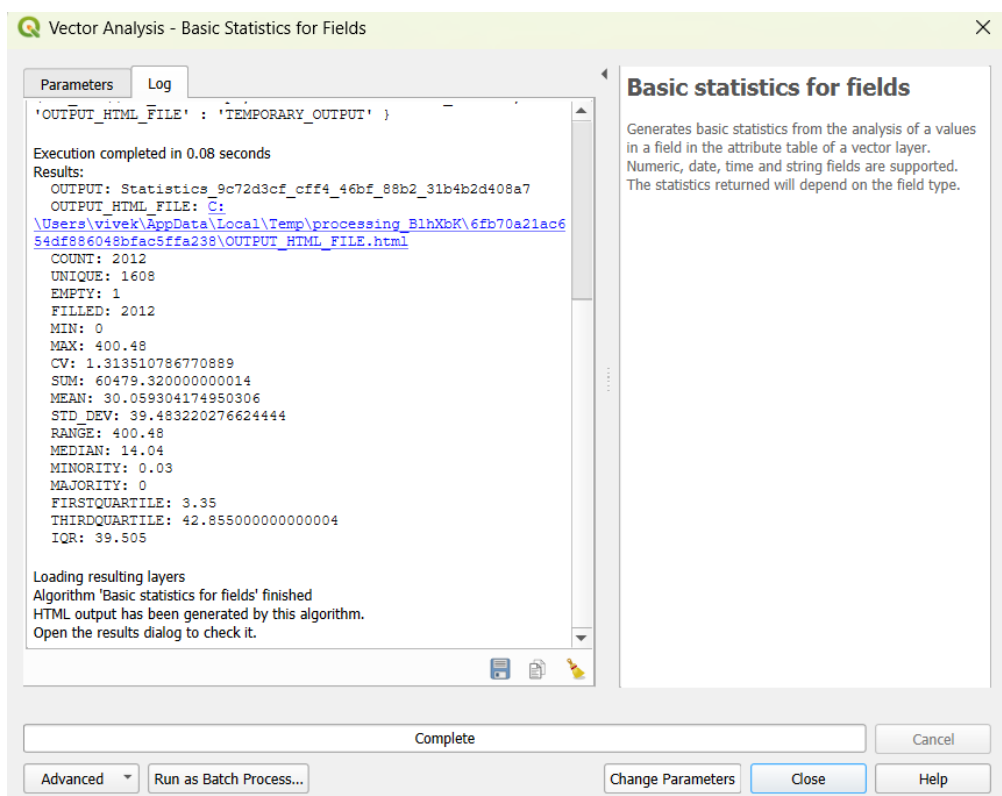
abc ISOCOUNTRY

1.2 route_len

1.2 Route_Le_1

1.2 route

1.2 ROUTE_LE_2





Results Viewer  

 Statistics report [05:16:50PM]

Algorithm: Statistics report
File path: C:\Users\vivek\AppData\Local\Temp\processing_BlhXbk\6fb70a21ac654df886048bfac5ffa238\OUTPUT_HTML_FILE.html

FINAL OUTPUT :

Analyzed field: route

Count: 2012

Unique values: 1608

NULL (missing) values: 1

NOT NULL (filled) values: 2012

Minimum value: 0

Maximum value: 400.48

Range: 400.48

Sum: 60479.320000

Mean value: 30.059304

Median value: 14.040000

Standard deviation: 39.483220276624

Coefficient of Variation: 1.313511

Minority (rarest occurring value): 0.03

Majority (most frequently occurring value): 0

First quartile: 3.350000

Third quartile: 42.855000

Interquartile Range (IQR): 39.505

CONCLUSION : Line lengths and statistics are calculated and implemented successfully .

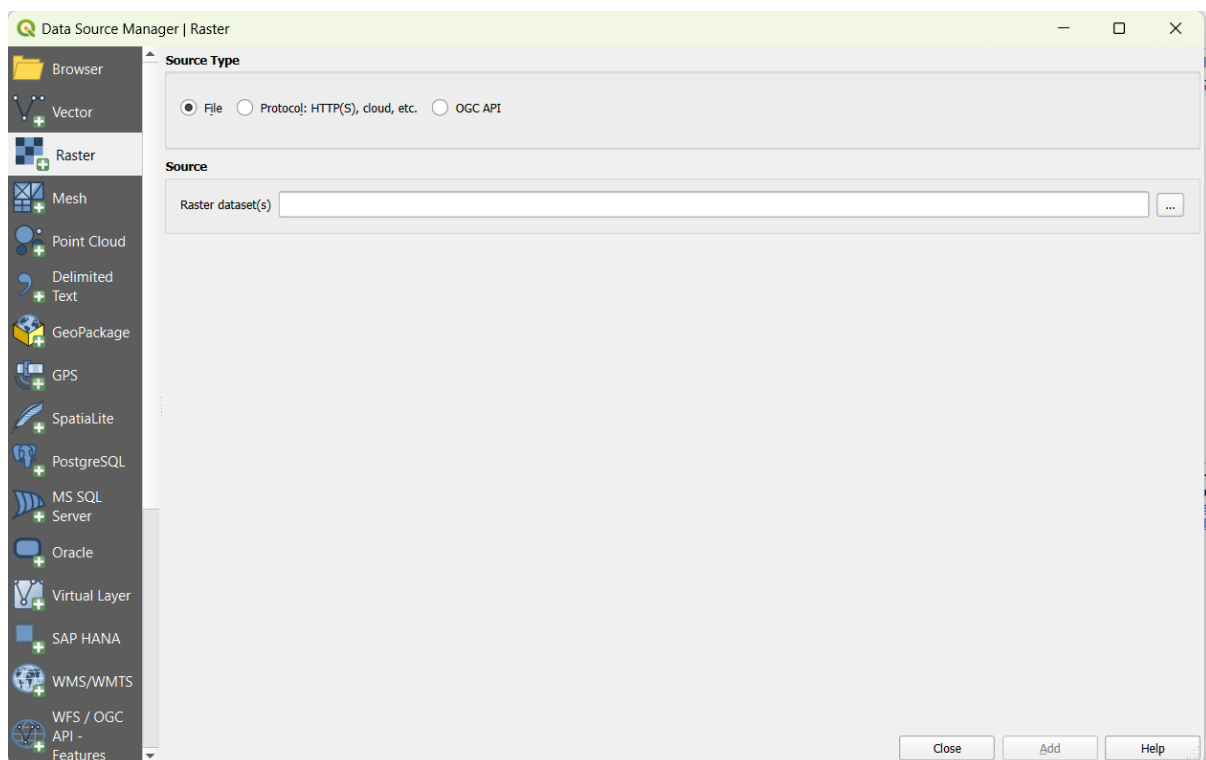
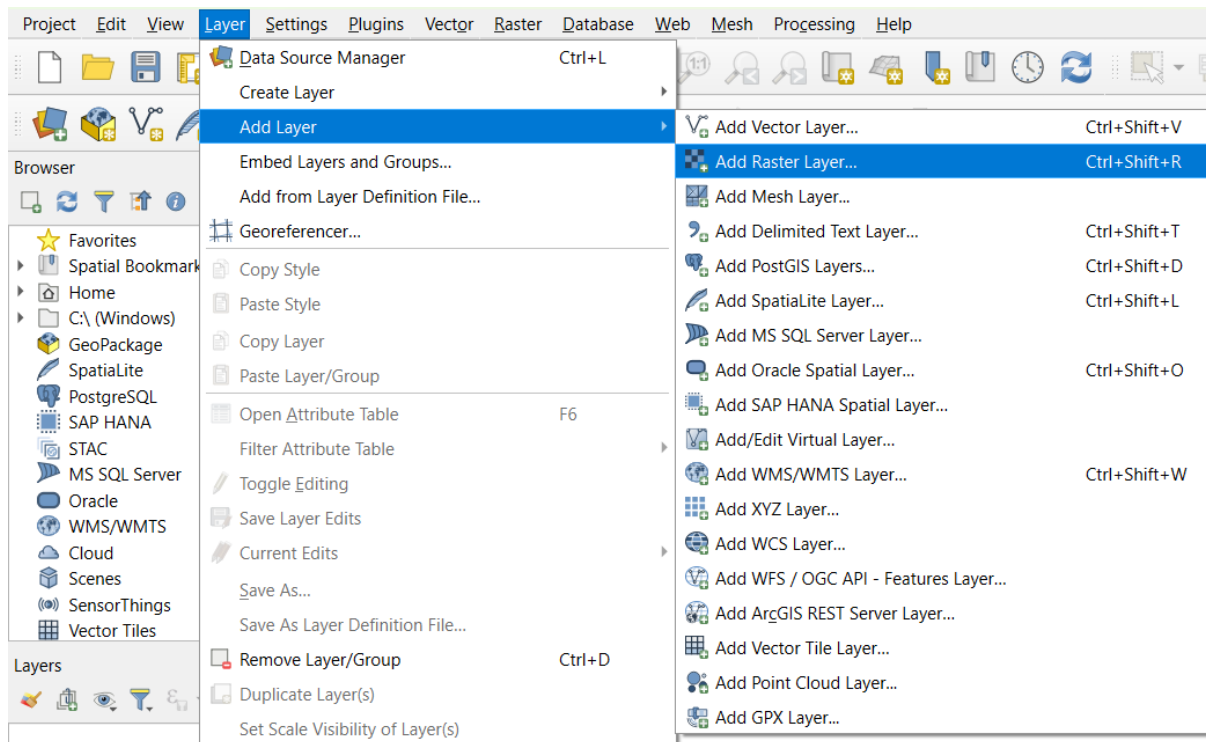
PRACTICAL NO. 3

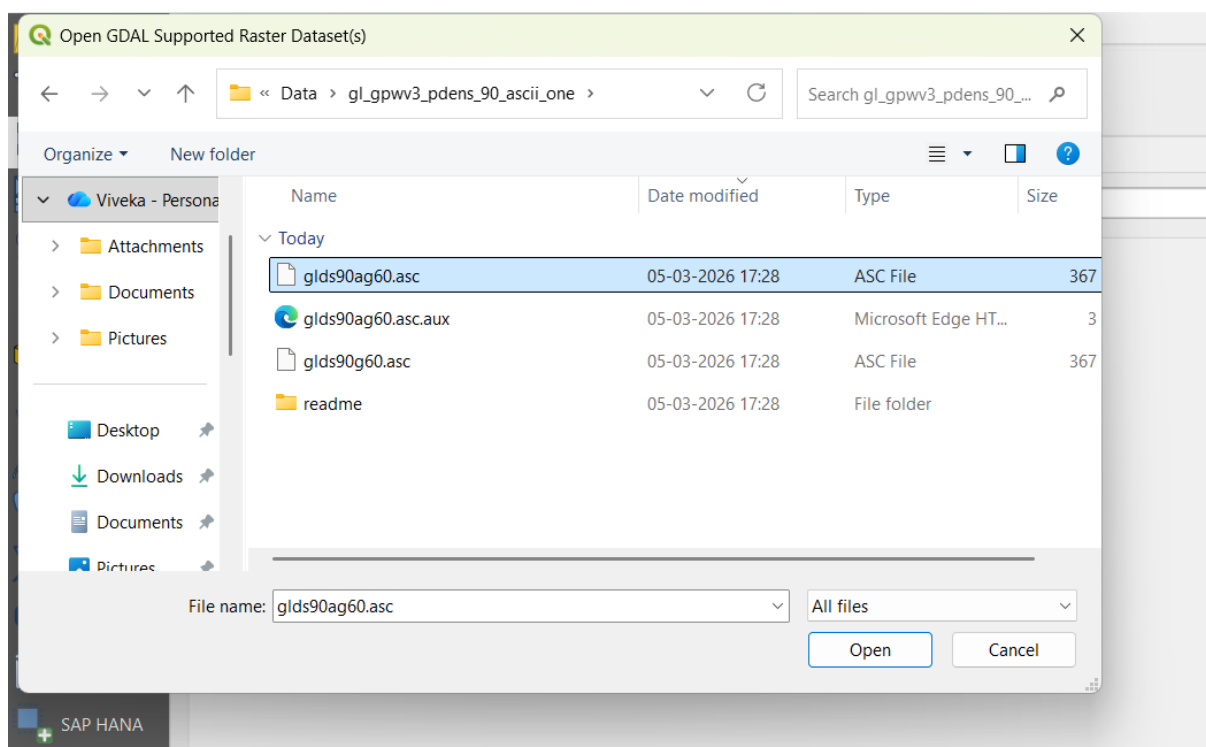
AIM : EXPLORING AND MANAGING RASTER DATA:

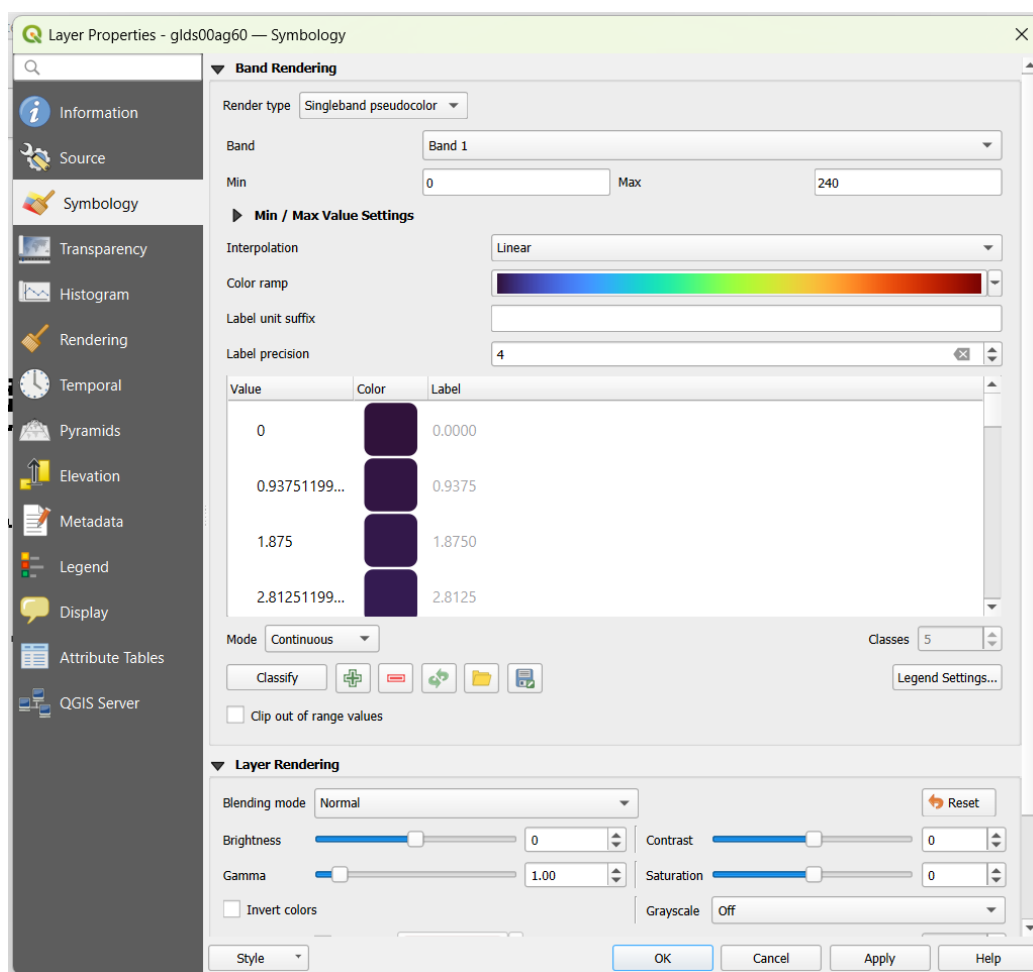
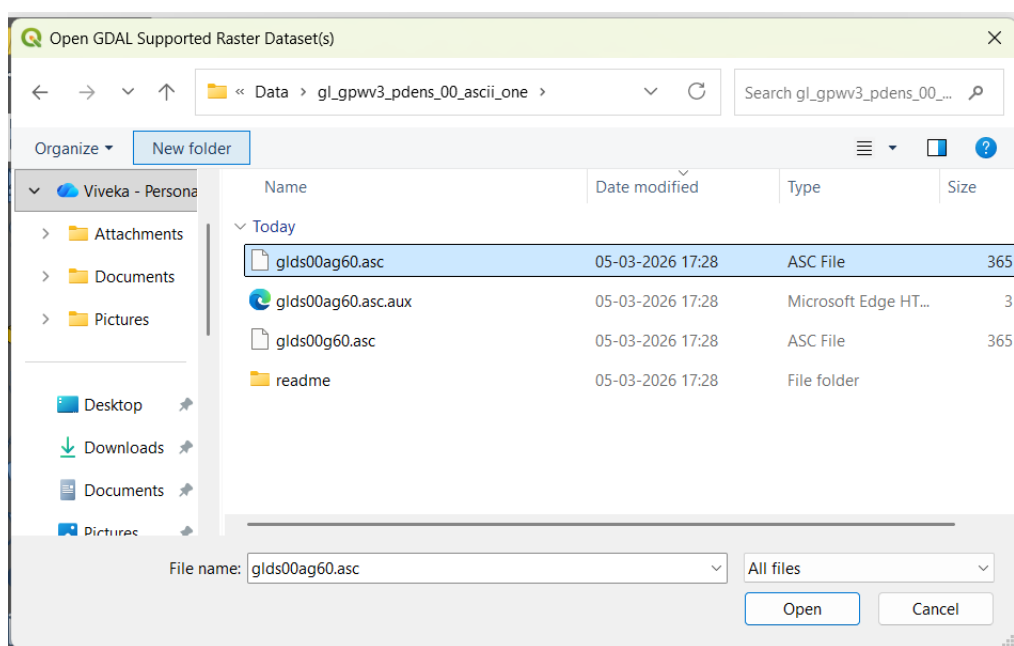
a) ADDING RASTER LAYERS

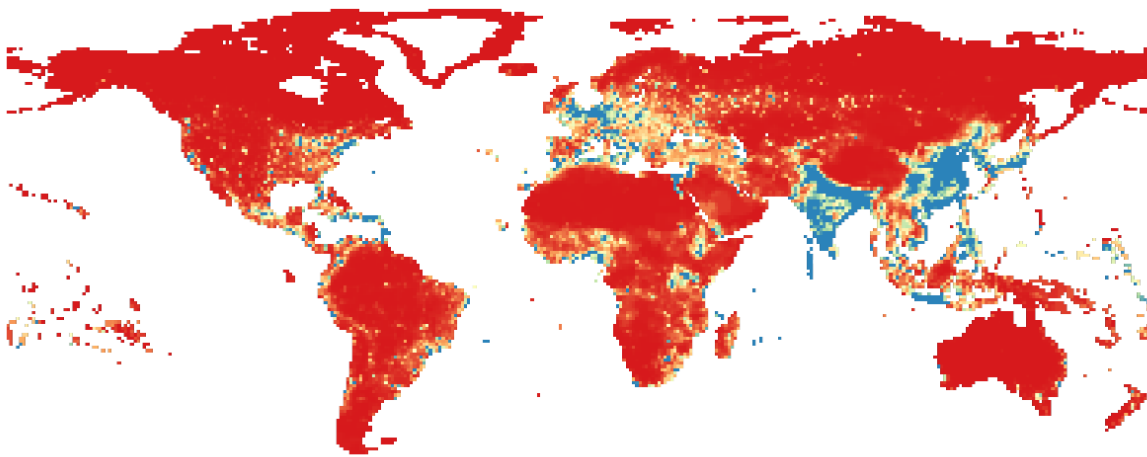
b) RASTER STYLING AND ANALYSIS

Steps:









Layer Properties - glds90ag60 — Symbology

Information
Source
Symbology
Transparency
Histogram
Rendering
Temporal
Pyramids
Elevation
Metadata
Legend
Display
Attribute Tables
QGIS Server

Band Rendering

Render type: Multiband color, Paletted/Unique values, Singleband gray, **Singleband pseudocolor**, Single color, Hillshade, Contours

Gray band: [Dropdown]

Color gradient: [Dropdown]

Max: 13760

Contrast enhancement: Stretch to MinMax

Legend Settings...

Layer Rendering

Blending mode: Normal [Reset]

Brightness: [Slider] 0 [Dropdown]

Gamma: [Slider] 1.00 [Dropdown]

Contrast: [Slider] 0 [Dropdown]

Saturation: [Slider] 0 [Dropdown]

☐ Invert colors

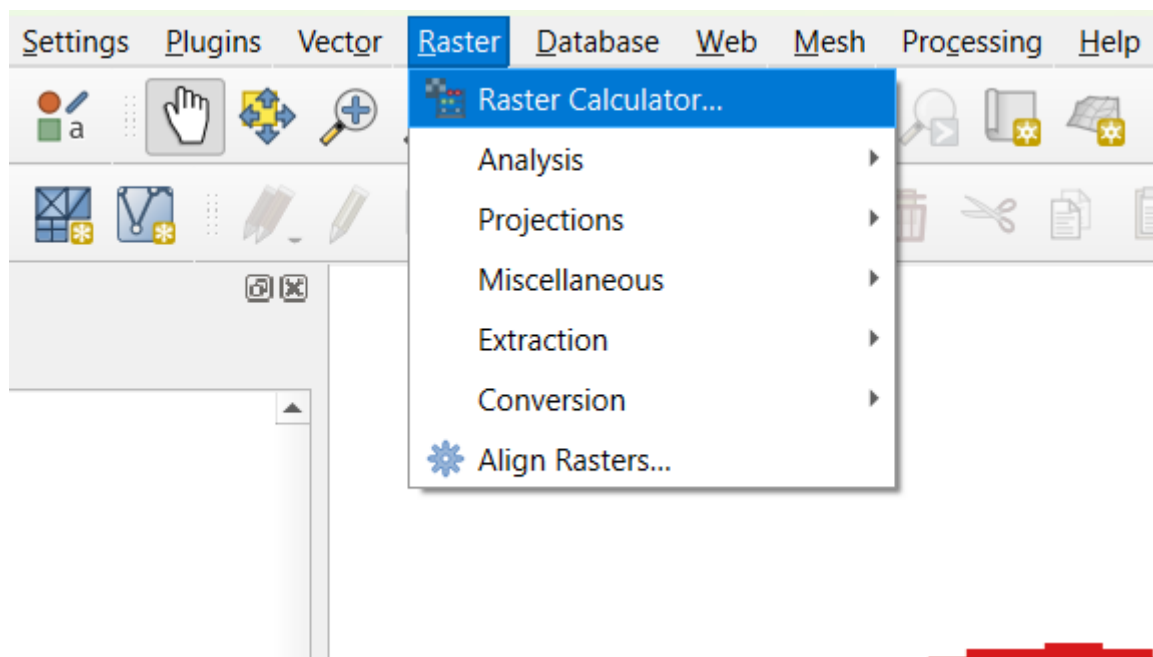
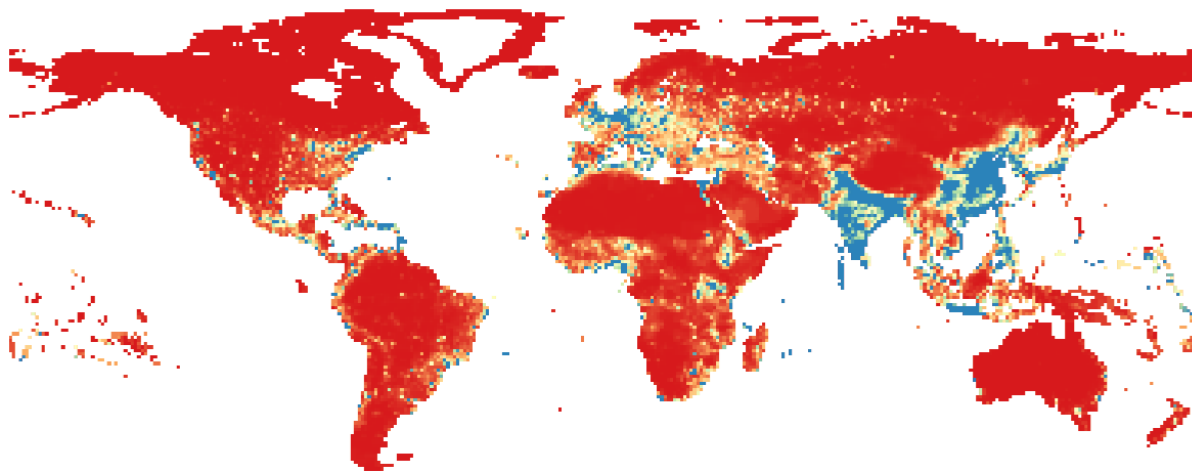
Grayscale: Off [Dropdown]

Hue: ☐ Colorize [Dropdown] Strength: [Slider] 100% [Dropdown]

Resampling

Zoomed: in Nearest Neighbour out Nearest Neighbour Oversampling: 2.00 [Dropdown] ☐ Early resampling

Style [Dropdown] OK Cancel Apply Help



Raster Bands

glds00ag60@1
glds90ag60@1

Result Layer

☐ Create on-the-fly raster instead of writing layer to disk

Output layer

Output format
GeoTIFF

Spatial Extent

Use Selected Layer Extent

X min
-180.00000
X max
180.00000
Y min
-58.00000
Y max
85.00000

Resolution

Columns
360
Rows
143

Output CRS
invalid projection

☒ Add result to project

Operators

+
-
<
<=
*
/
>
>=
(
)
=
!=
min
max
abs
^
IF
AND
OR
sqrt
cos
sin
tan
log10
acos
asin
atan
ln

Raster Calculator Expression

OK

Cancel

Help

Raster Bands

glds00ag60@1
glds90ag60@1

Result Layer

☐ Create on-the-fly raster instead of writing layer to disk

Output layer
pupulation_difference

Output format
GeoTIFF

Spatial Extent

Use Selected Layer Extent

X min
-180.00000
X max
180.00000
Y min
-58.00000
Y max
85.00000

Resolution

Columns
360
Rows
143

Output CRS
invalid projection

☒ Add result to project

Operators

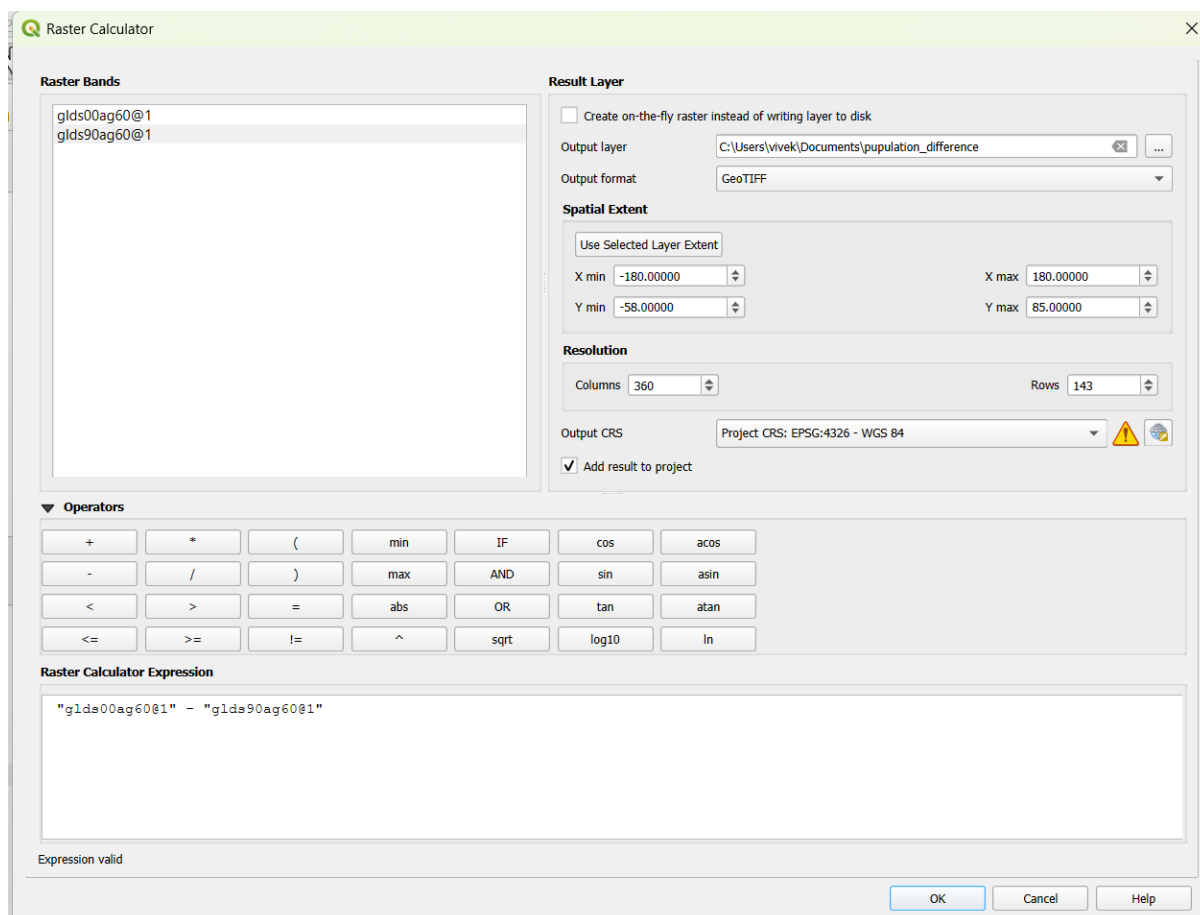
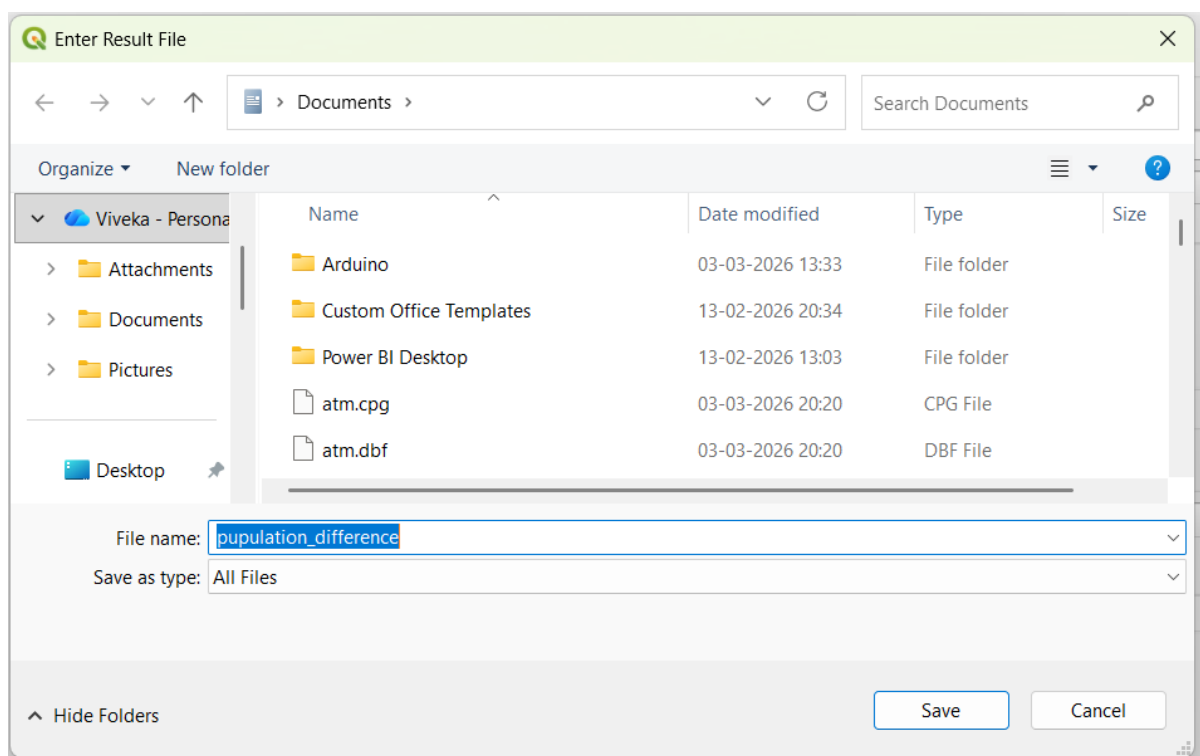
+
-
<
<=
*
/
>
>=
(
)
=
!=
min
max
abs
^
IF
AND
OR
sqrt
cos
sin
tan
log10
acos
asin
atan
ln

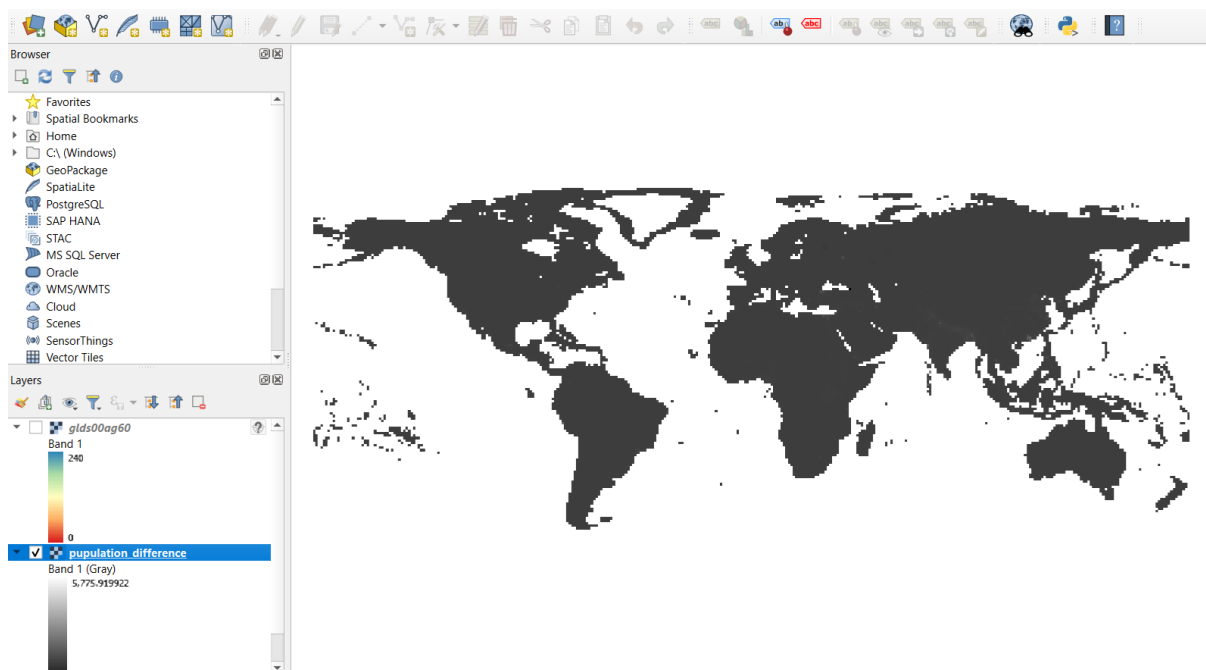
Raster Calculator Expression

OK

Cancel

Help





Layer Properties - population_difference — Symbology

Band Rendering

Render type: Singleband gray

Gray band: Band 1 (Gray)

Color gradient: Black to White

Min: -1840.73 Max: 5775.92

Contrast enhancement: Stretch to MinMax

Min / Max Value Settings

Layer Rendering

Blending mode: Normal

Brightness: 0

Gamma: 1.00

Invert colors: ☐

Hue: ☐ Colorize ☐ Strength: 100%

Contrast: 0

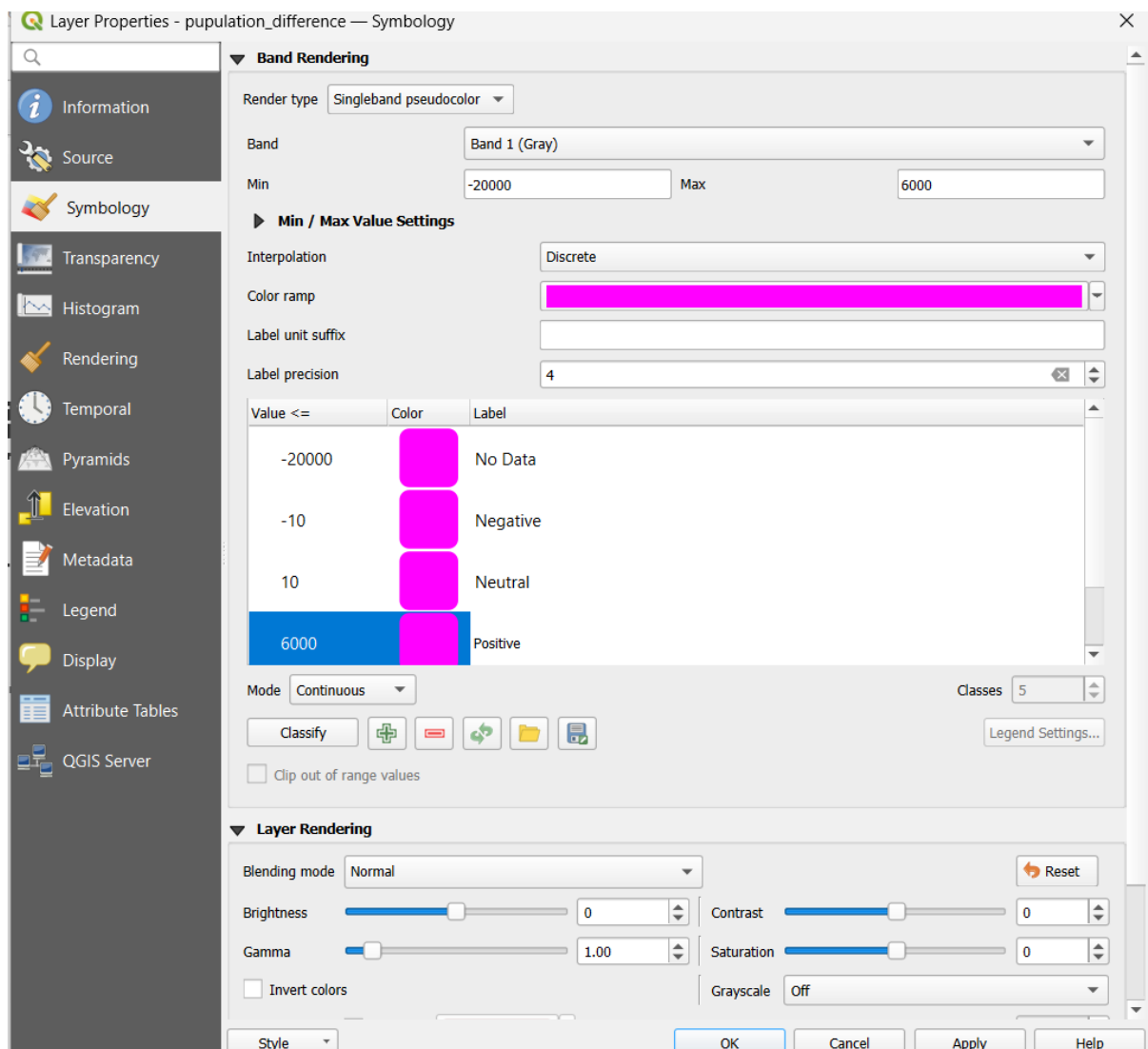
Saturation: 0

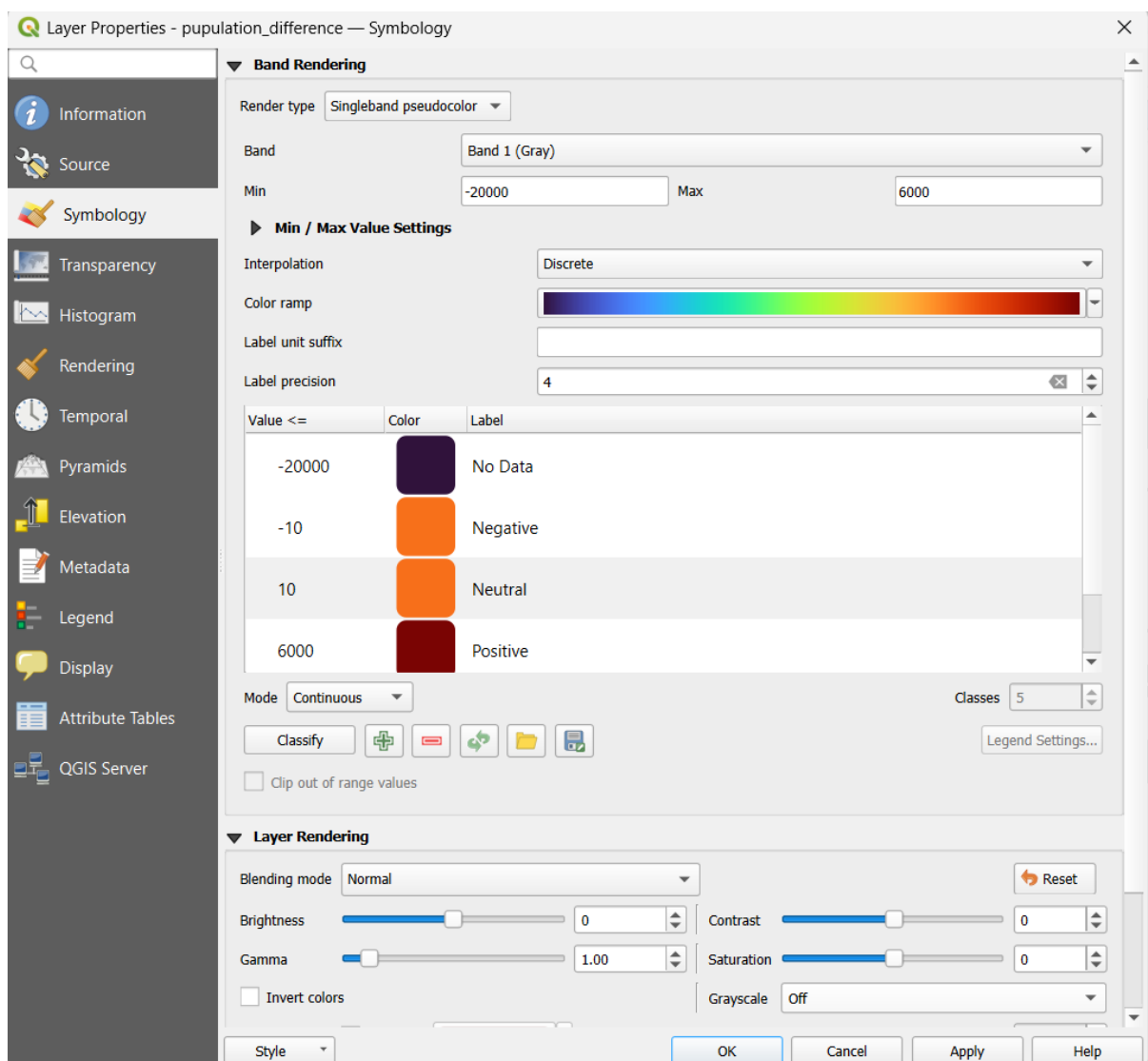
Grayscale: Off

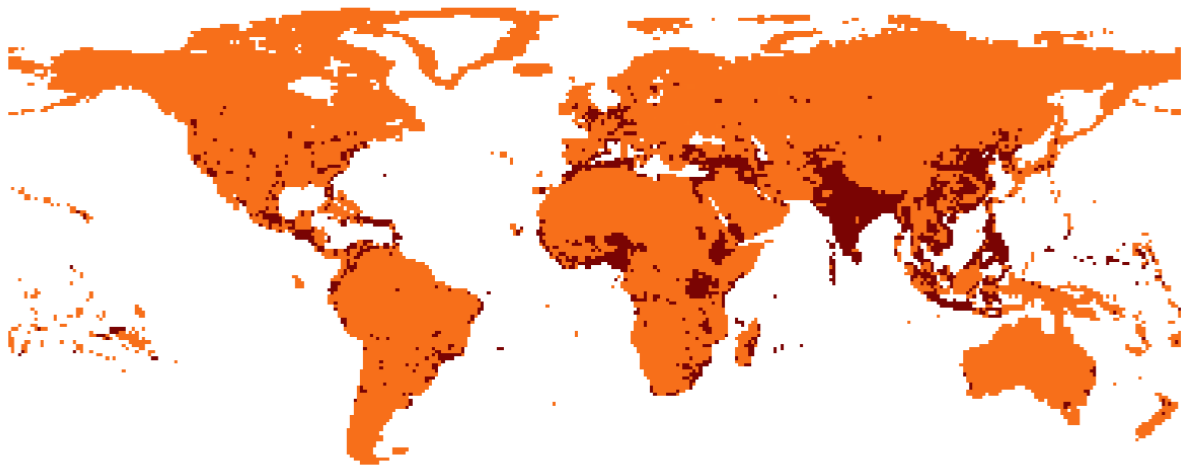
Resampling

Zoomed: in Nearest Neighbour out Nearest Neighbour Oversampling: 2.00 Early resampling: ☐

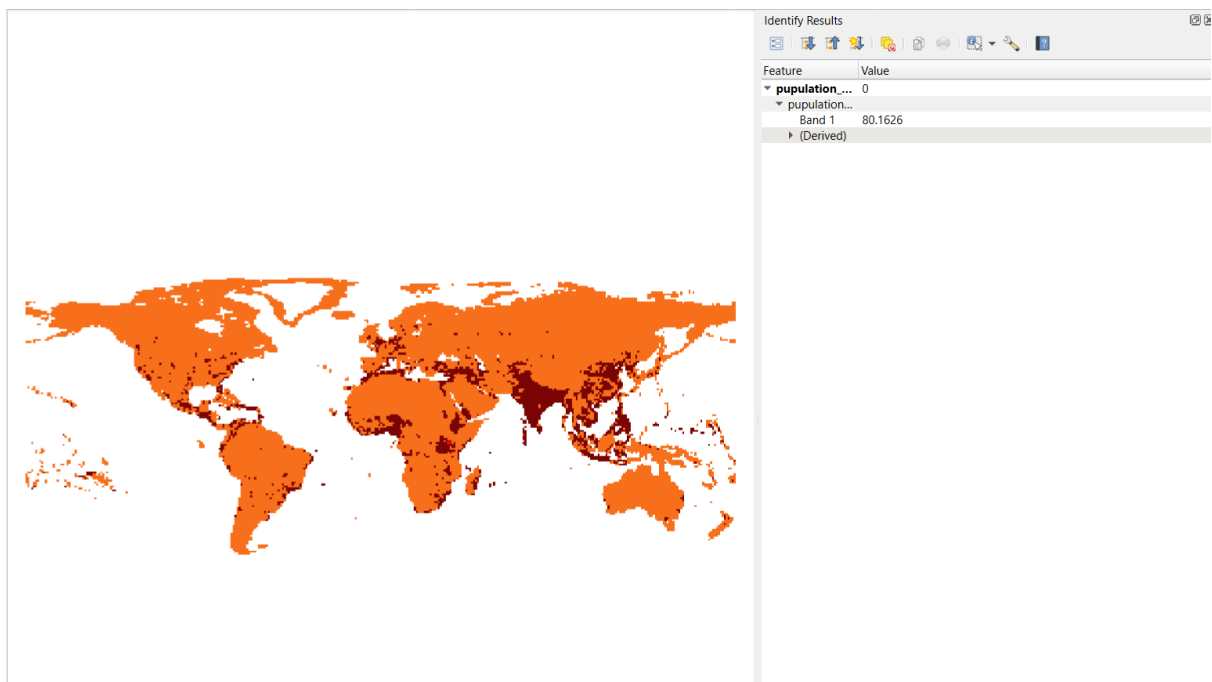
Style OK Cancel Apply Help







FINAL OUTPUT :

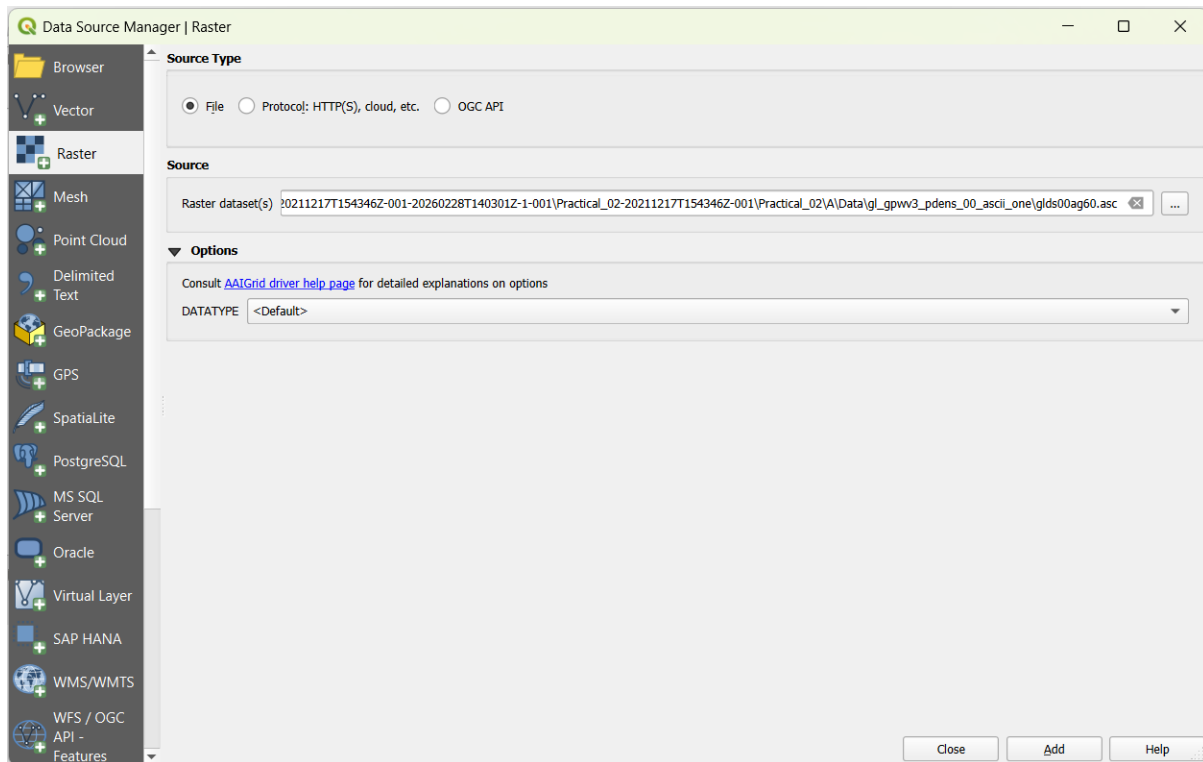
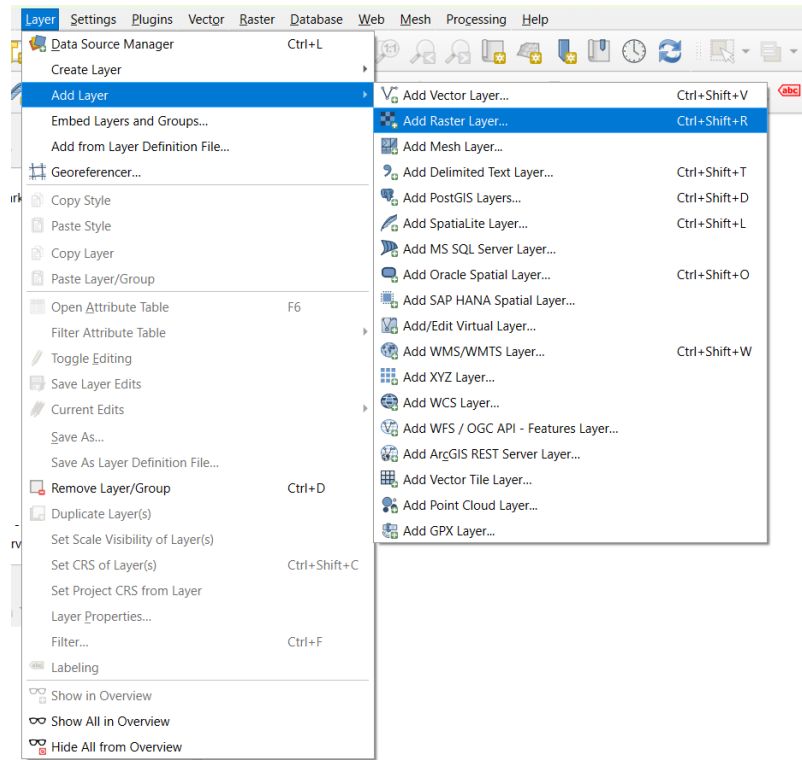


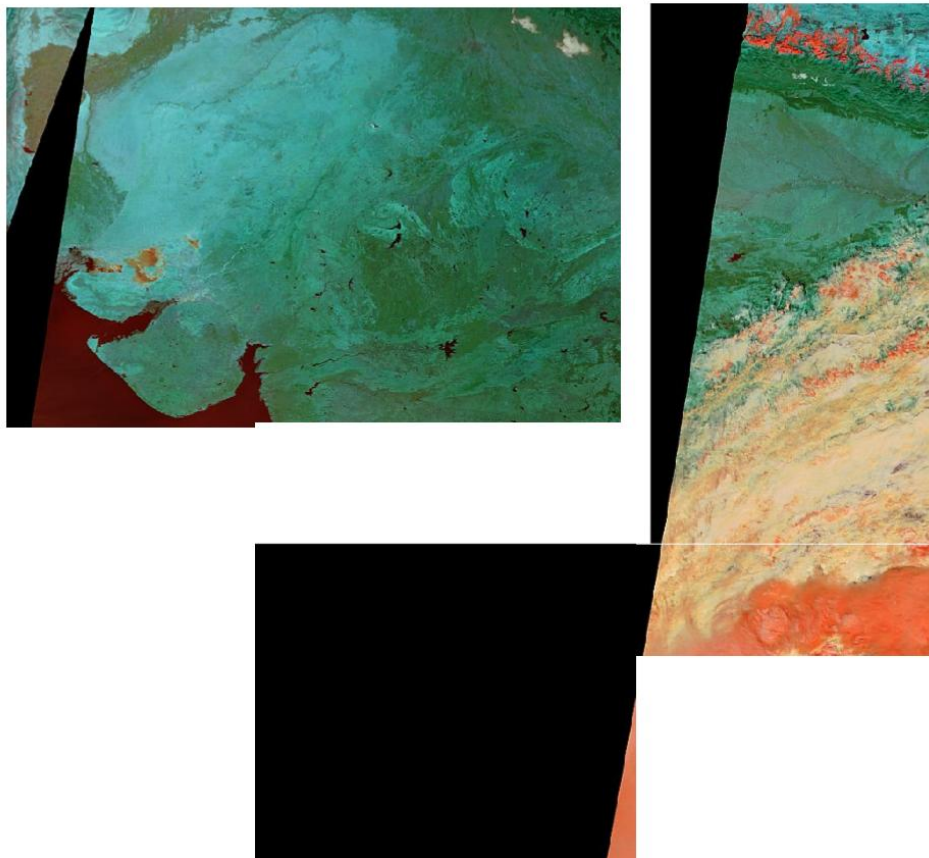
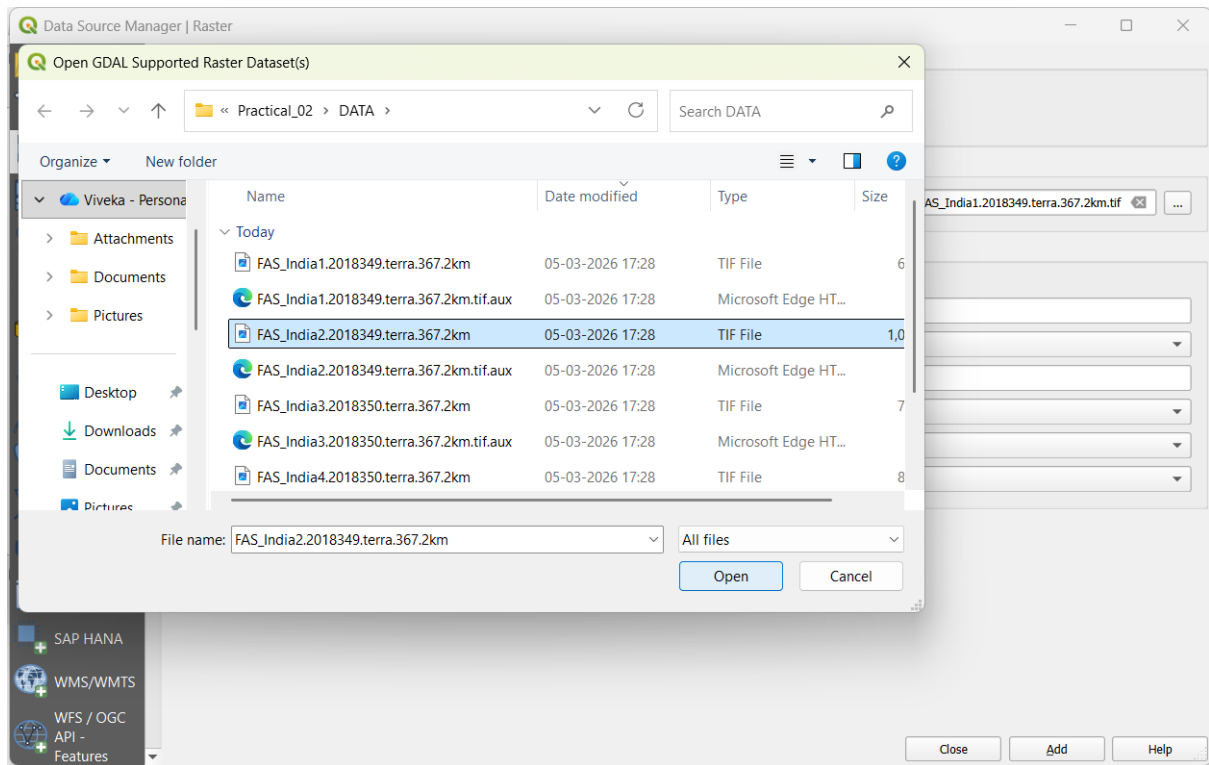
CONCLUSION : Raster data is explored and managed successfully .

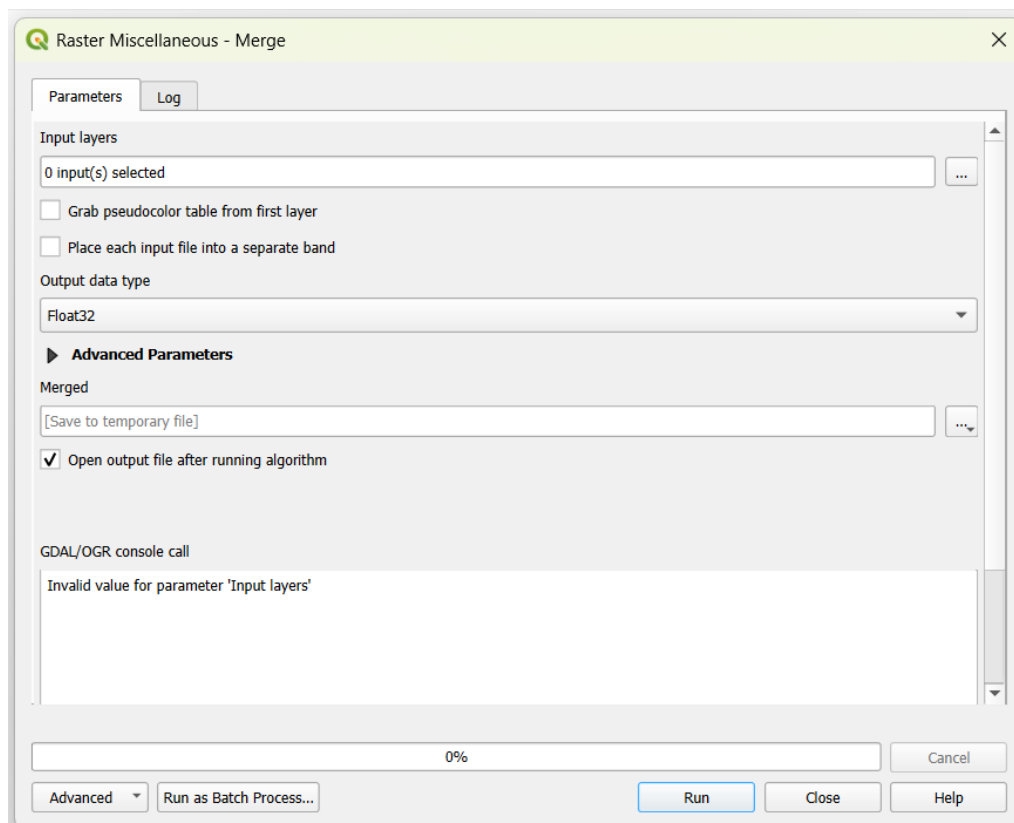
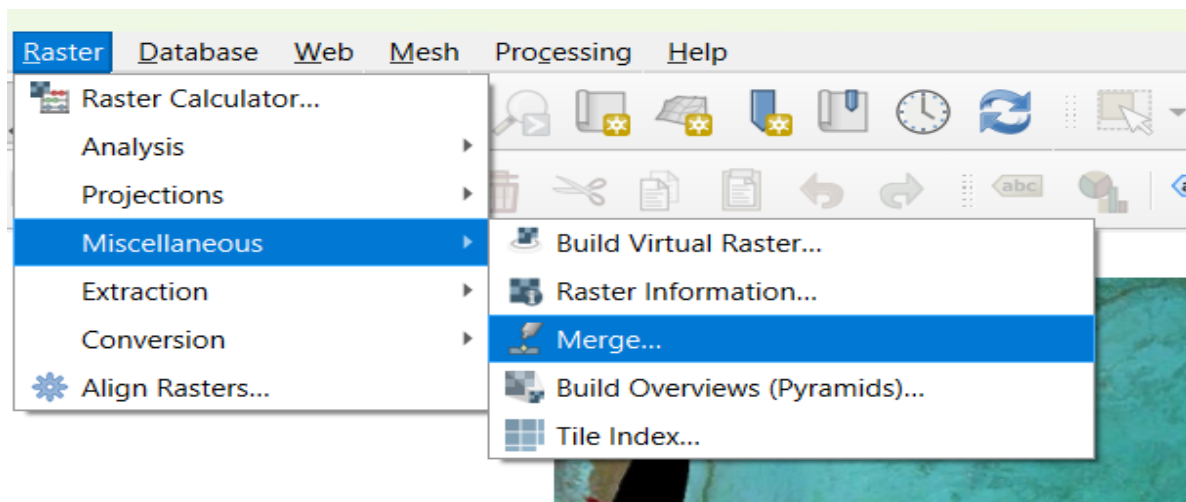
PRACTICAL NO. 4

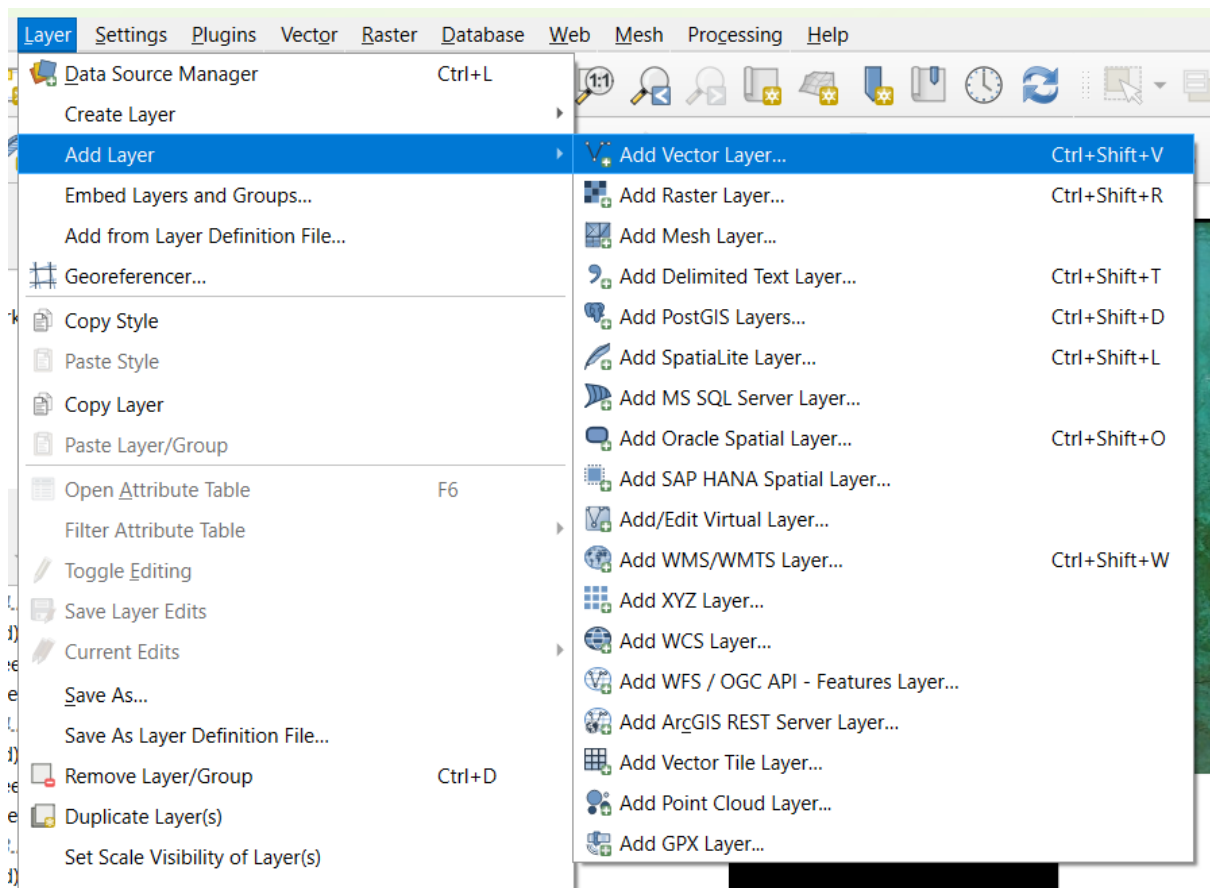
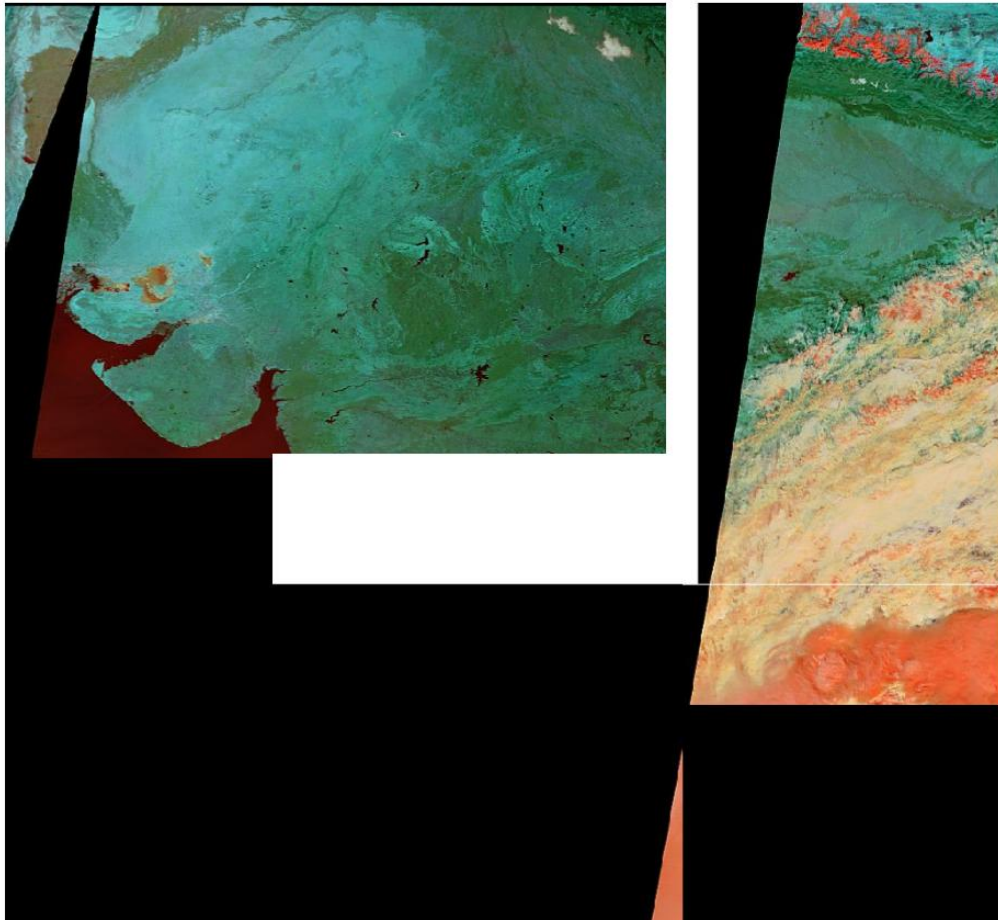
AIM : RASTER MOSAICKING AND CLIPPING

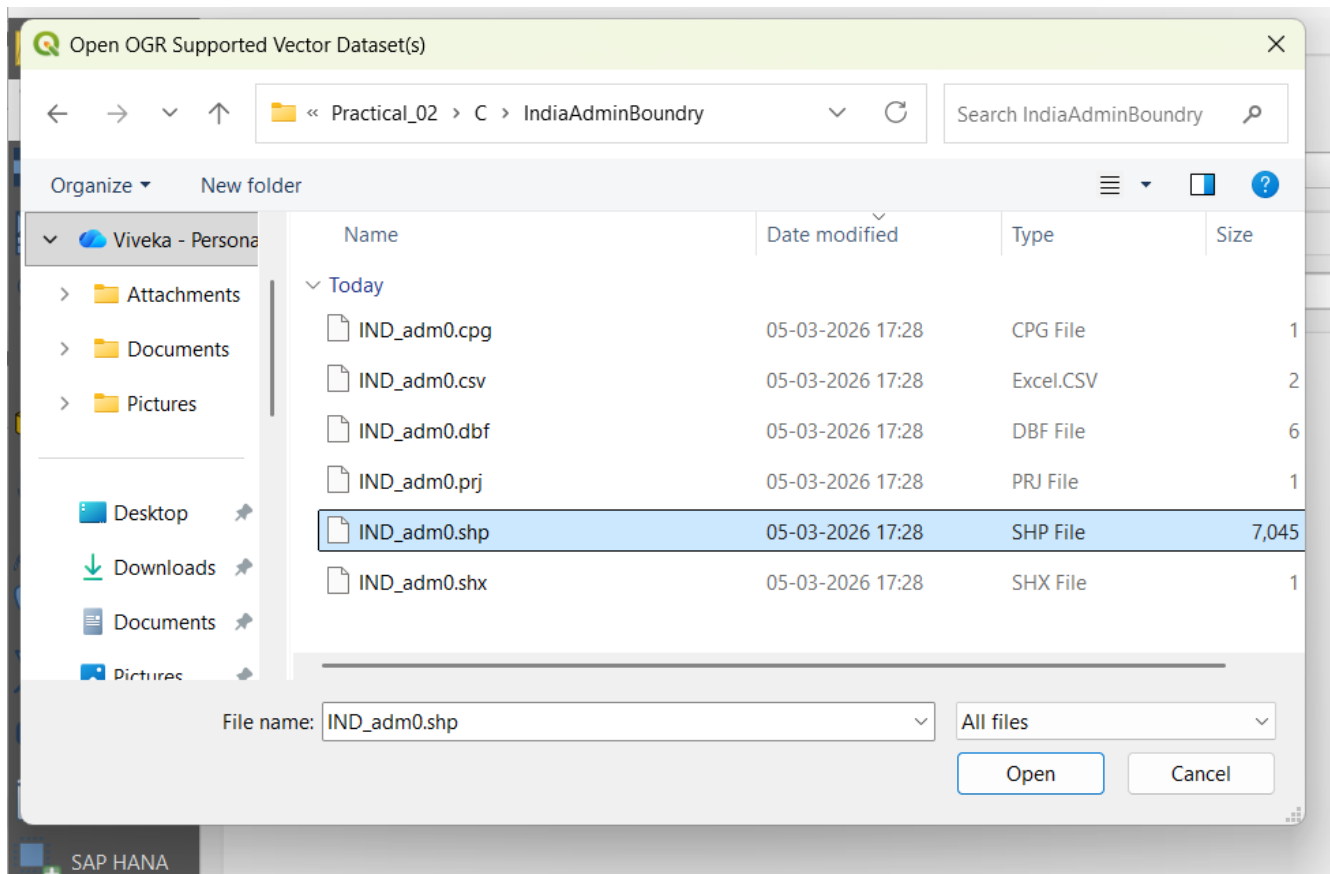
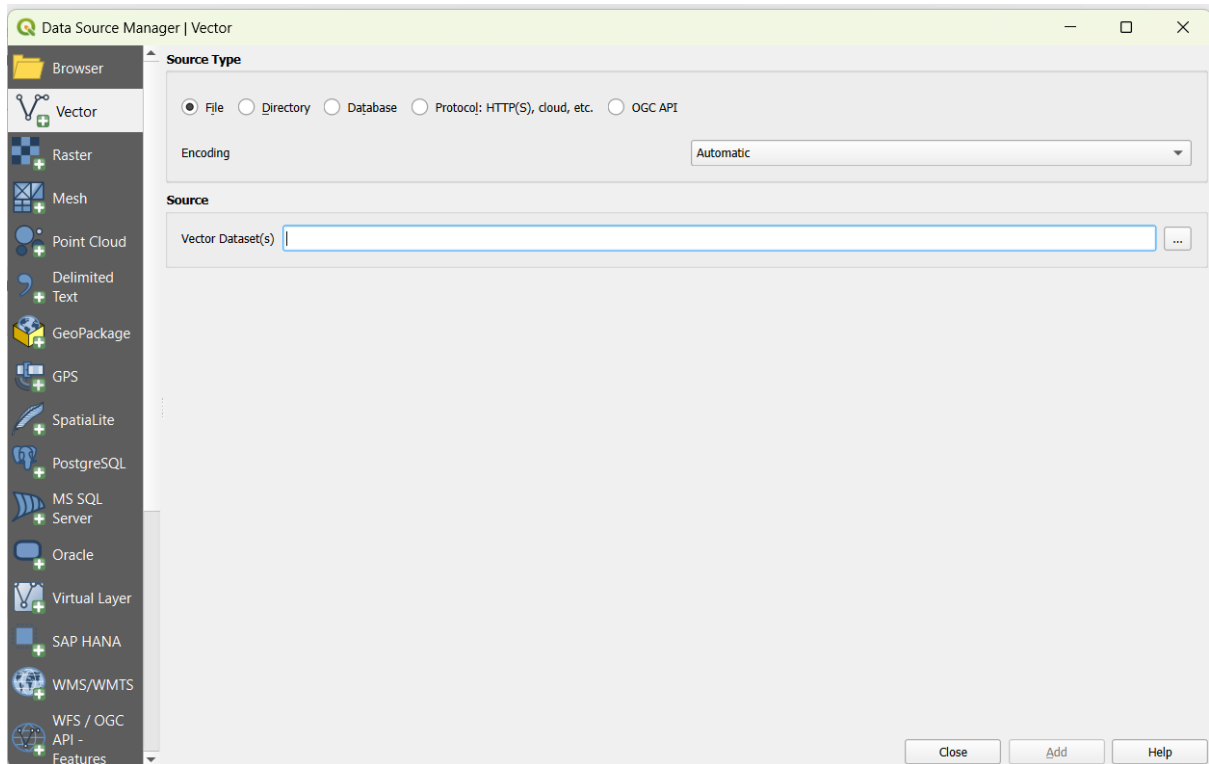
Steps:

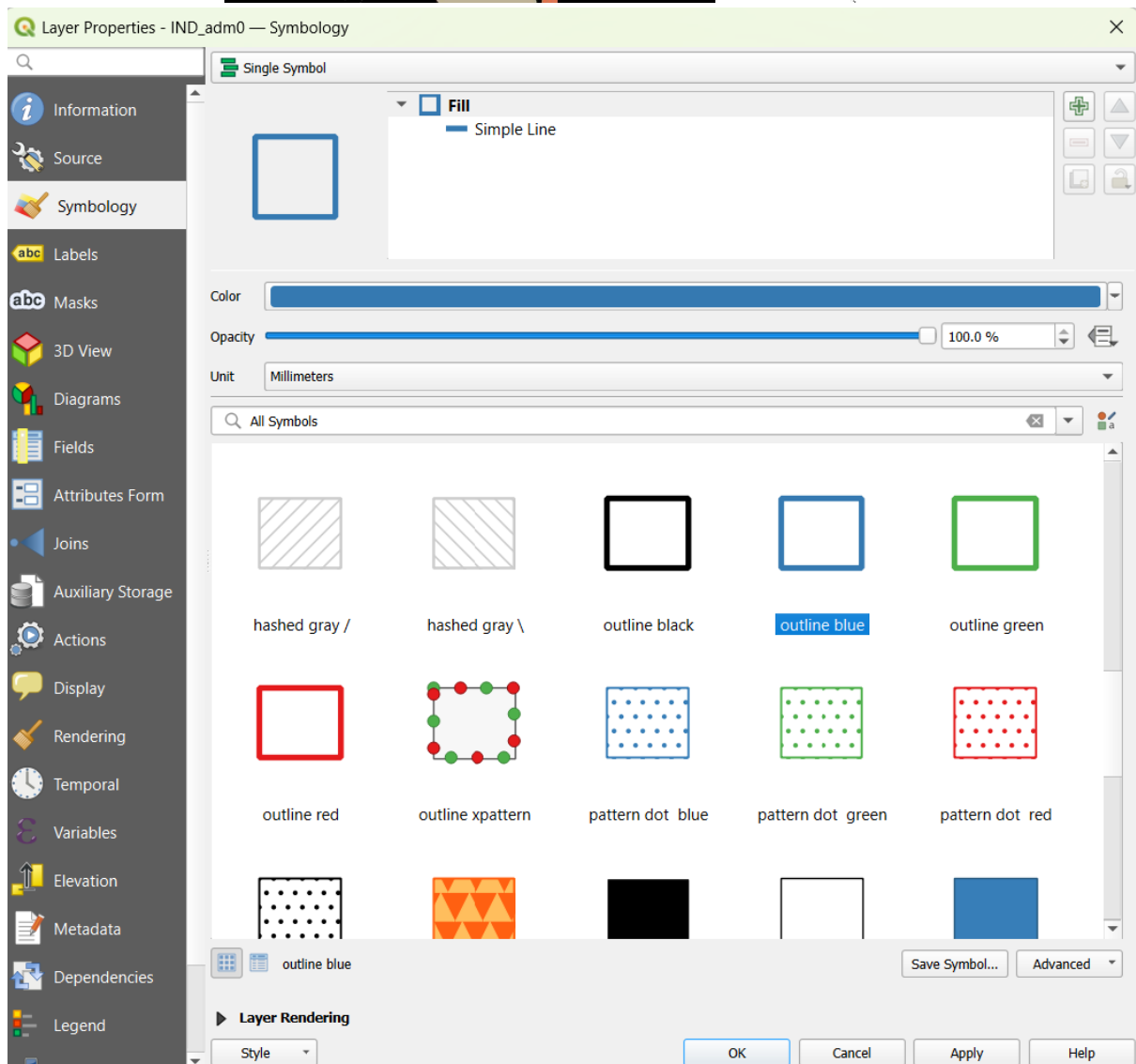


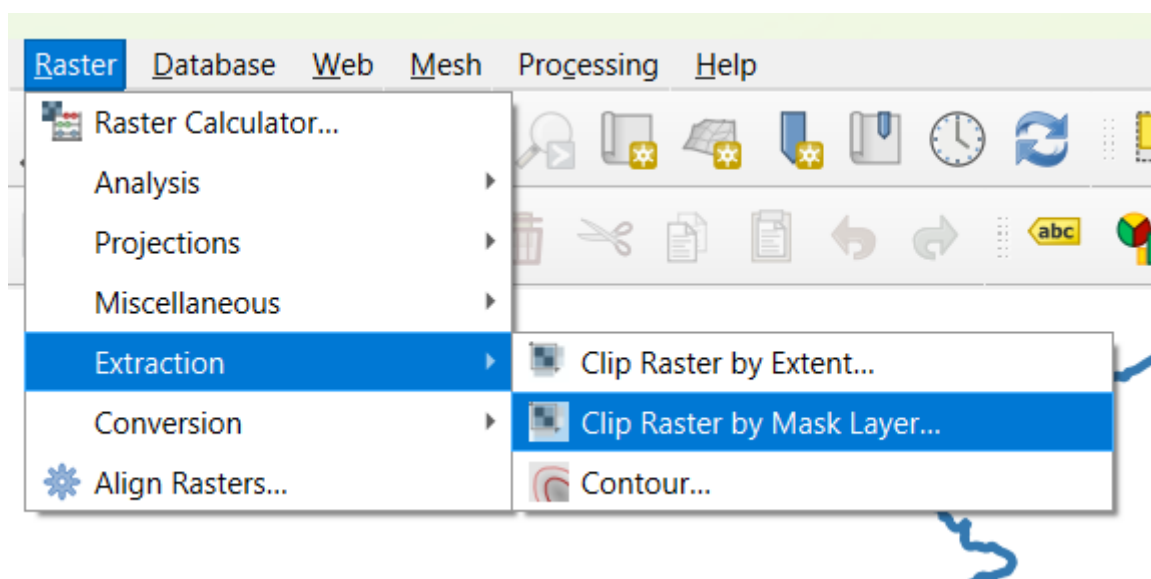
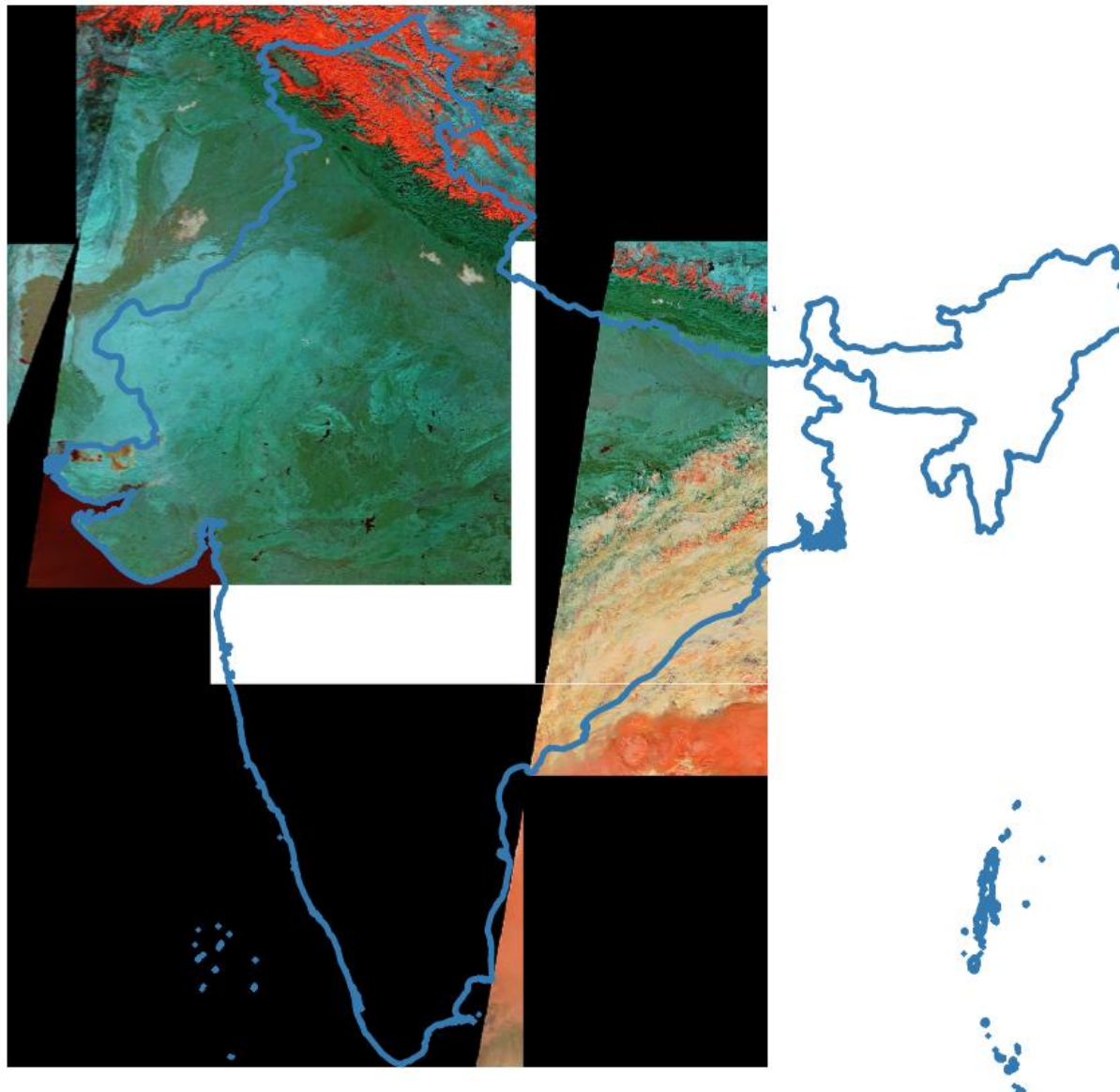












Raster Extraction - Clip Raster by Mask Layer

Parameters Log

Input layer

Mask layer

IND_adm0 [EPSG:4326]

☐ Selected features only

Source CRS [optional]

Target CRS [optional]

Target extent [optional]

Not set

Assign a specified NoData value to output bands [optional]

Not set

☐ Create an output alpha band

☒ Match the extent of the clipped raster to the extent of the mask layer

☐ Keep resolution of input raster

0%

Cancel

Advanced Run as Batch Process... Run Close Help

Raster Extraction - Clip Raster by Mask Layer

Parameters Log

Input layer

Merged [EPSG:4326]

Mask layer

IND_adm0 [EPSG:4326]

☐ Selected features only

Source CRS [optional]

Target CRS [optional]

Target extent [optional]

Not set

Assign a specified NoData value to output bands [optional]

Not set

☐ Create an output alpha band

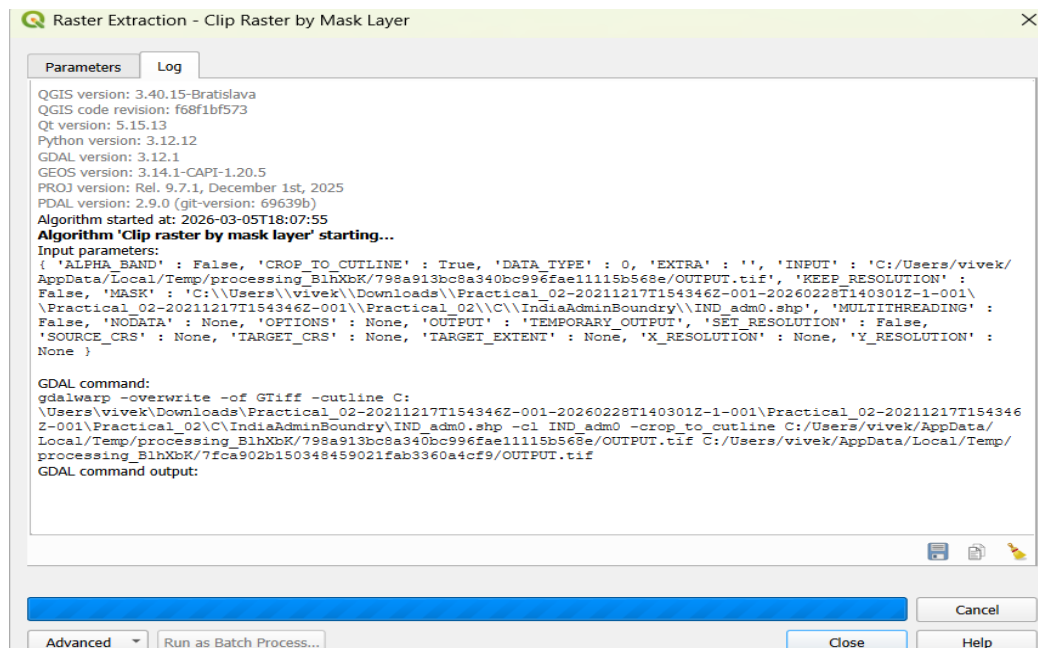
☒ Match the extent of the clipped raster to the extent of the mask layer

☐ Keep resolution of input raster

0%

Cancel

Advanced Run as Batch Process... Run Close Help



FINAL OUTPUT :

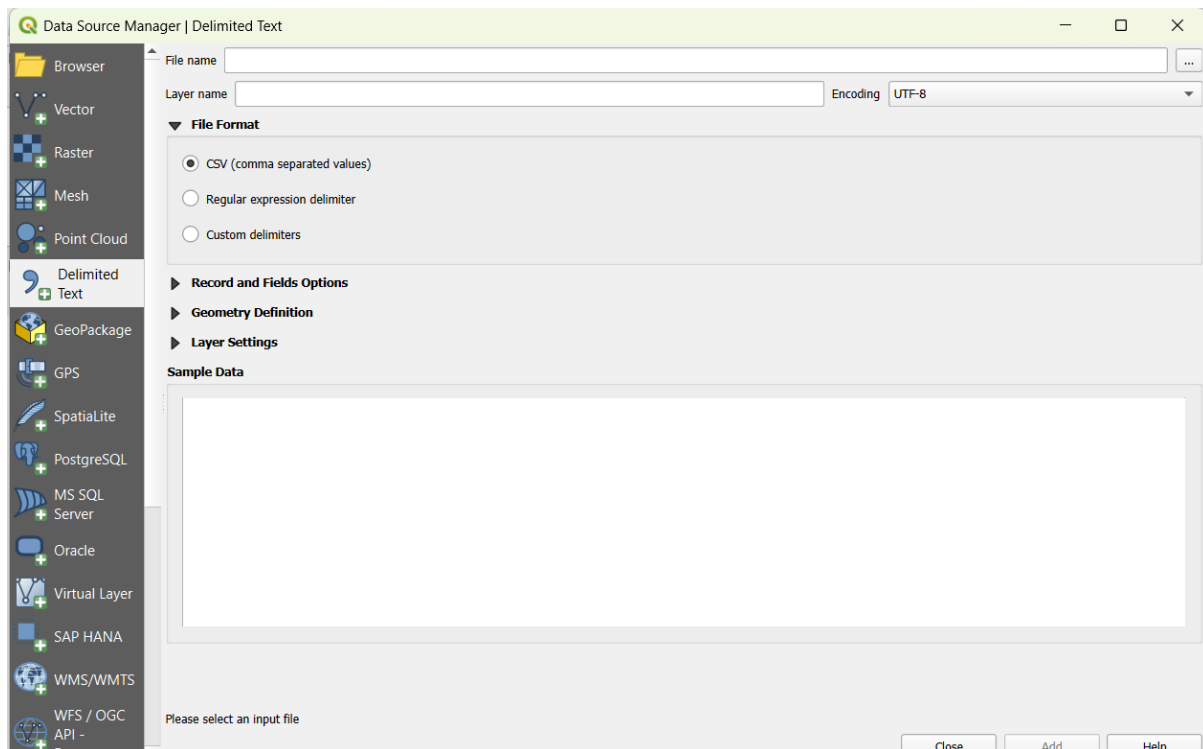
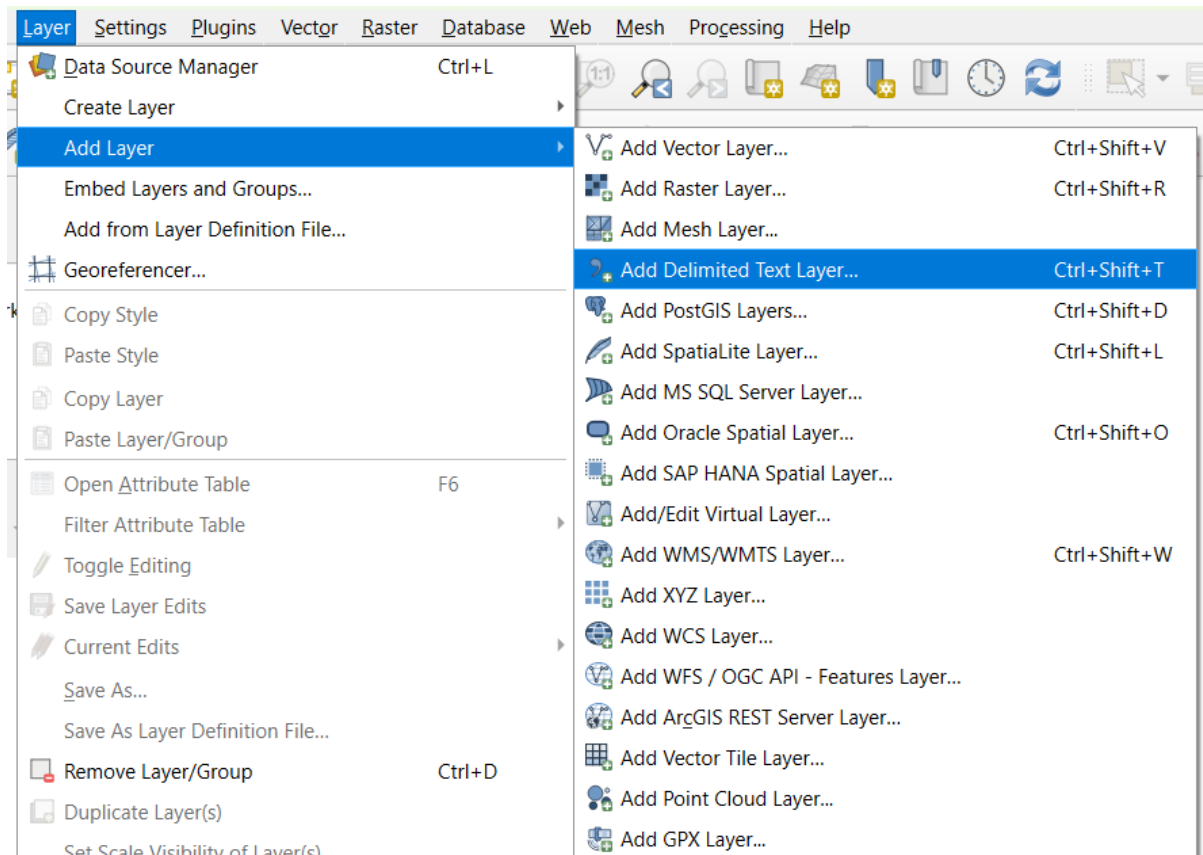


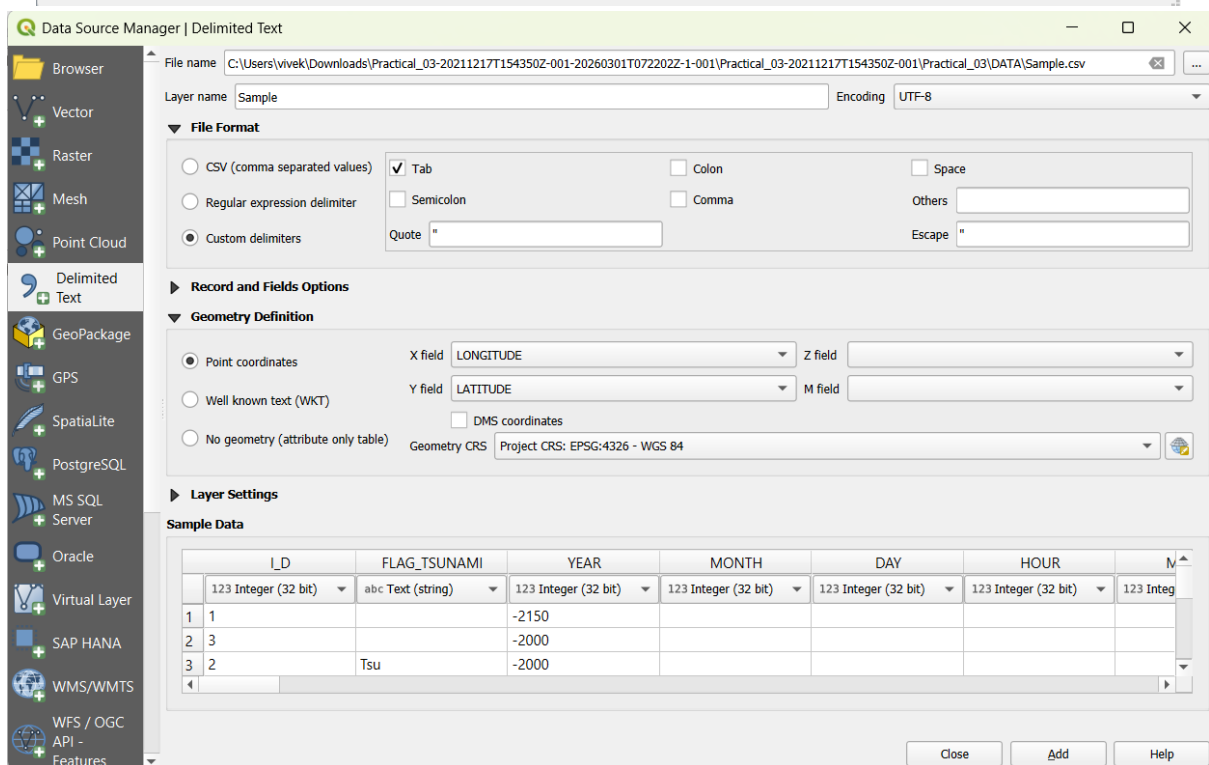
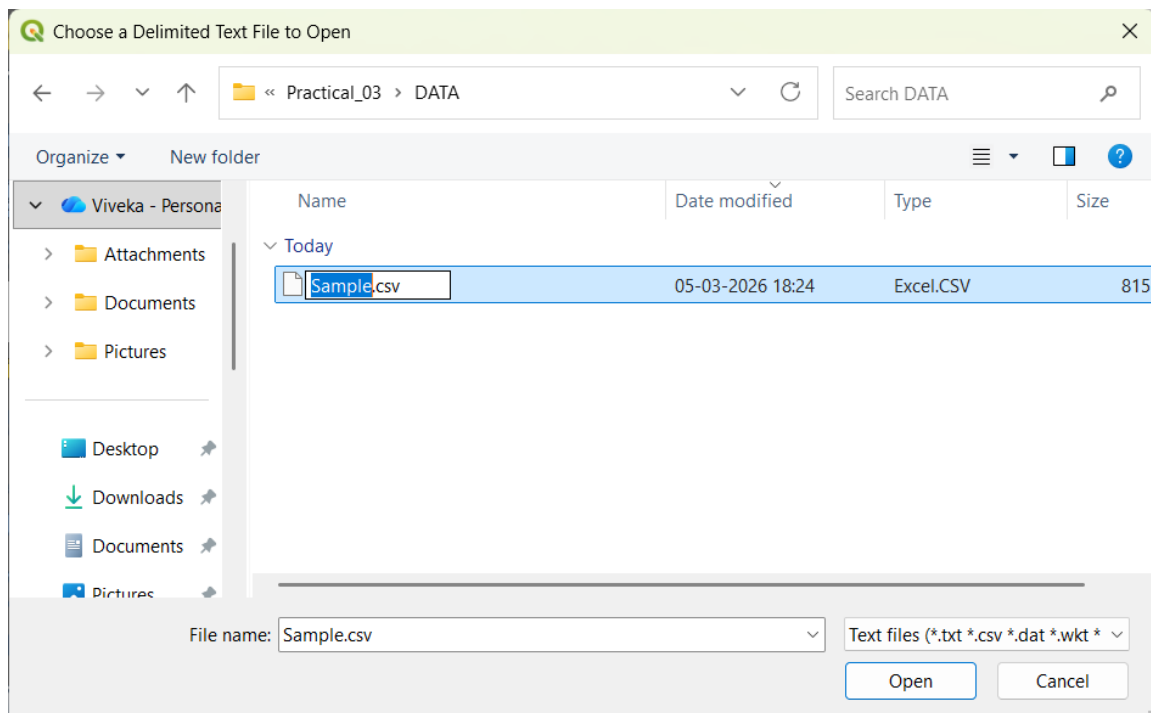
CONCLUSION : Raster Mosaicking and Clipping is implemented successfully .

PRACTICAL NO. 5

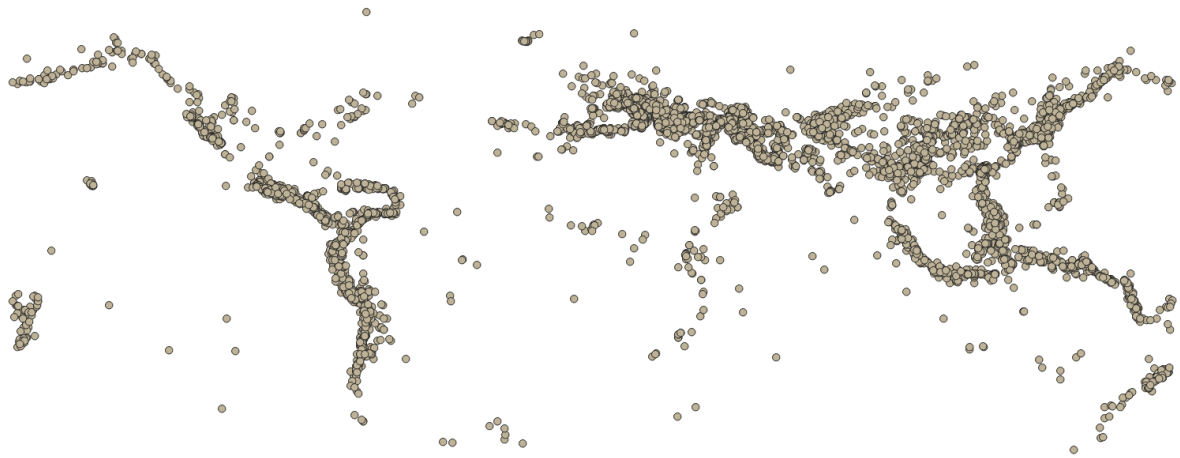
AIM :IMPORTING SPREAD SHEETS OR CSV FILE .

Steps:





FINAL OUTPUT :



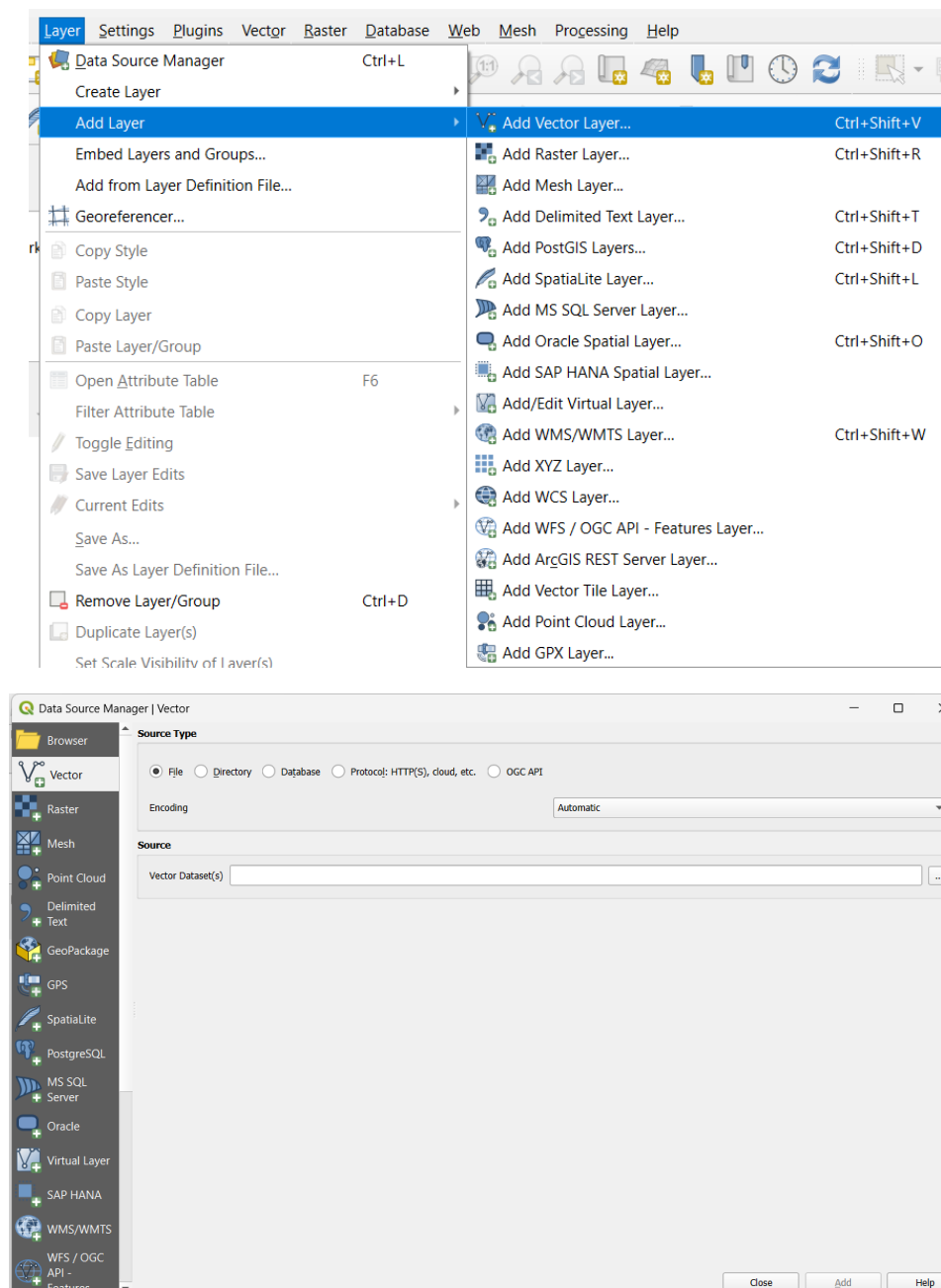
CONCLUSION : spreadsheets or csv file is imported successfully .

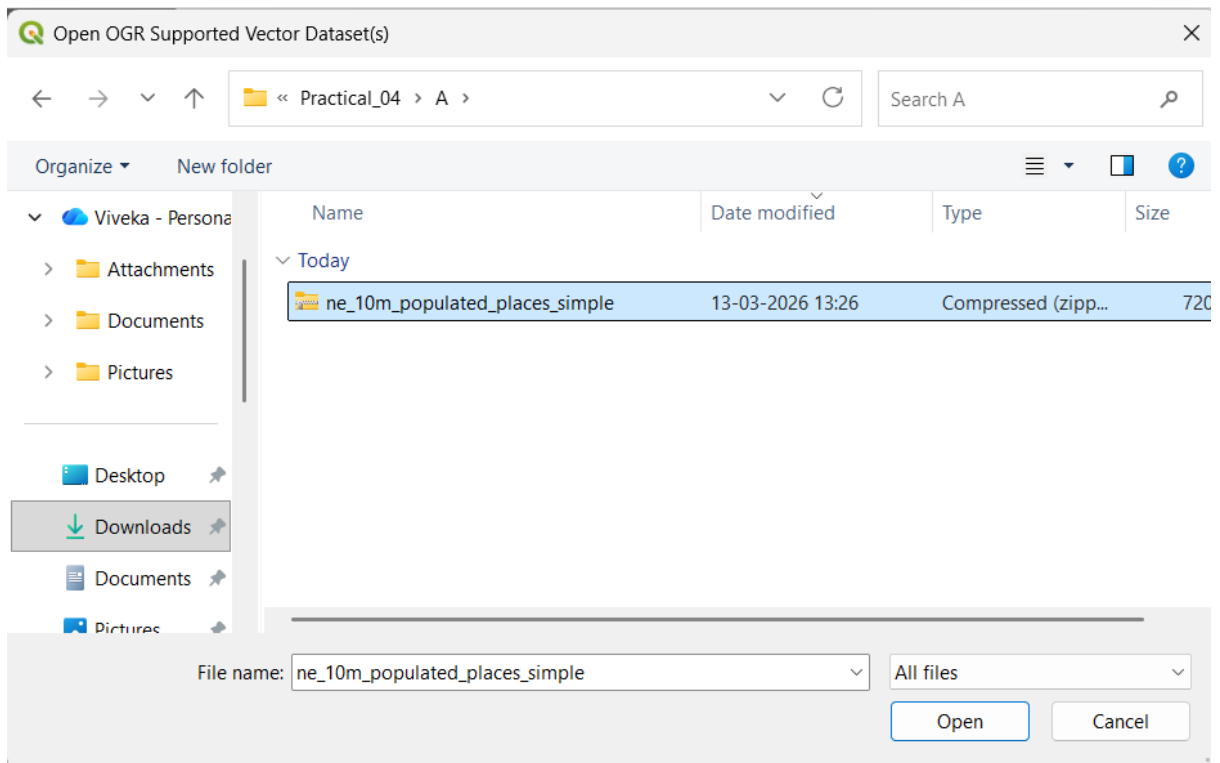
PRACTICAL NO.6

AIM : WORKING WITH ATTRIBUTES AND TERRAIN DATA AND HILL SHADE ANALYSIS.

a) WORKING WITH ATTRIBUTES:

Steps:





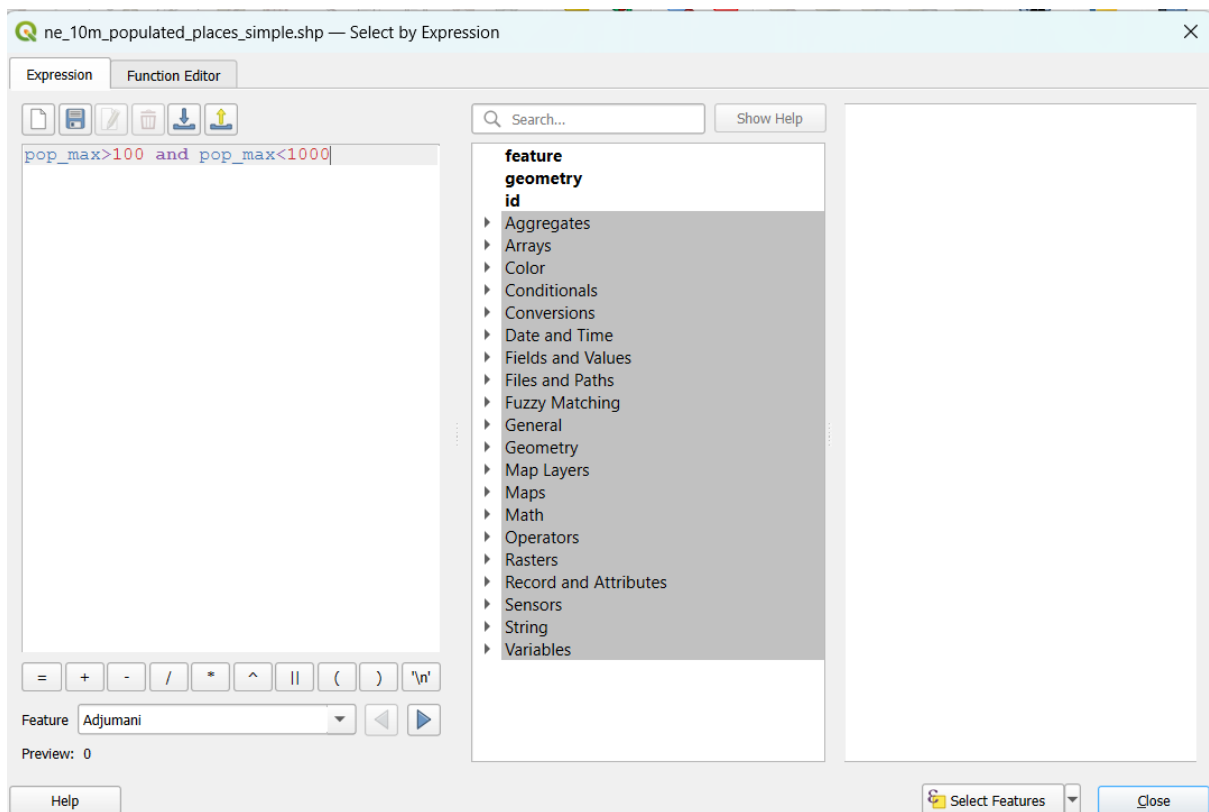
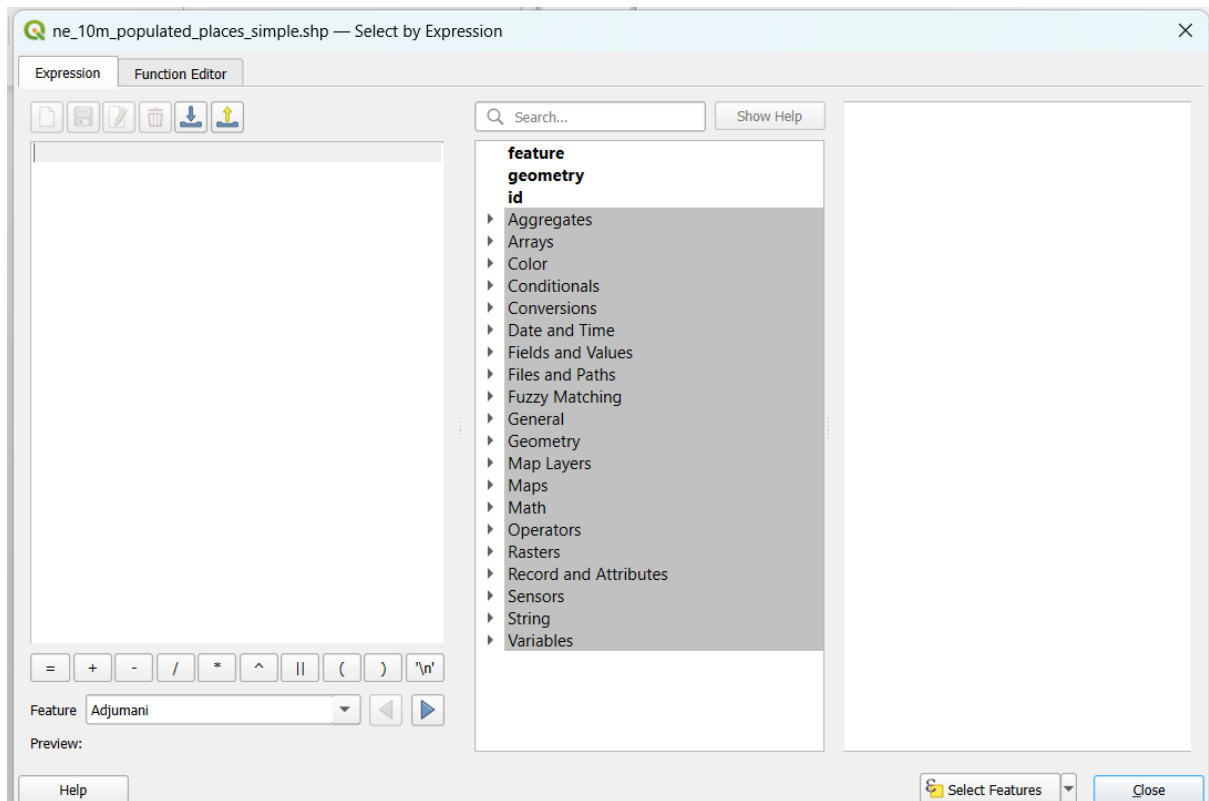
ne_10m_populated_places_simple.shp

- Zoom to Layer(s)
- Zoom to Selection
- Show in Overview
- Show Feature Count
- Show Labels
- Copy Layer
- Rename Layer
- Duplicate Layer
- Remove Layer...
- Open Attribute Table
- Filter...
- Change Data Source...
- Set Layer Scale Visibility...
- Layer CRS
- Export
- Styles
- Add Layer Notes...
- Properties...

ne_10m_populated_places_simple.shp — Features Total: 7322, Filtered: 7322, Selec...

	iff	diffnote	Select features using an expression	pop_other	rank_max	r
1	1	Added missing ...	21714	21714	0	7
2	1	Added missing ...	21093	21093	0	7
3	1	Added missing ...	23279	23279	0	7
4	1	Added missing ...	19698	19698	0	6
5	1	Added missing ...	32234	32234	0	7
6	1	Added missing ...	61845	61845	0	8
7	1	Added missing ...	21054	21054	0	7
8	1	Added missing ...	61705	61705	0	8

Show All Features



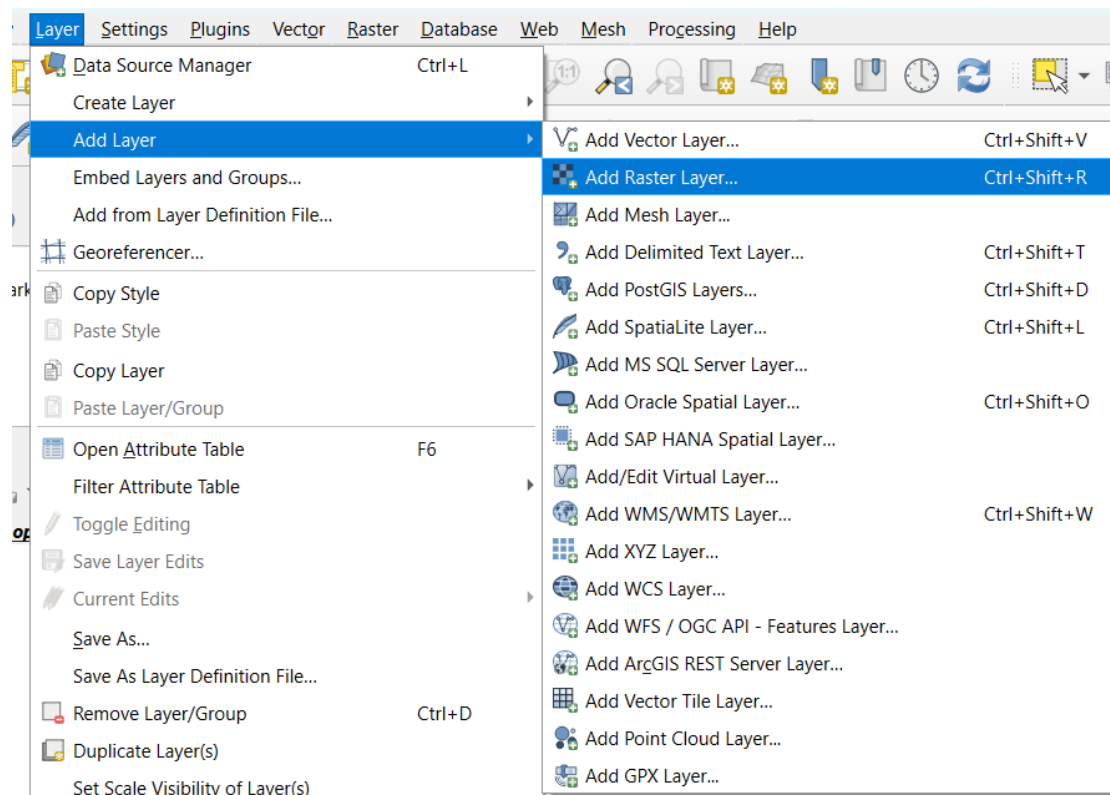
WORKING WITH ATTRIBUTES :

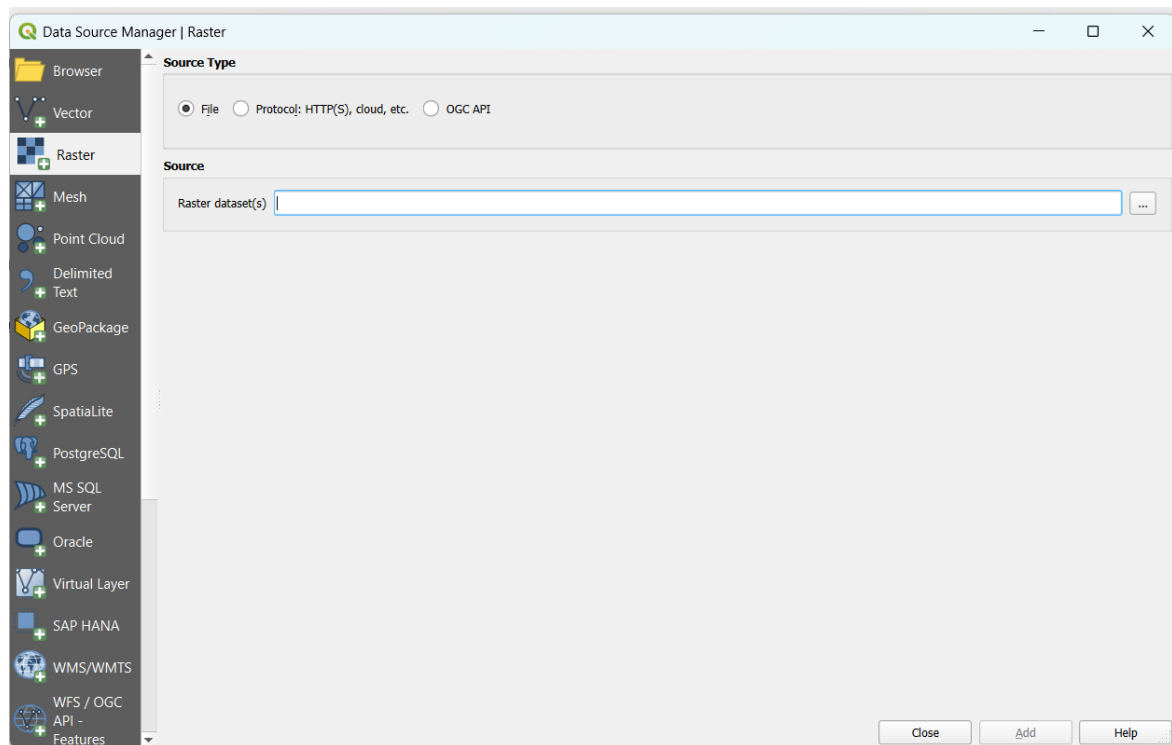
Final Output :

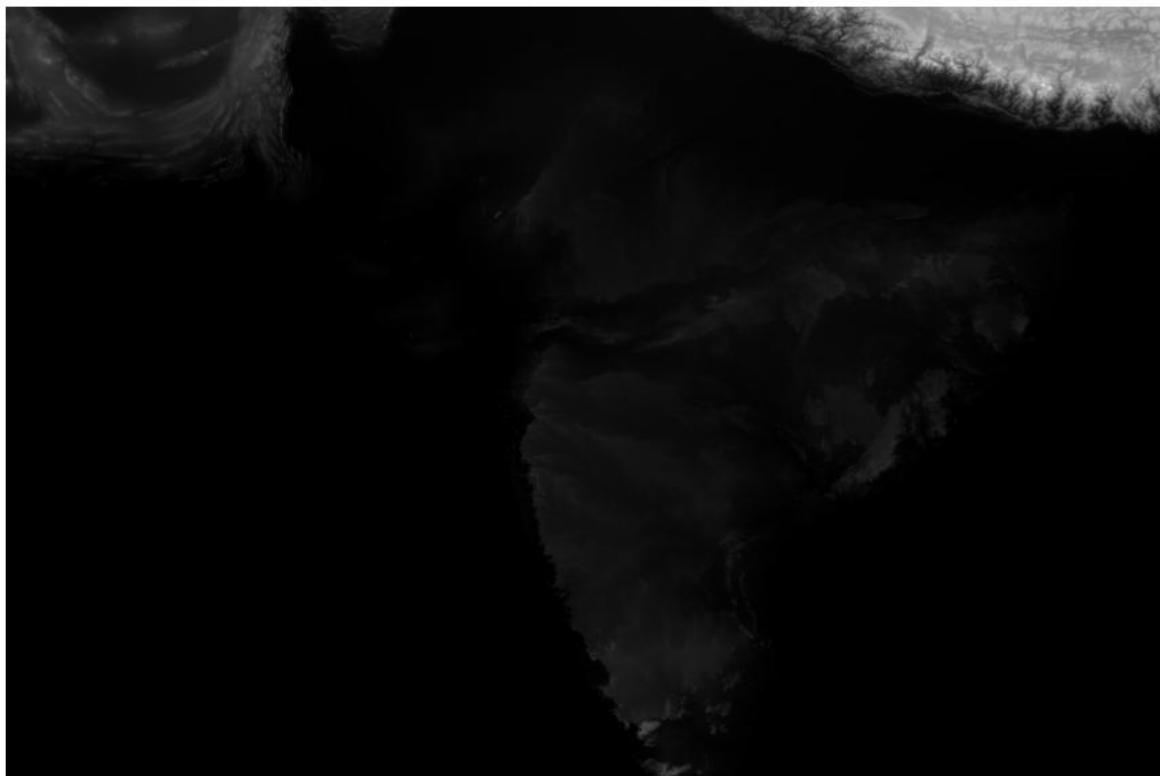
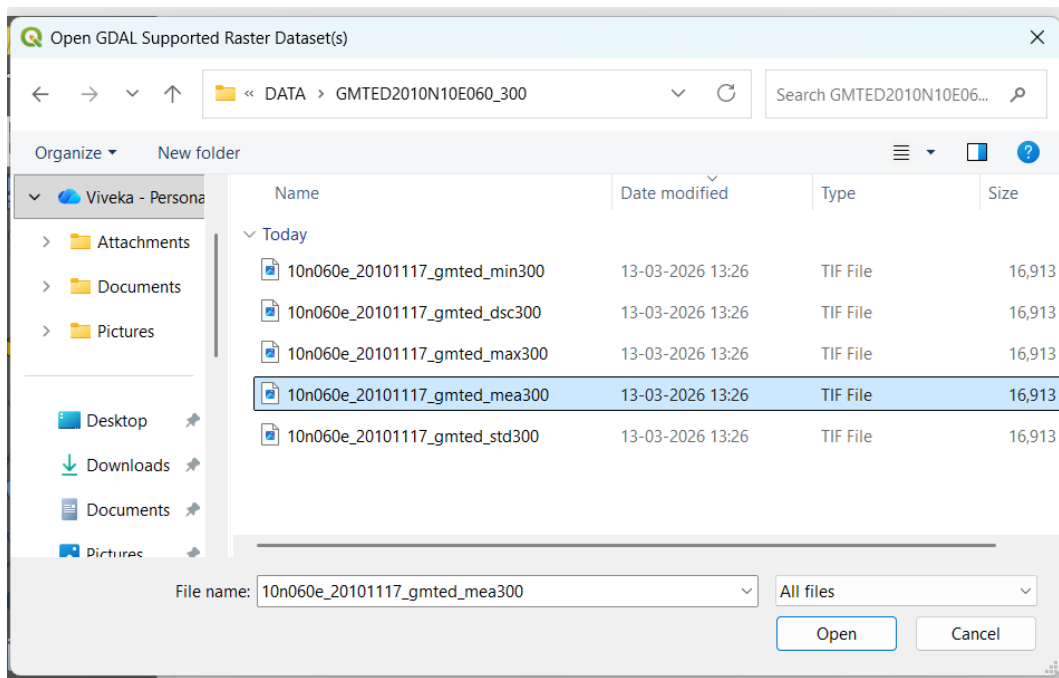


b) TERRAIN DATA AND HILL SHADE ANALYSIS:

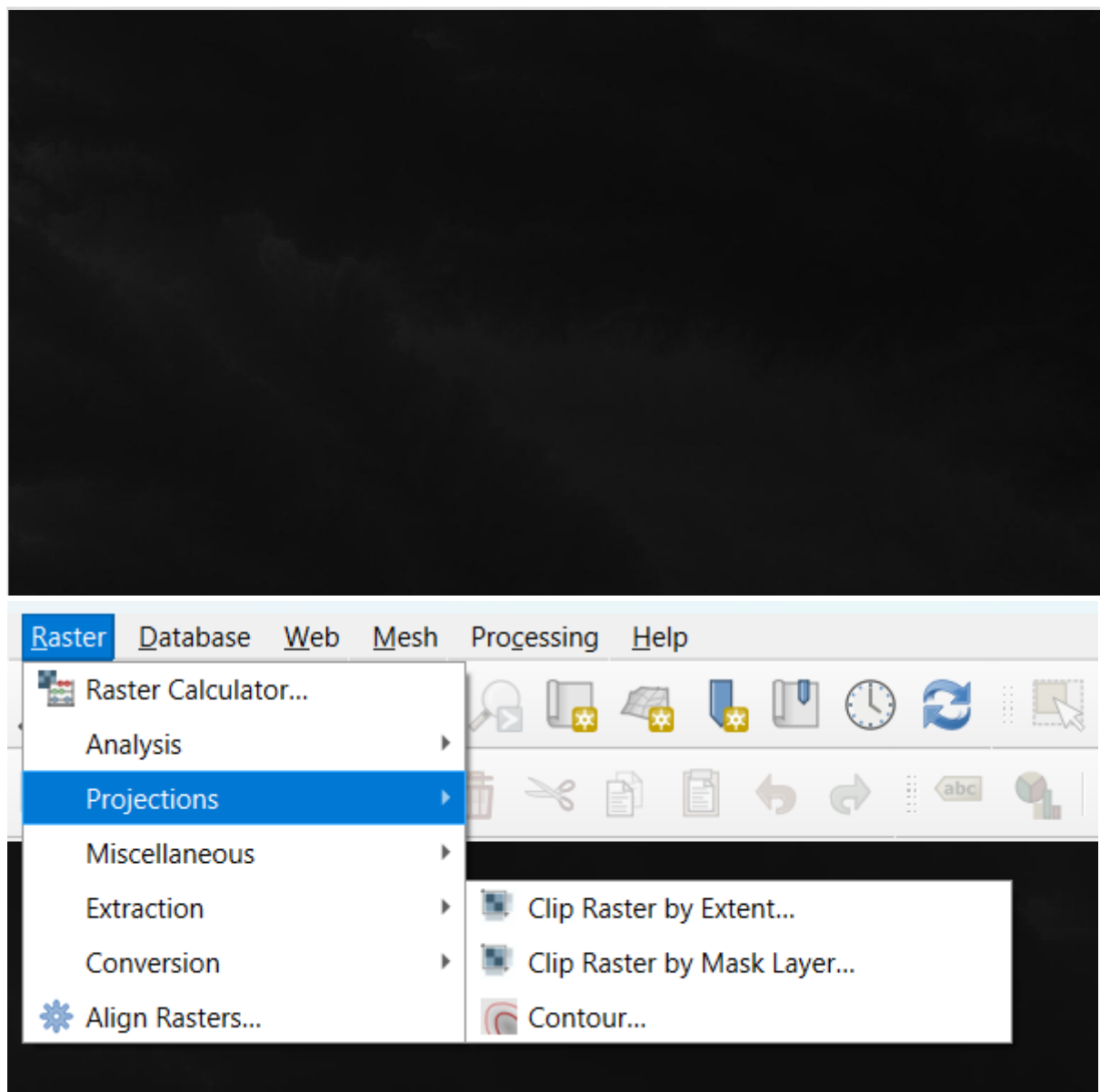
Steps:

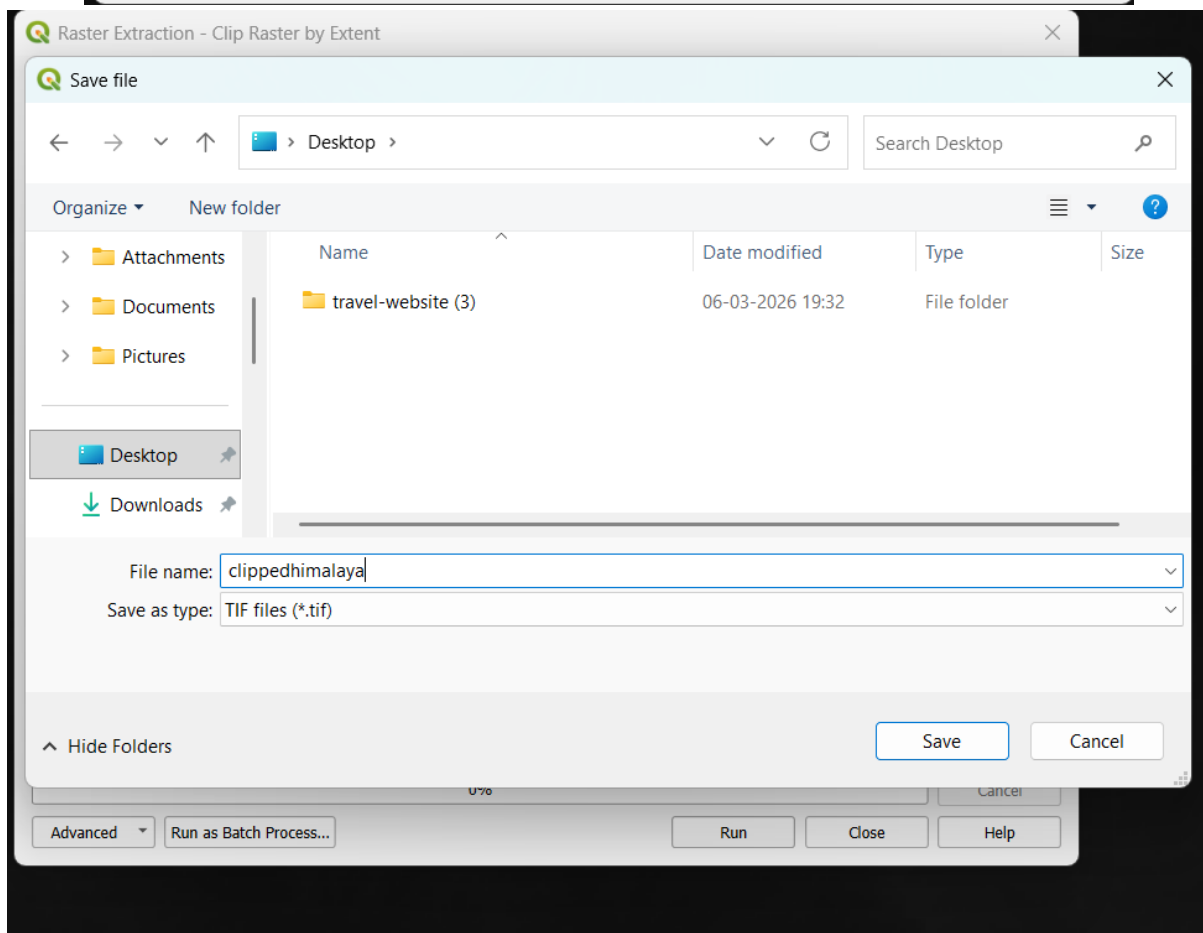
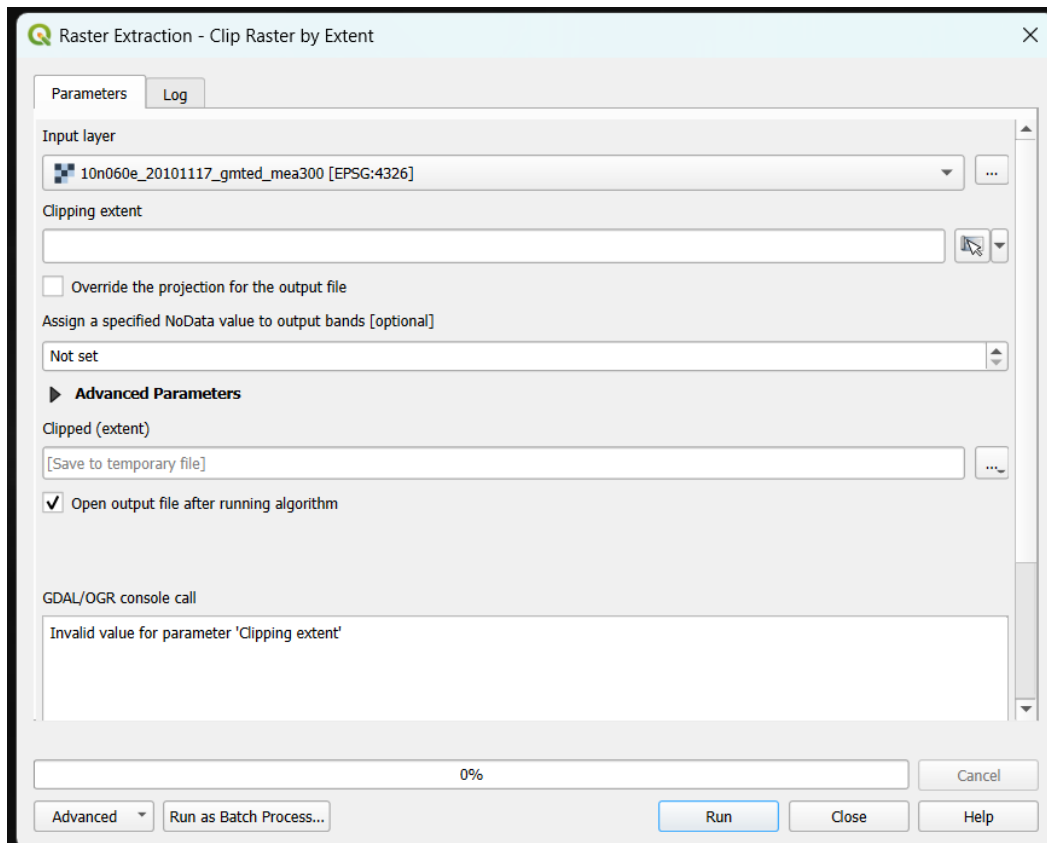






Coordinate 4.63°, 81.37° Scale 1:900000





 Raster Extraction - Clip Raster by Extent

ParametersLog

Input layer

10n060e_20101117_gmted_mea300 [EPSG:4326]

Clipping extent

74.052983524,77.416723867,18.073512683,19.883499383 [EPSG:4326]

☐ Override the projection for the output file

Assign a specified NoData value to output bands [optional]

Not set

► Advanced Parameters

Clipped (extent)

C:/Users/vivek/Desktop/clippedhimalaya.tif

☒ Open output file after running algorithm

GDAL/OGR console call

gdal_translate -projwin 74.052983524 19.883499383 77.416723867 18.073512683 -of GTiff C:\Users\vivek\Downloads\Practical_04-20211217T154353Z-001-20260301T072222Z-1-001\Practical_04\DATA\GMTED2010N10E060_300\10n060e_20101117_gmted_mea300.tif C:/Users/vivek/Desktop/clippedhimalaya.tif

0%

Cancel

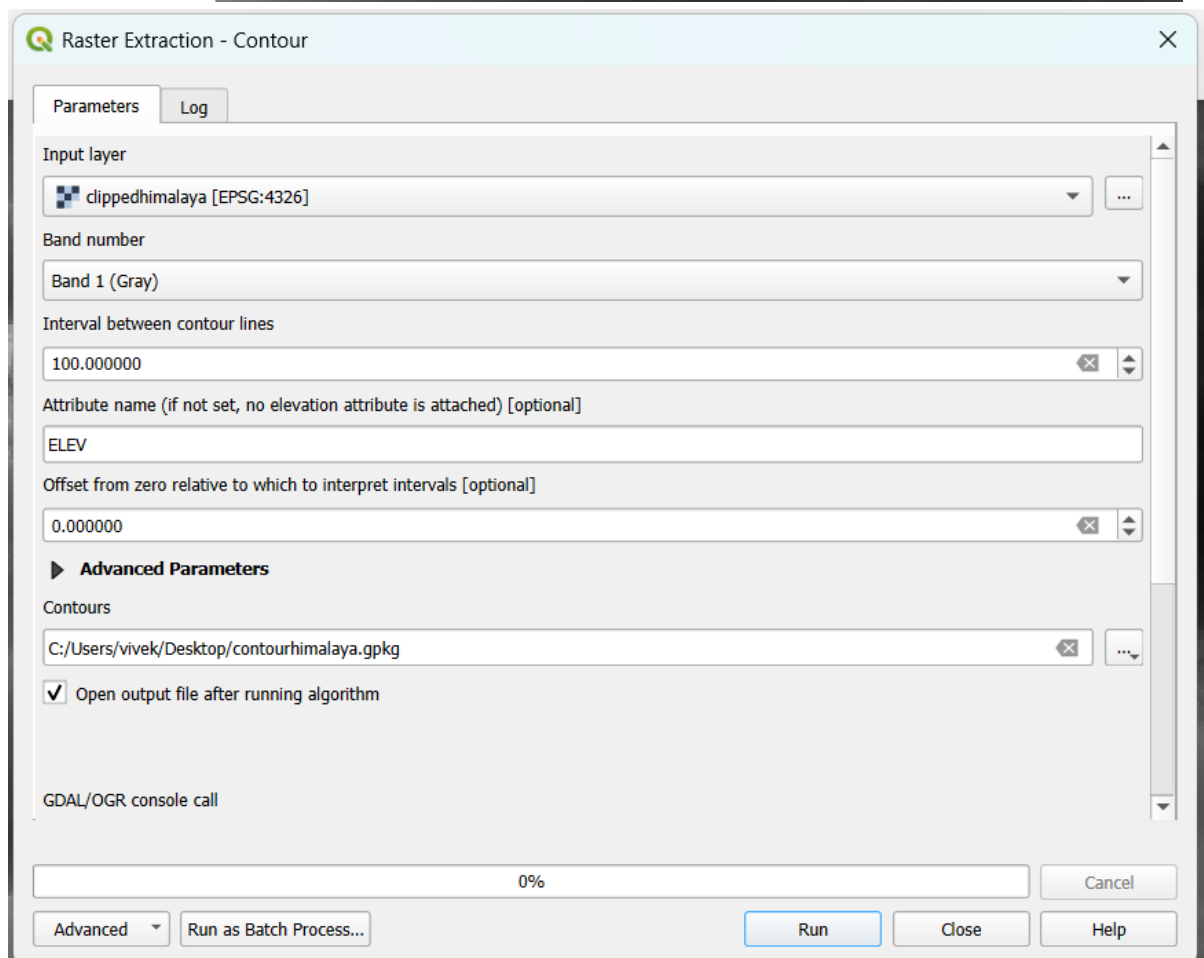
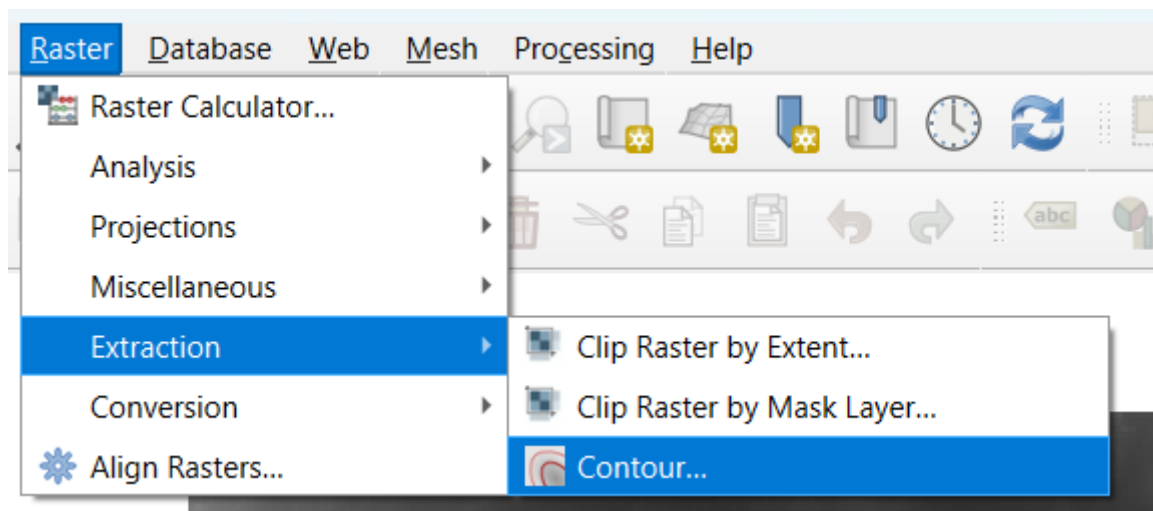
Advanced ▼

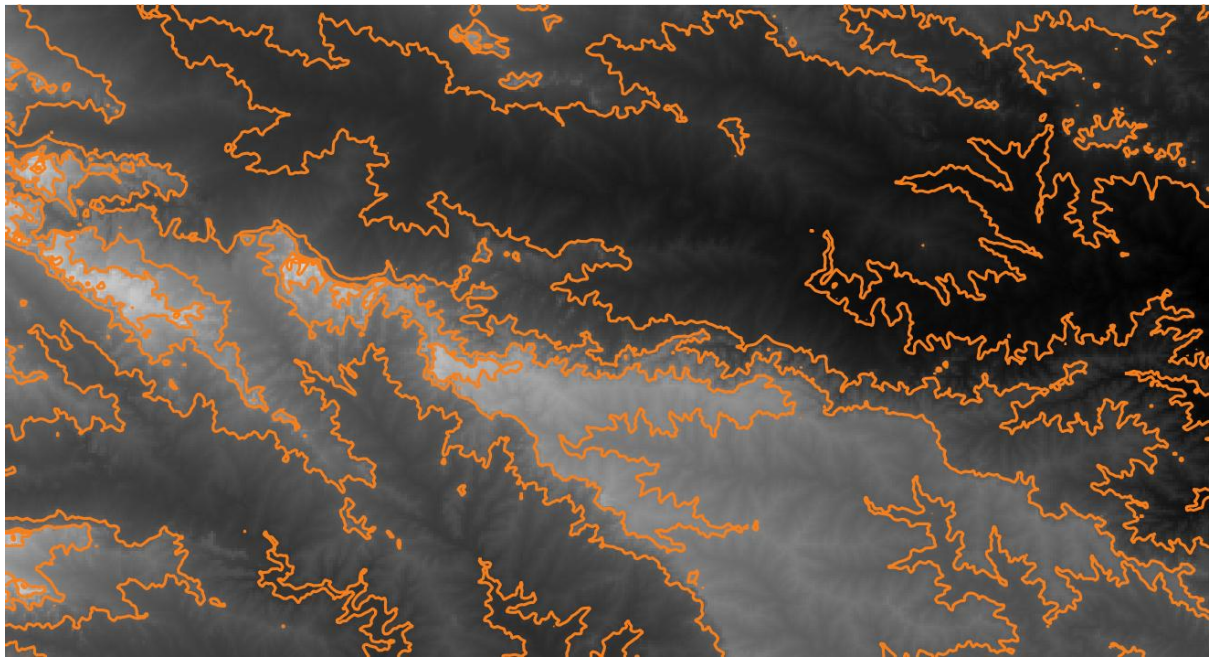
Run as Batch Process...

Run

Close

Help





Layer Properties - contourhimalaya — Labels

Single Labels

Value: 1.2 ELEV

Text Sample

Lorem Ipsum

Text

Font: Open Sans

Style: Regular

Size: 10.0000

Points

Color: [Red]

Opacity: 90.0 %

☐ Allow HTML formatting

Search: All Settings

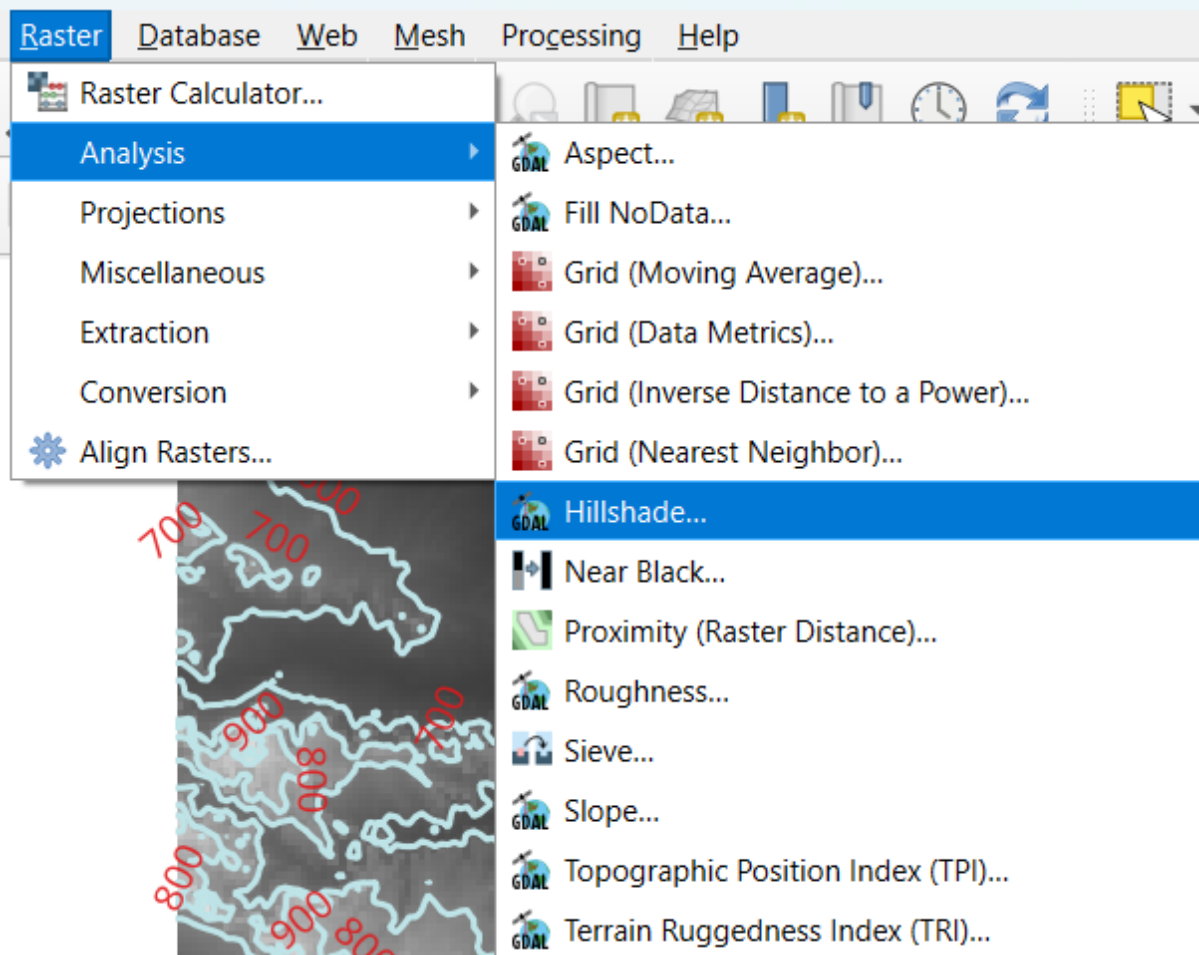
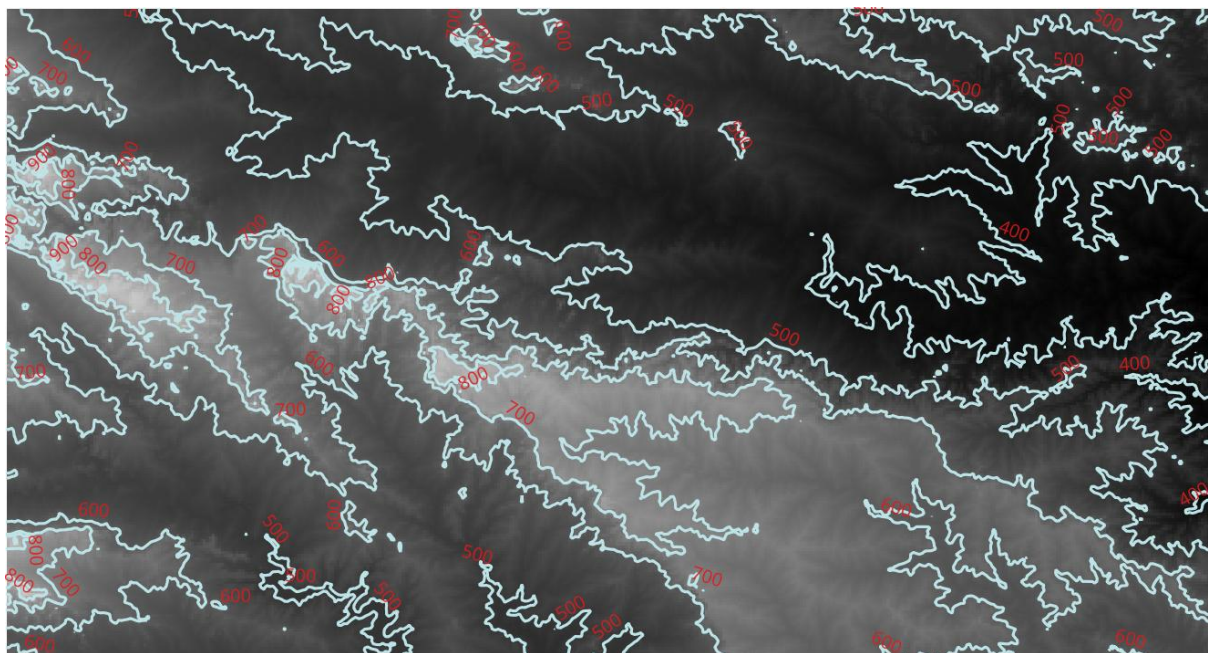
Default Project Styles

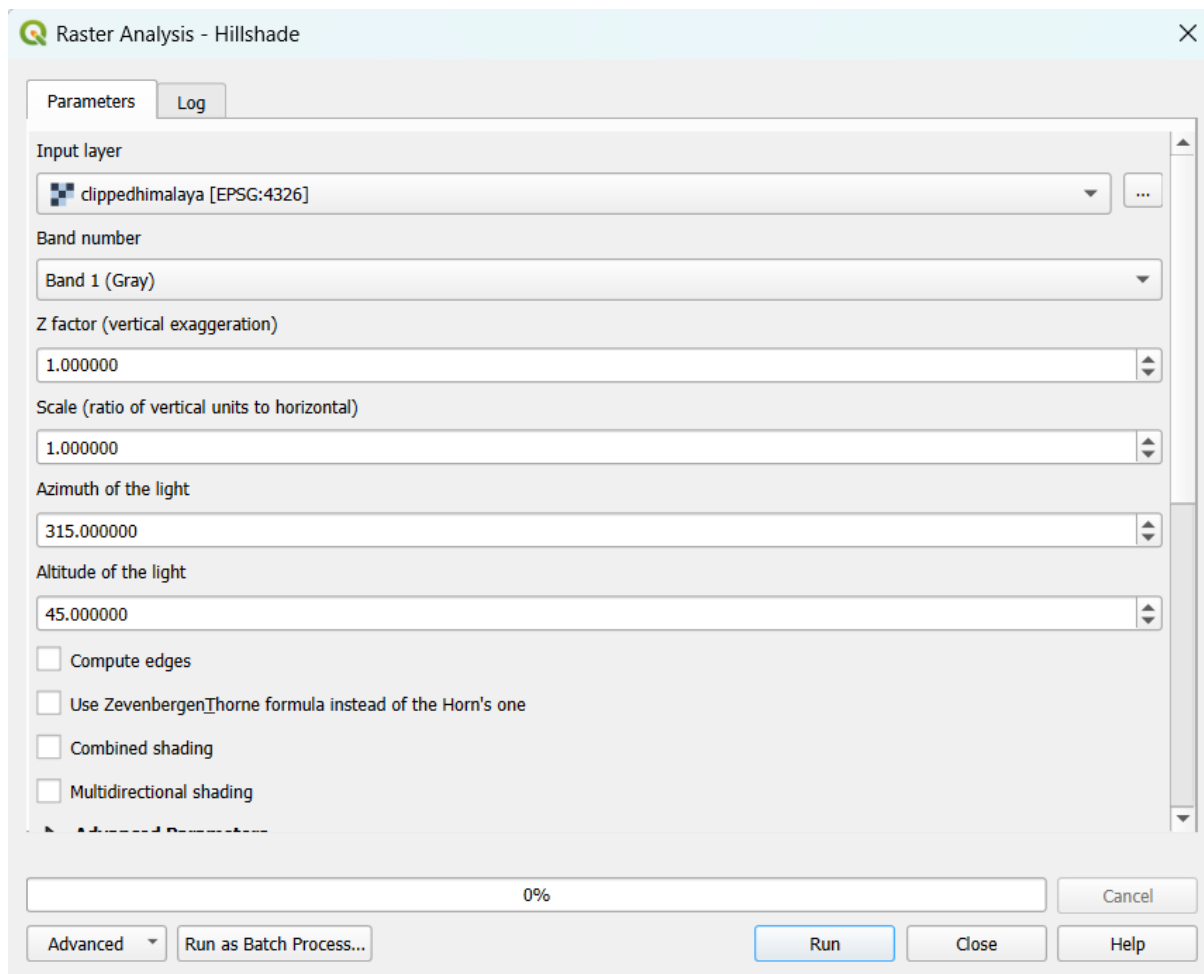
Default watercourses

Save Settings...

Style

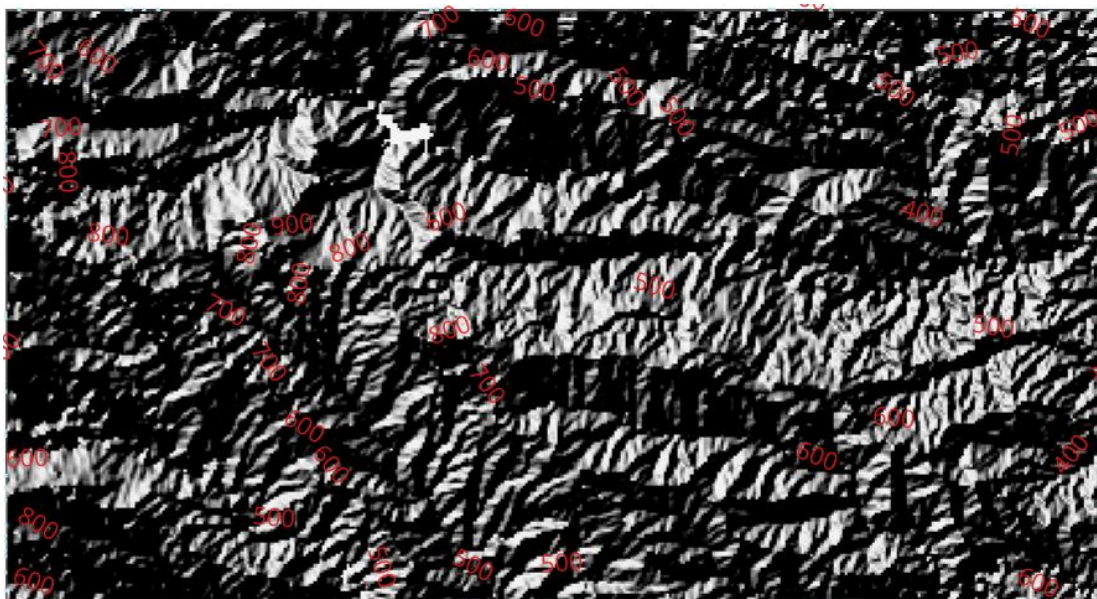
OK Cancel Apply Help





b) TERRAIN DATA AND HILL SHADE ANALYSIS

FINAL OUTPUT:

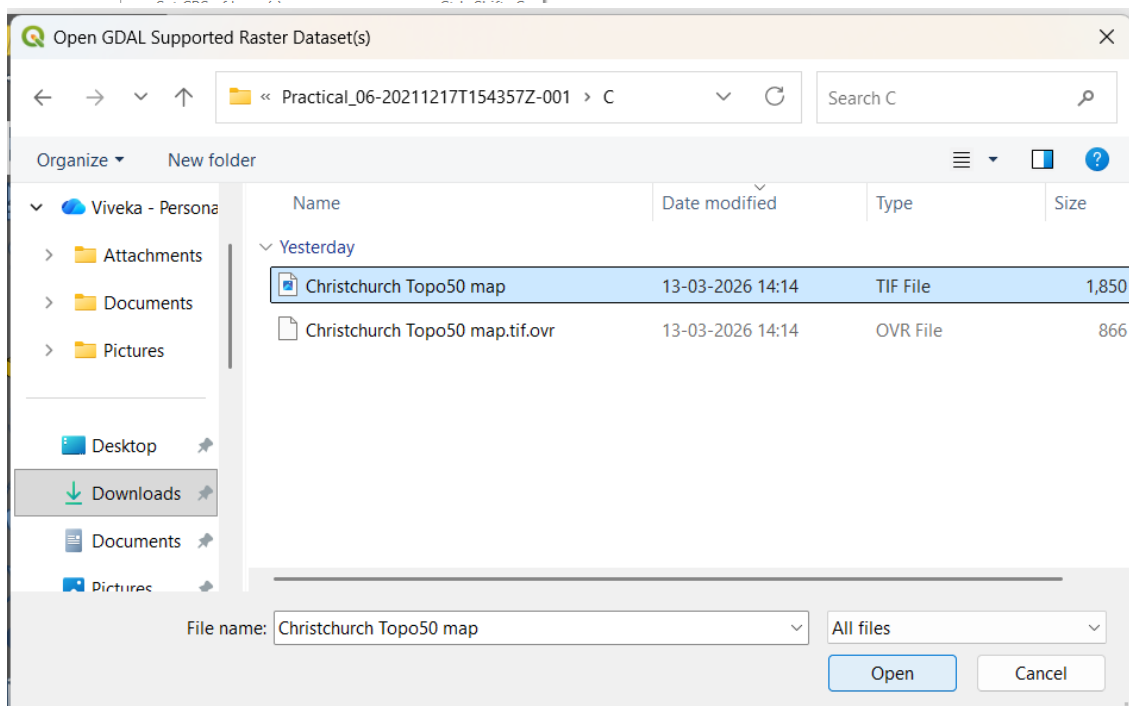
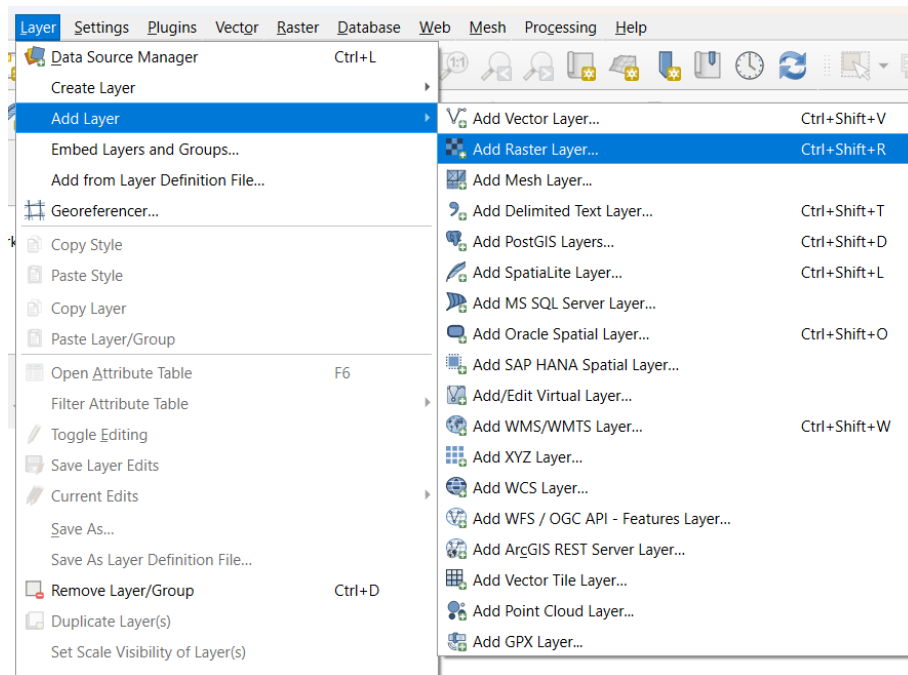


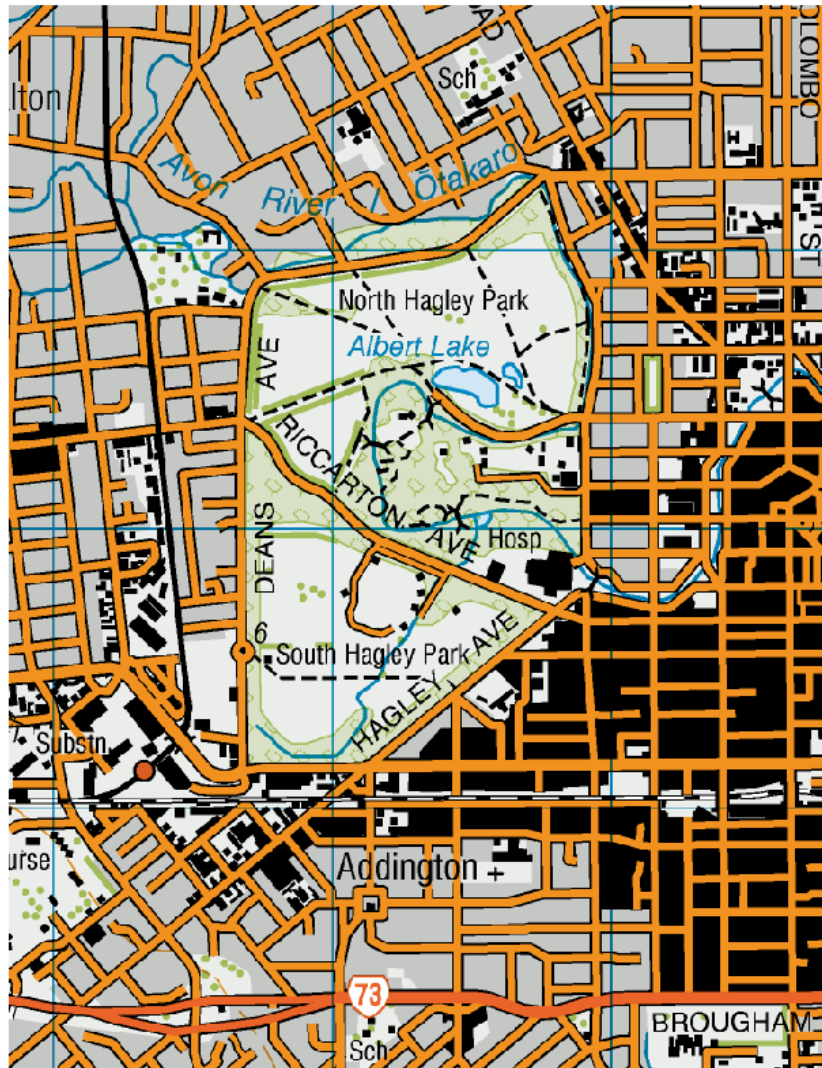
CONCLUSION : Working with attributes and terrain data and hill shade analysis is done successfully .

PRACTICAL NO.7

AIM : DIGITIZING MAP DATA

Steps:

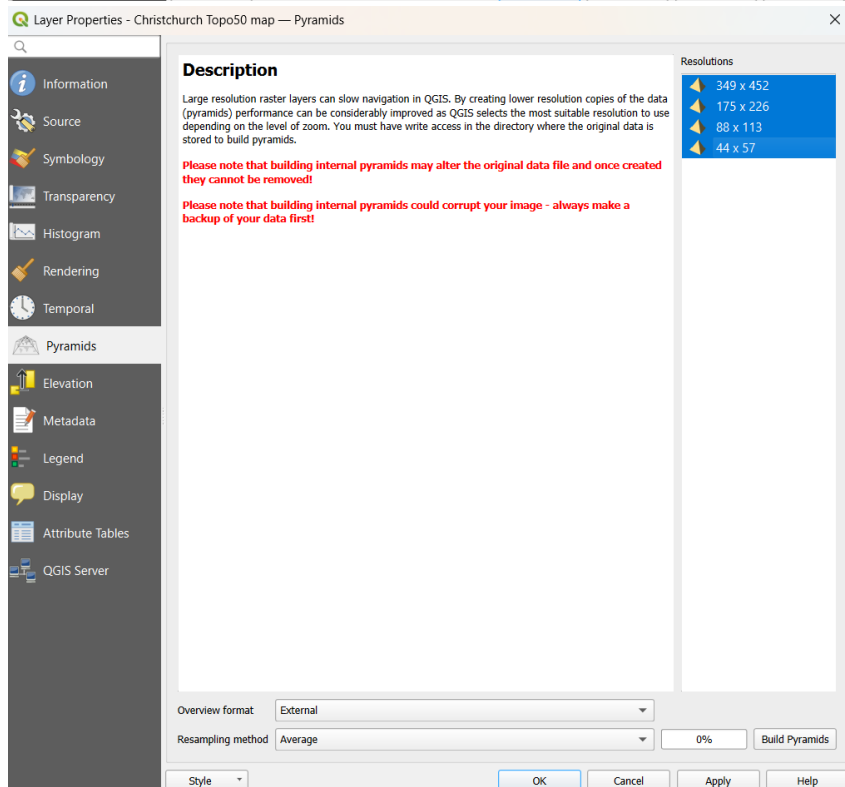
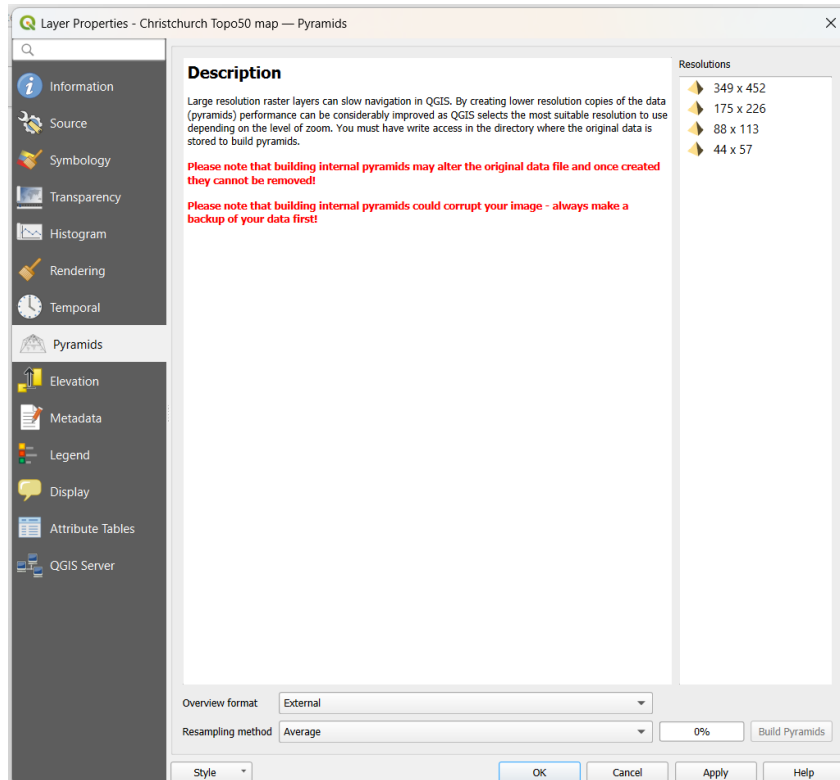


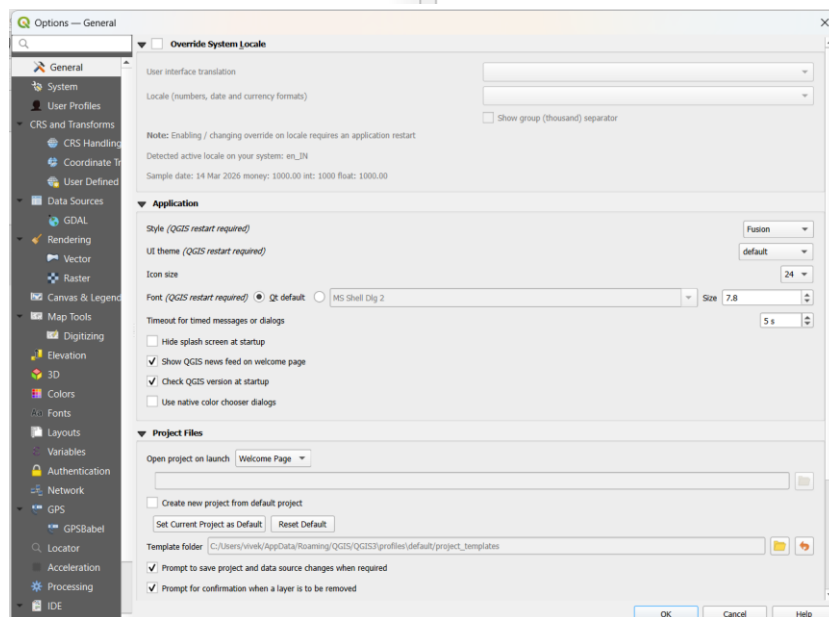
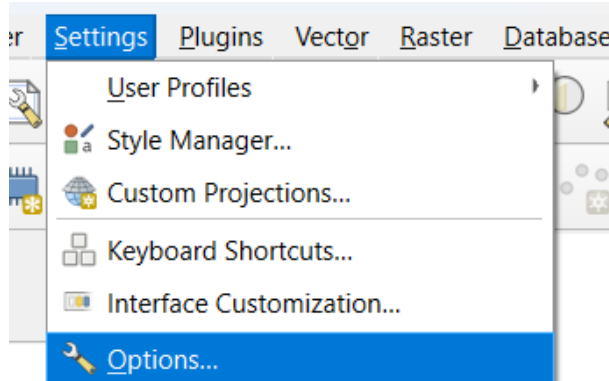
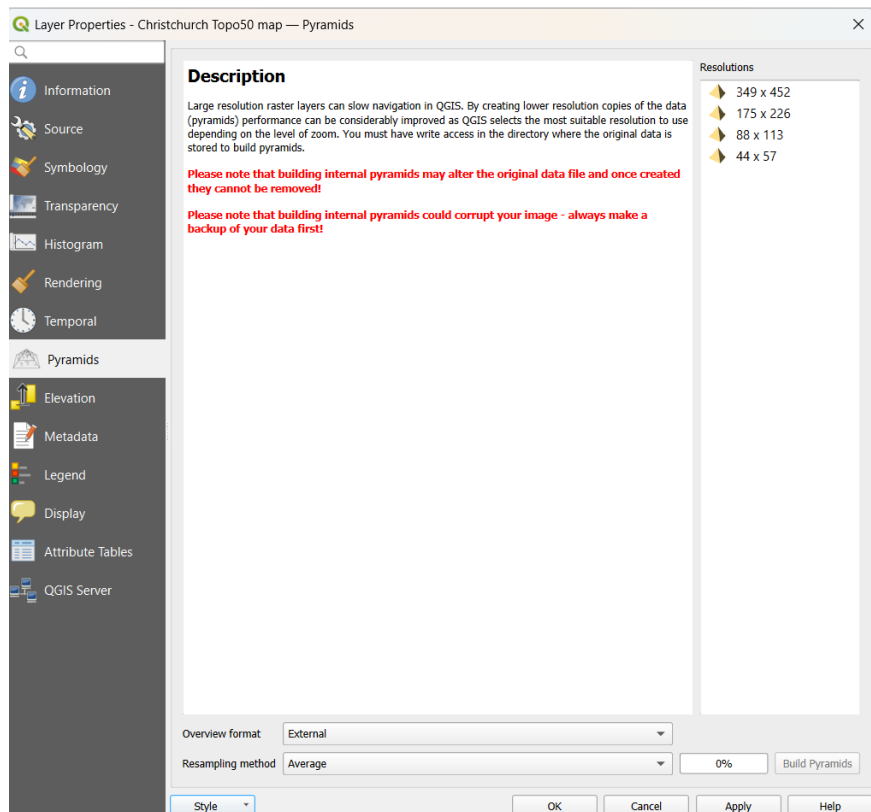


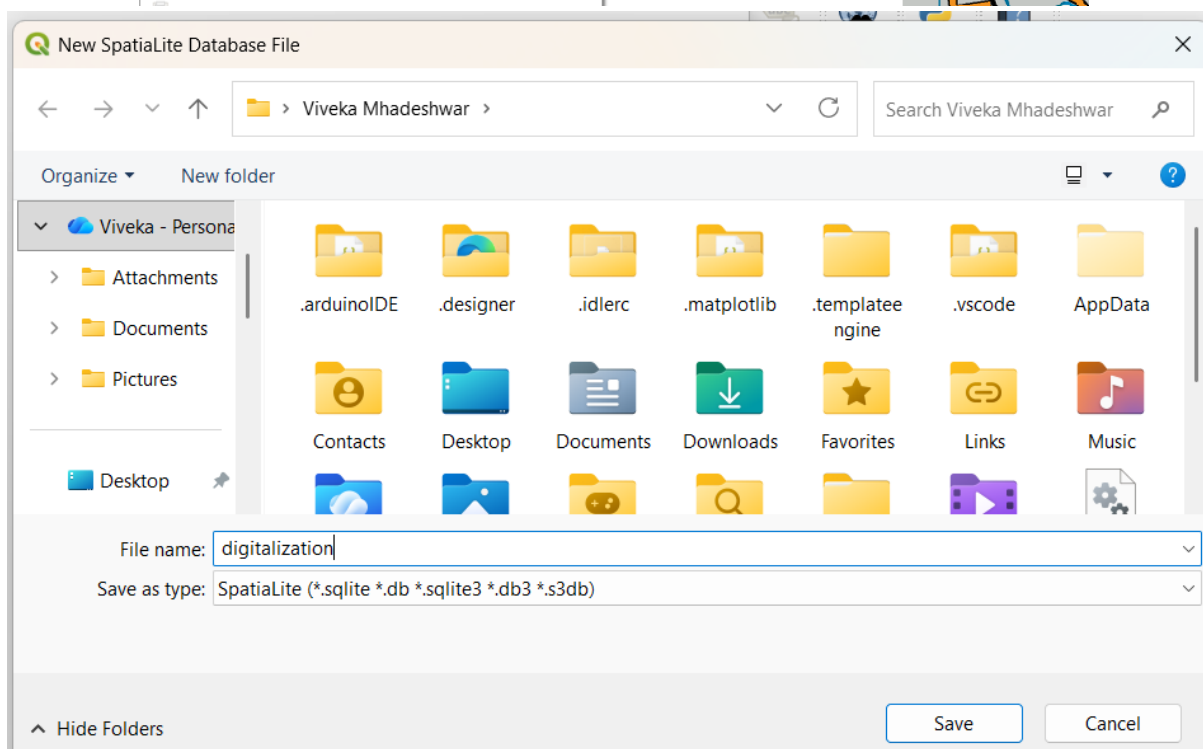
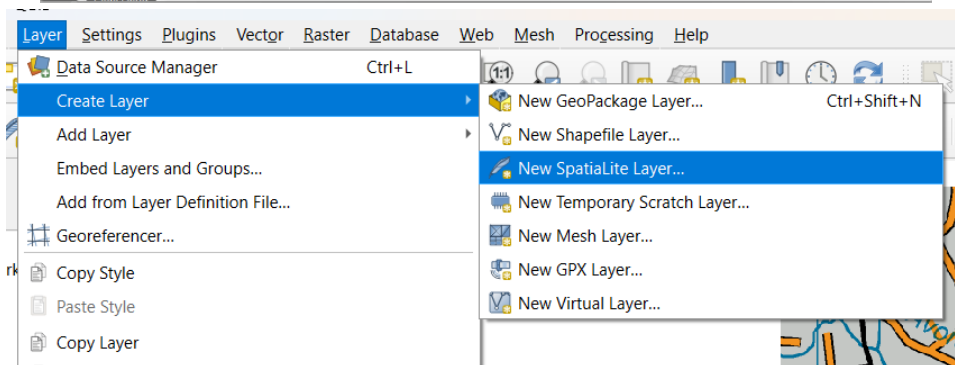
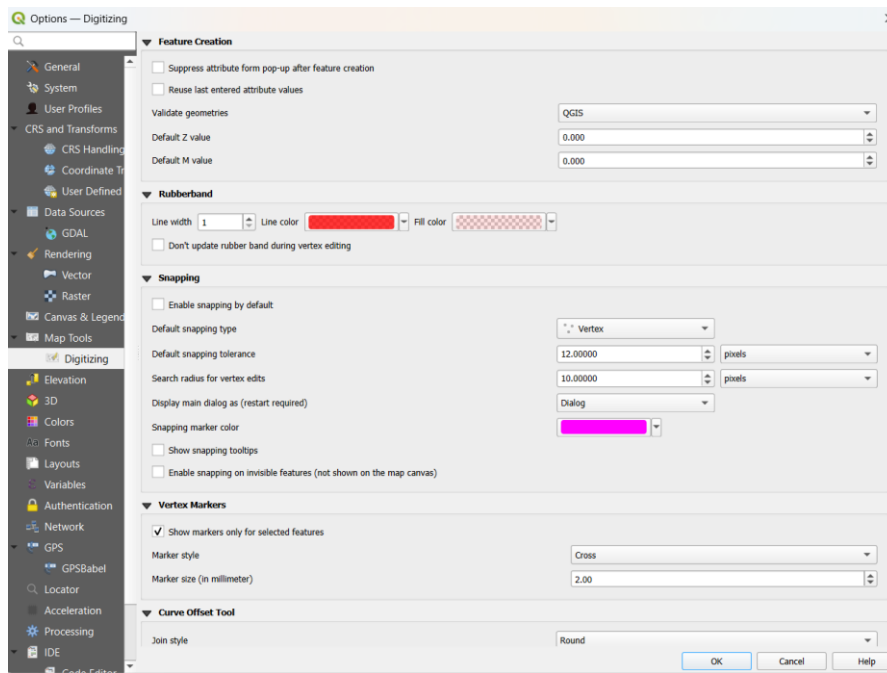
Christchurch Topo50 map

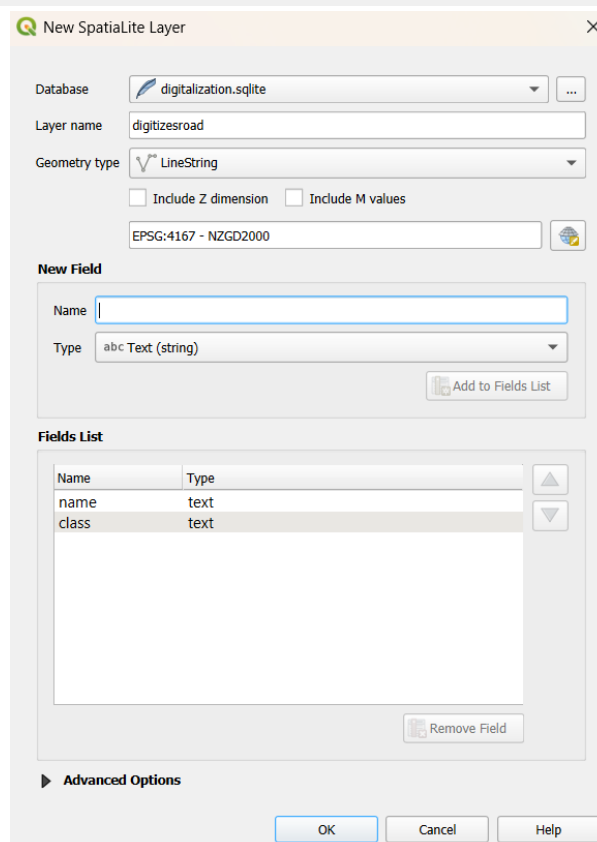
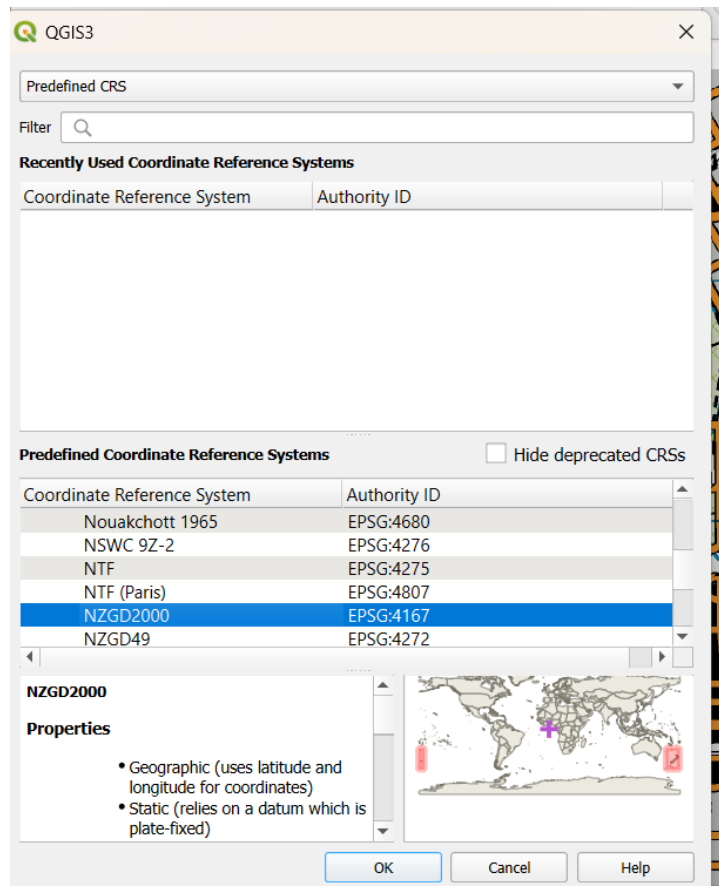
- Band 1 (Red)
- Band 2 (Green)
- Band 3 (Blue)

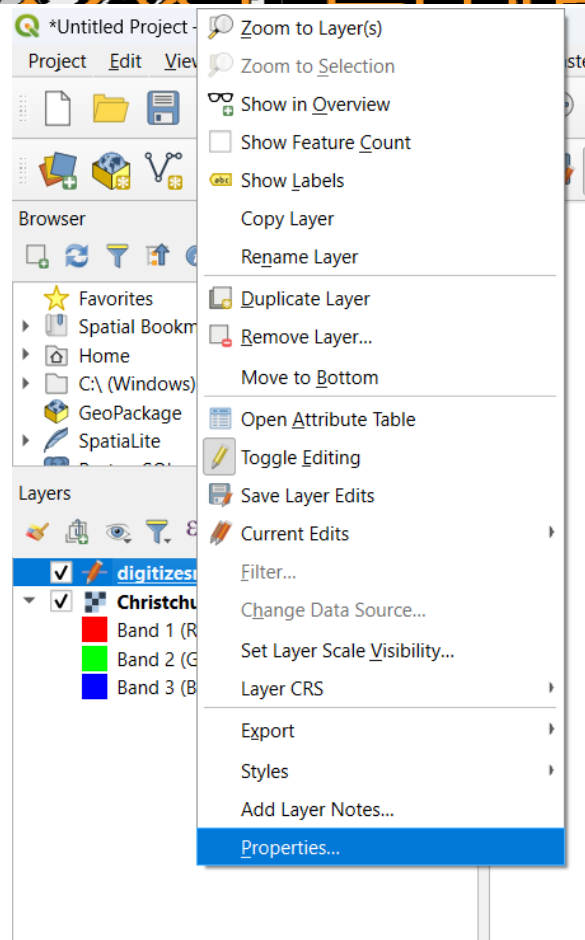
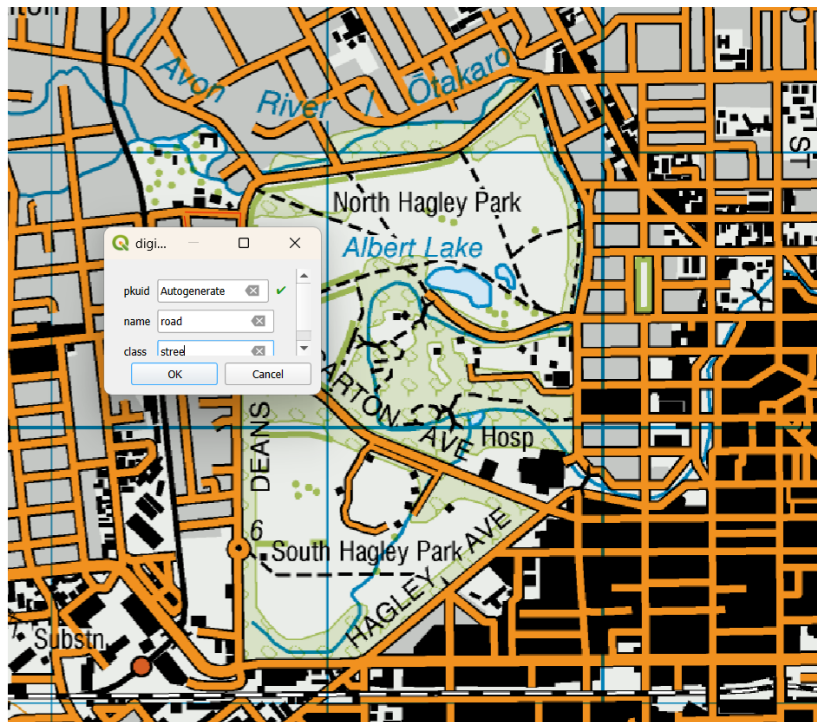
- Zoom to Layer(s)
- Show in Overview
- Copy Layer
- Rename Layer
- Zoom to Native Resolution (100%)
- Stretch Using Current Extent
- Load Raster Attribute Table from VAT.DBF
- Duplicate Layer
- Remove Layer...
- Change Data Source...
- Set Layer Scale Visibility...
- Layer CRS
- Export
- Styles
- Add Layer Notes...
- Properties...

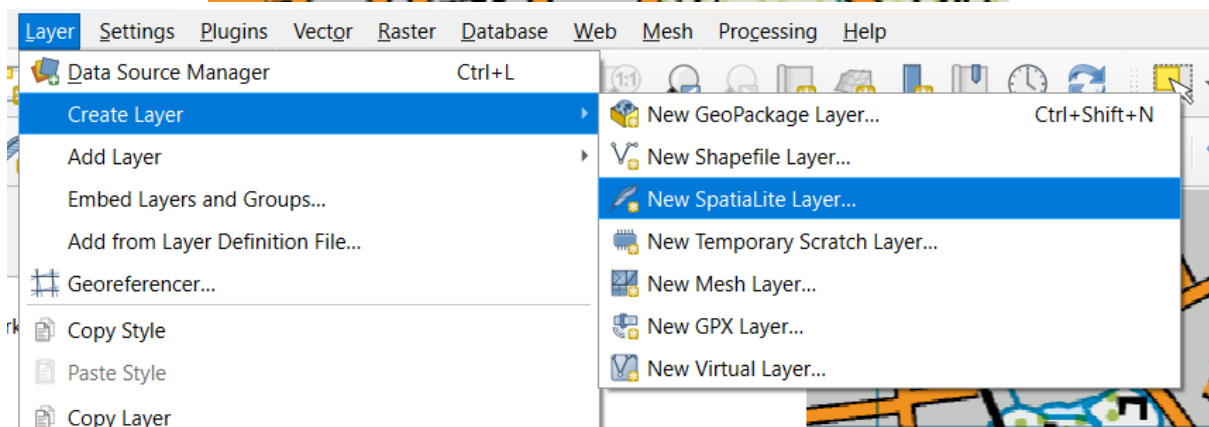
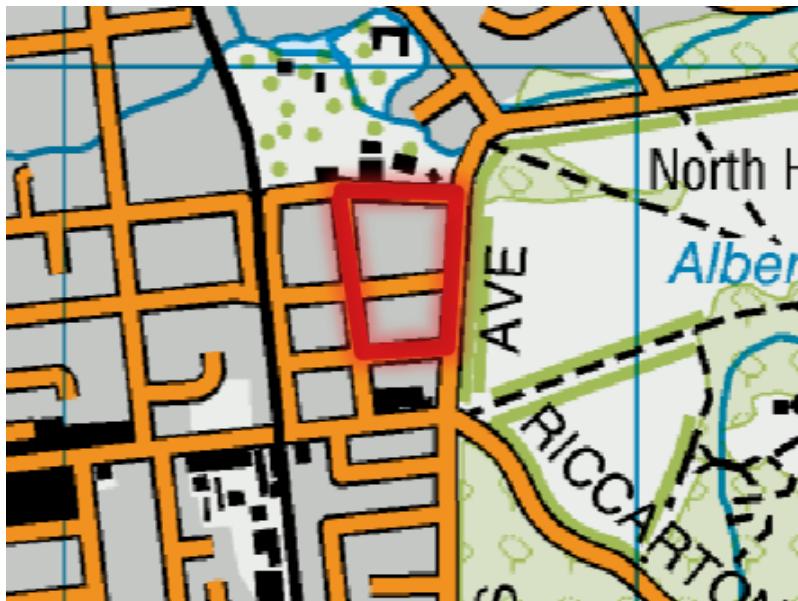
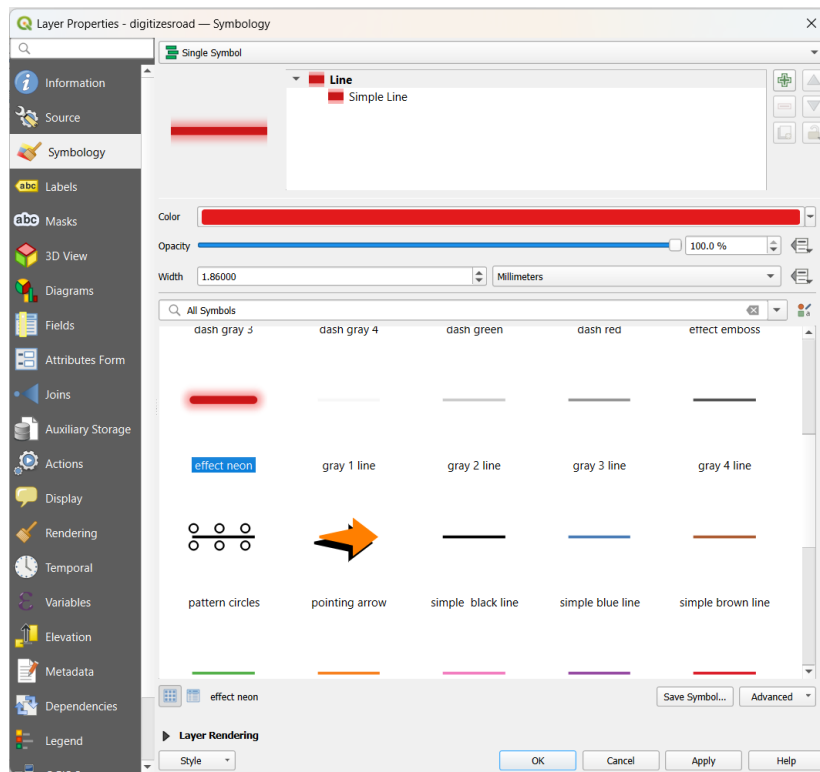












 New Spatialite Layer ✕

Database

 digitalization.sqlite ▾ ...

Layer name

digitized_garden

Geometry type

 Polygon ▾

☐ Include Z dimension

☐ Include M values

EPSG:4326 - WGS 84



New Field

Name

Type

abc Text (string) ▾

 Add to Fields List

Fields List

Name	Type
name	text




 Remove Field

► **Advanced Options**

OK

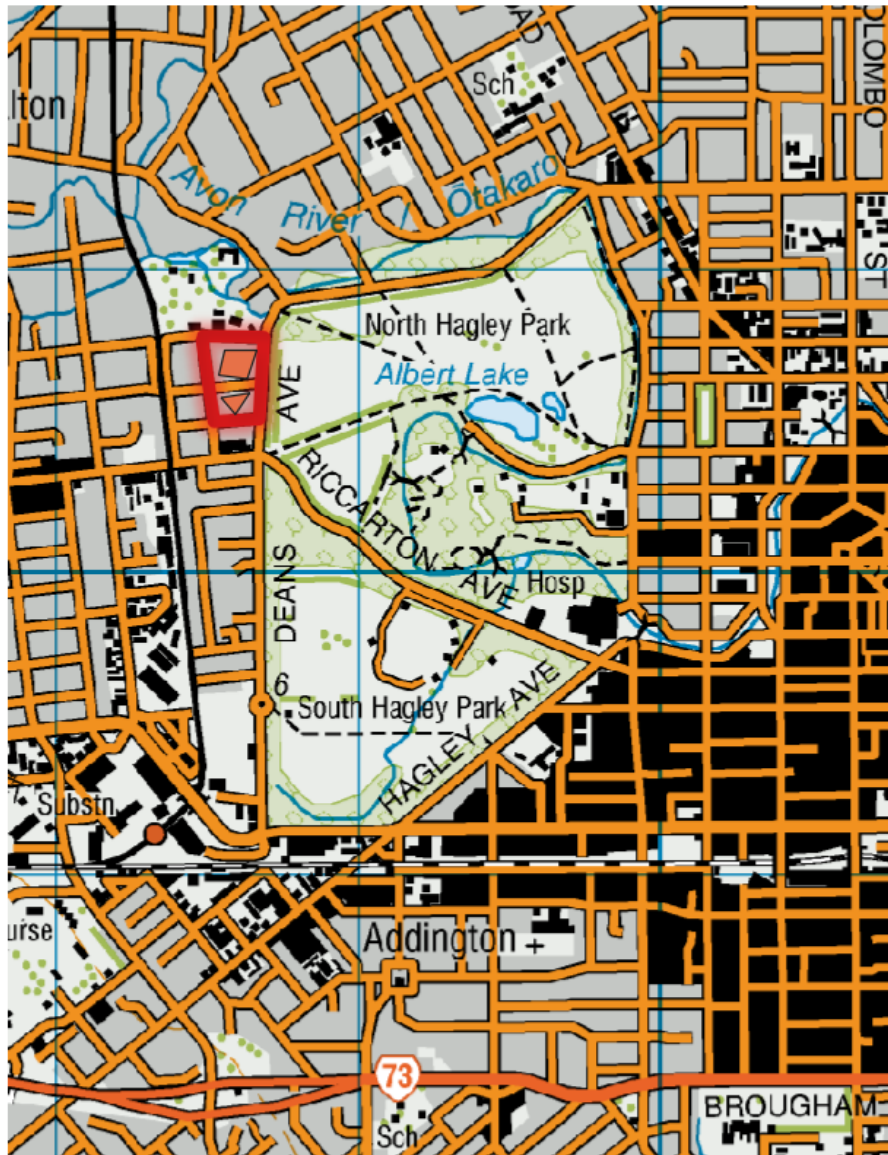
Cancel

Help



b) DIGITIZING MAP DATA

Final Output :

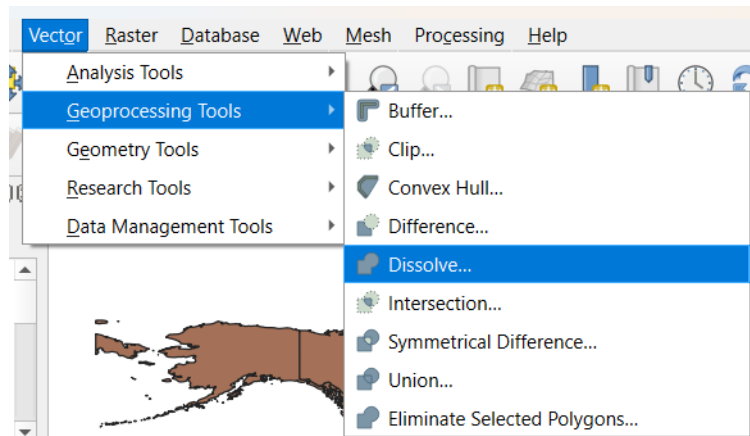
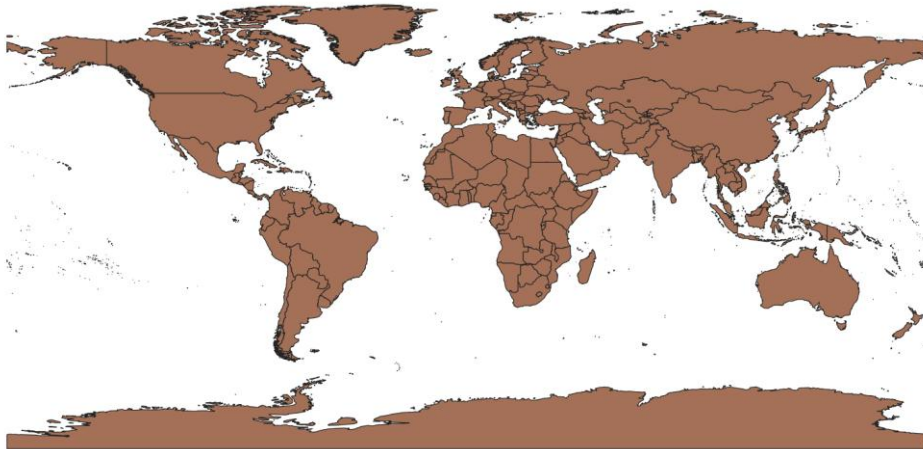
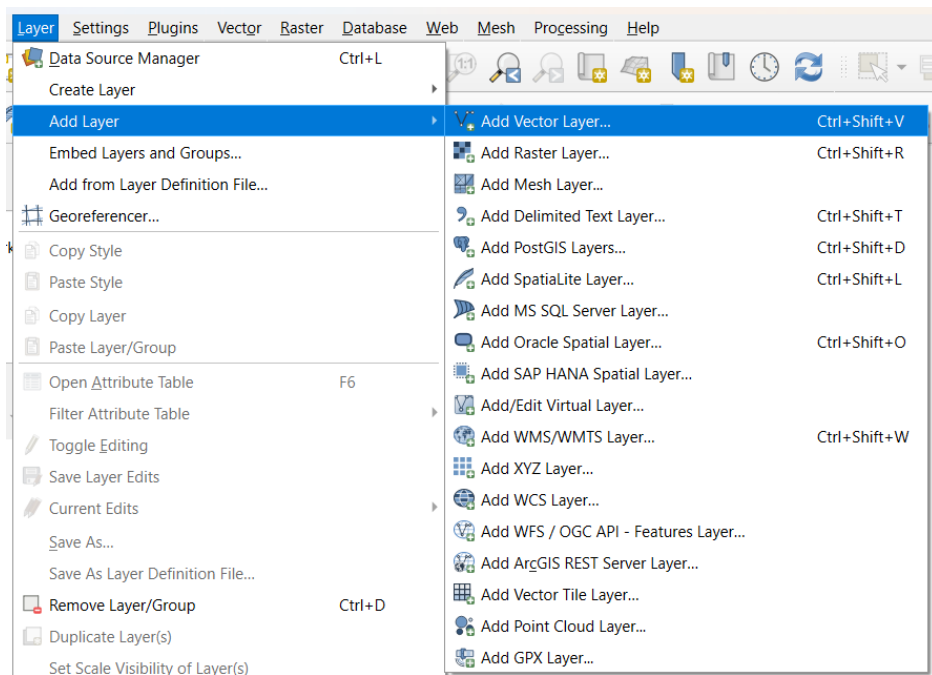


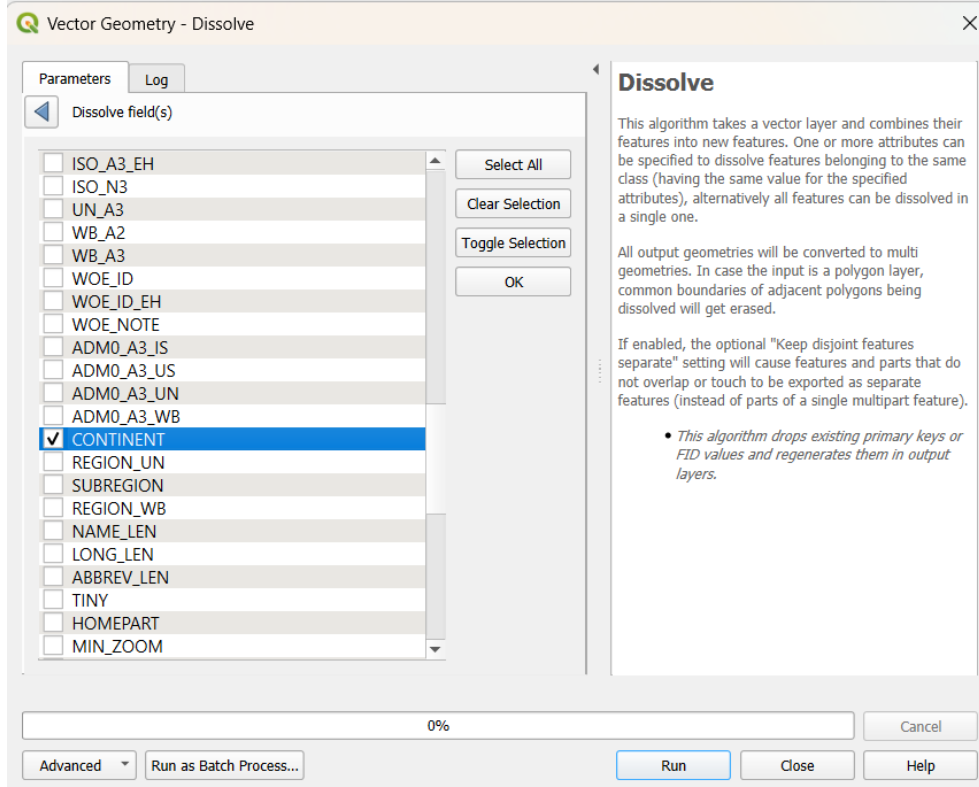
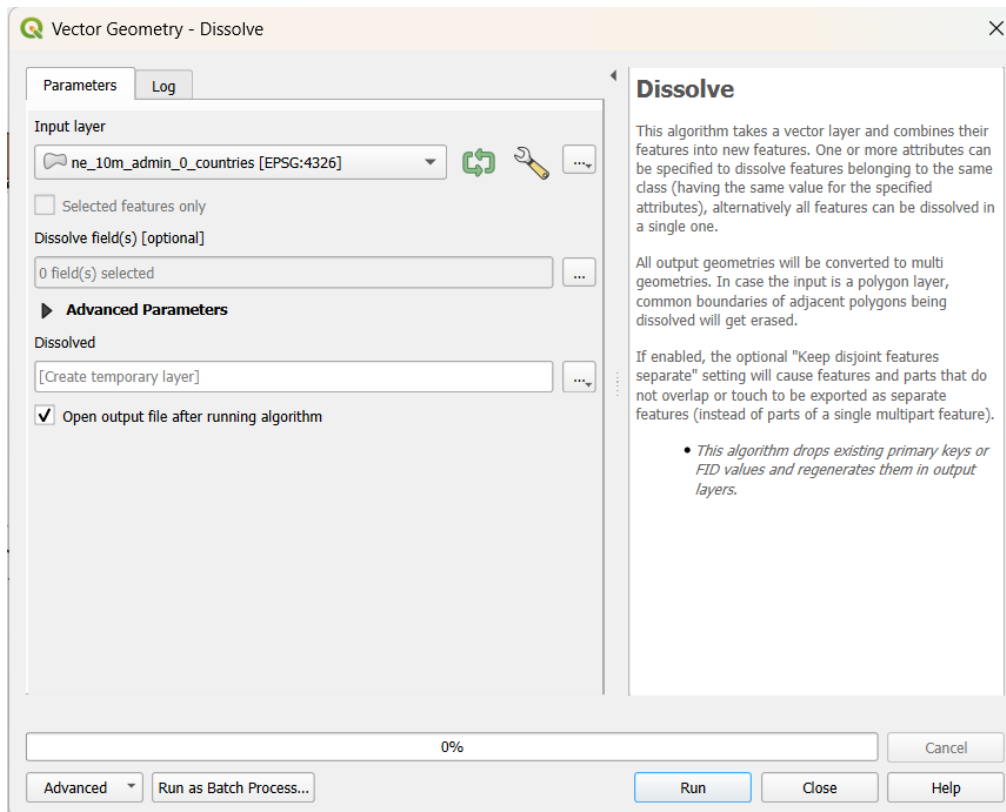
CONCLUSION : digitizing map data implemented successfully .

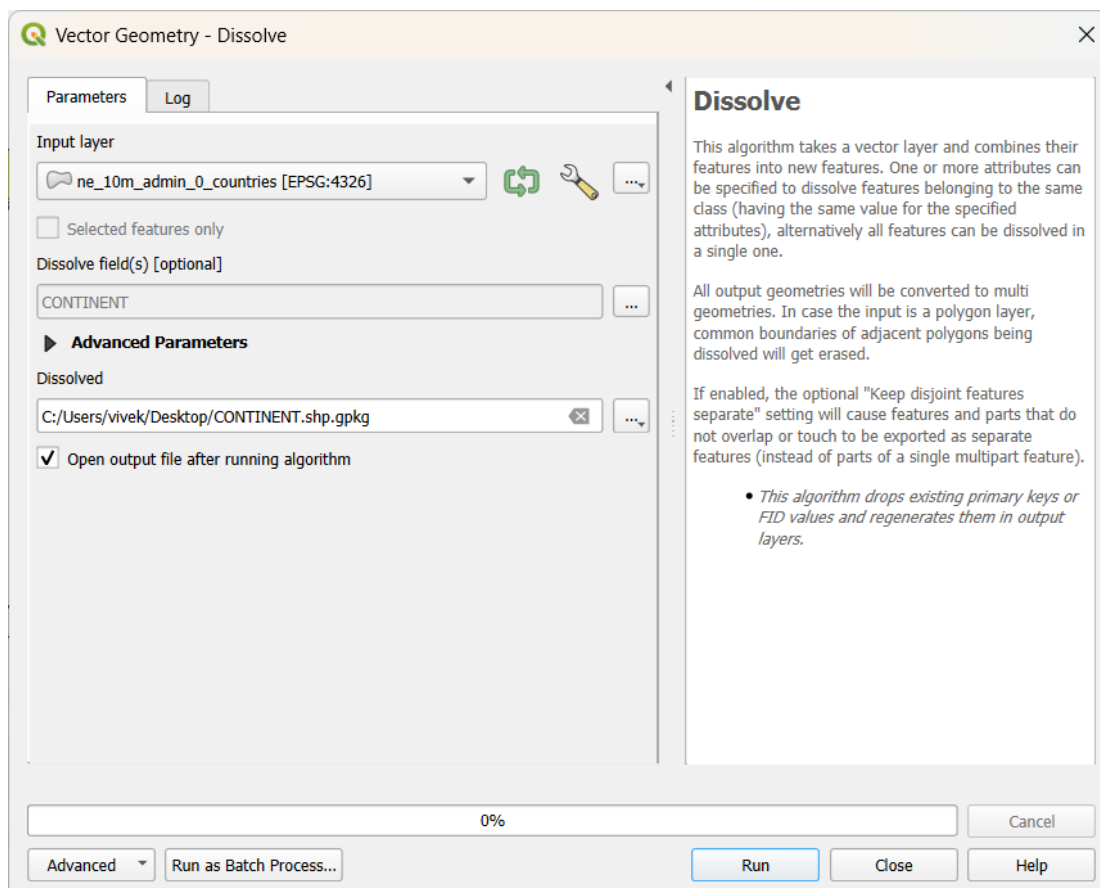
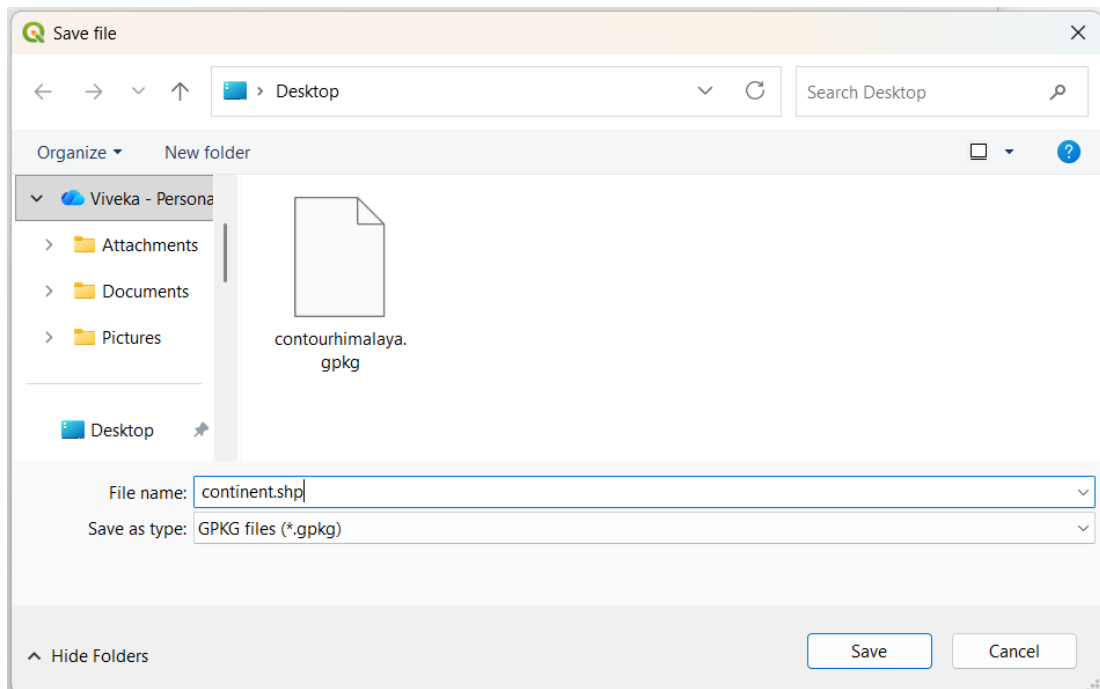
PRACTICAL NO.8

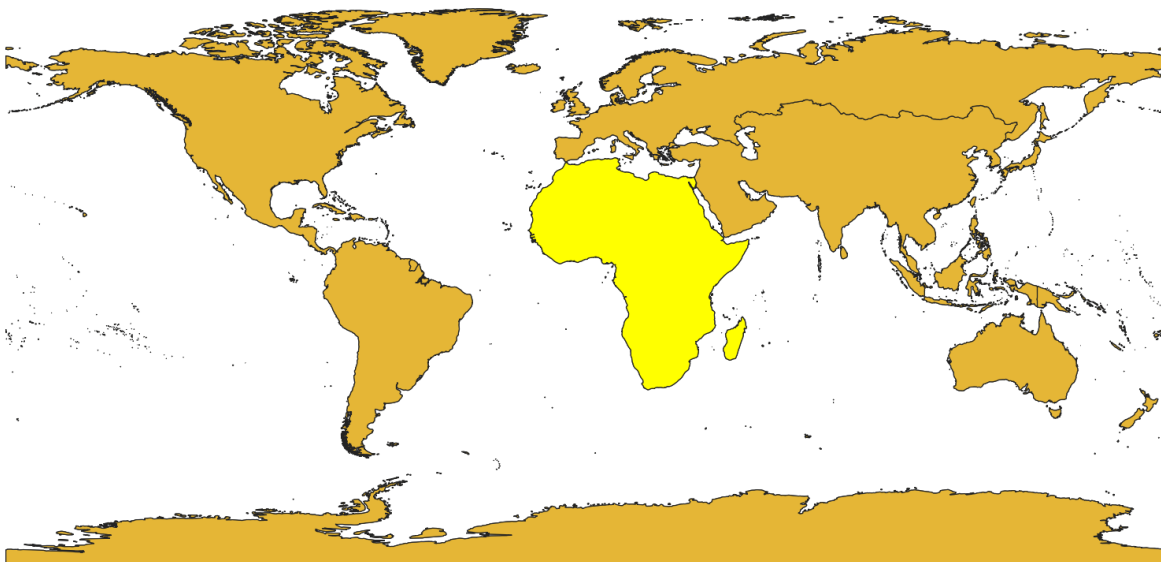
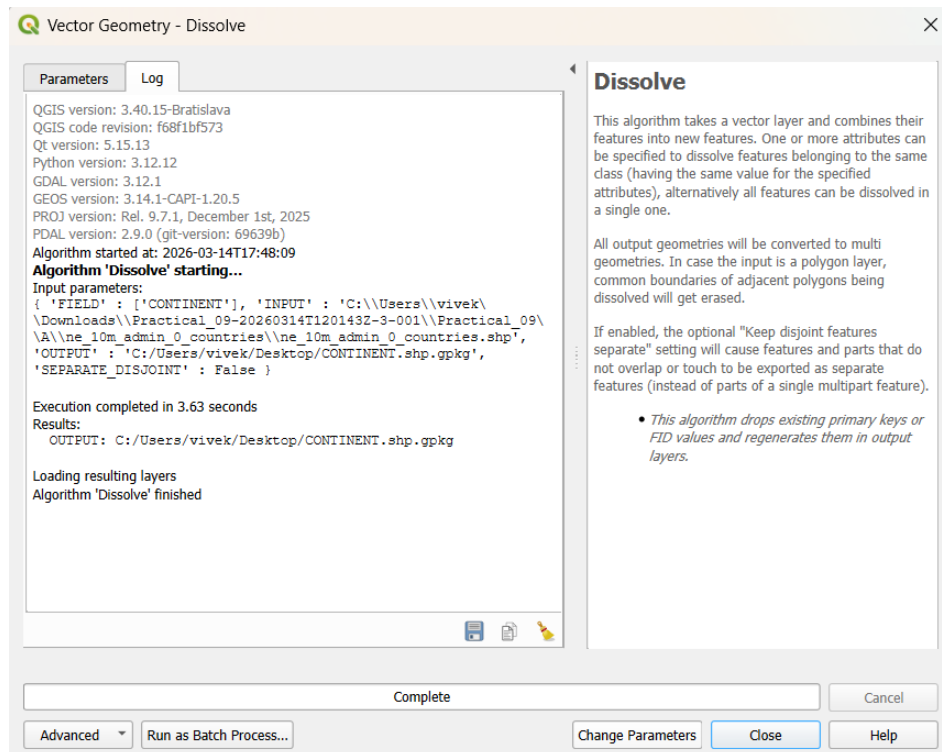
AIM : BATCH PROCESSING USING PROCESSING FRAMEWORK

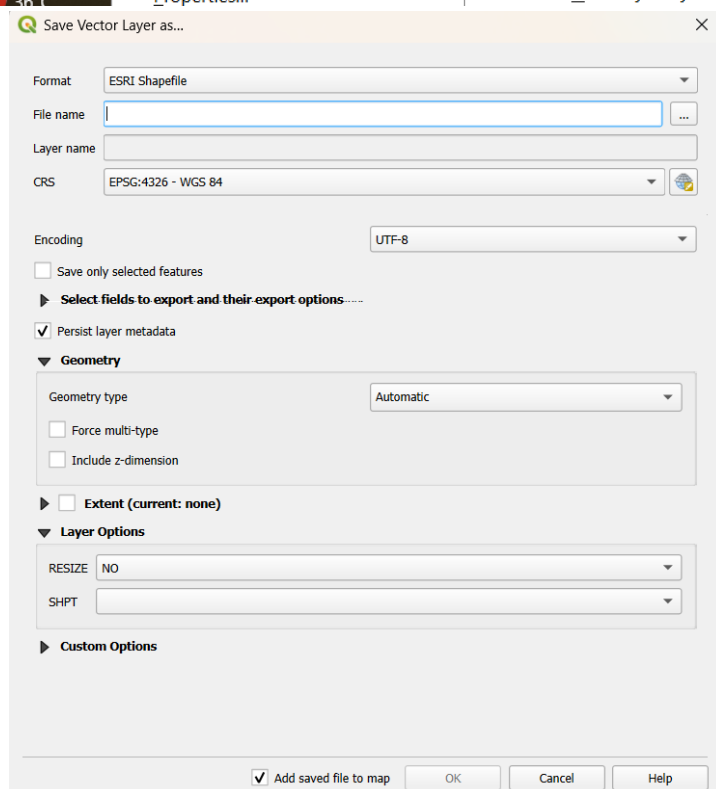
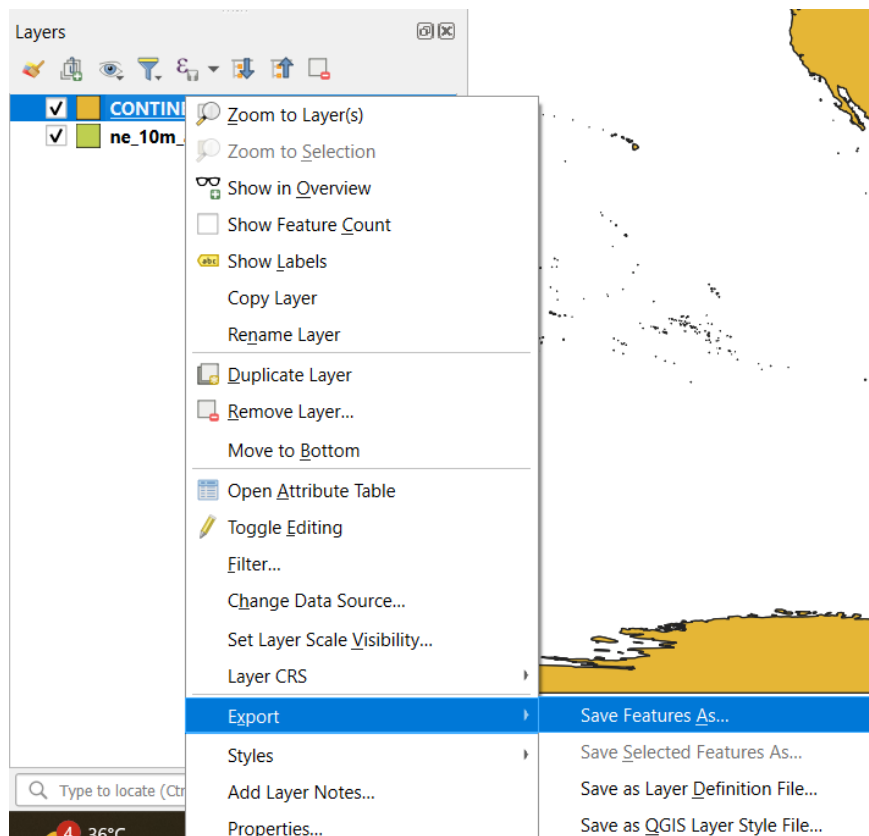
Steps:

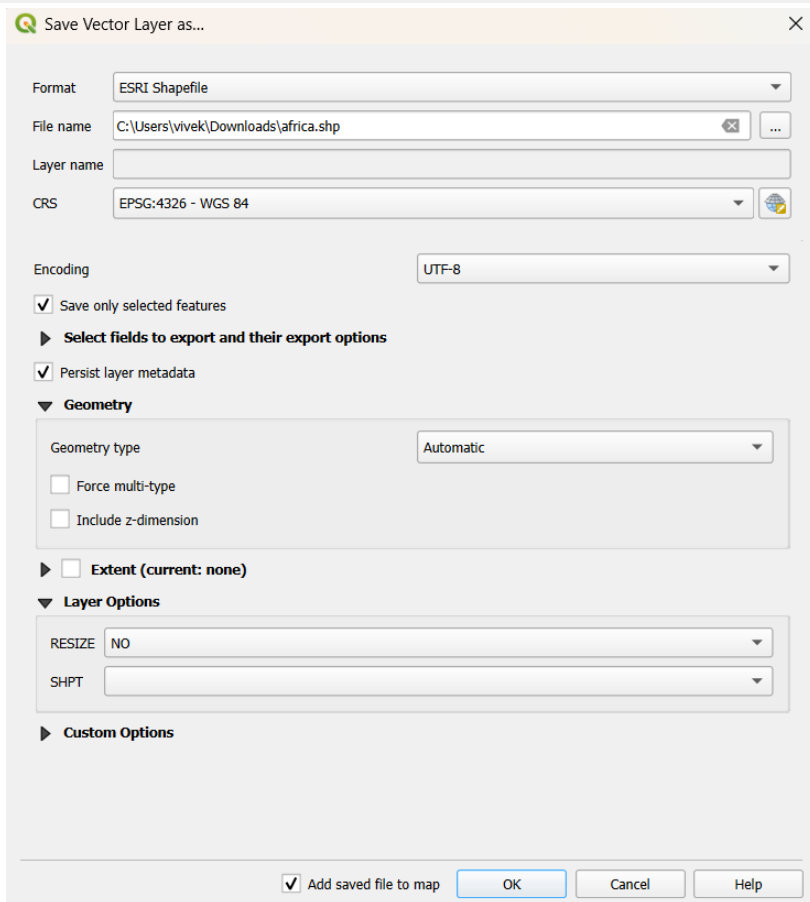
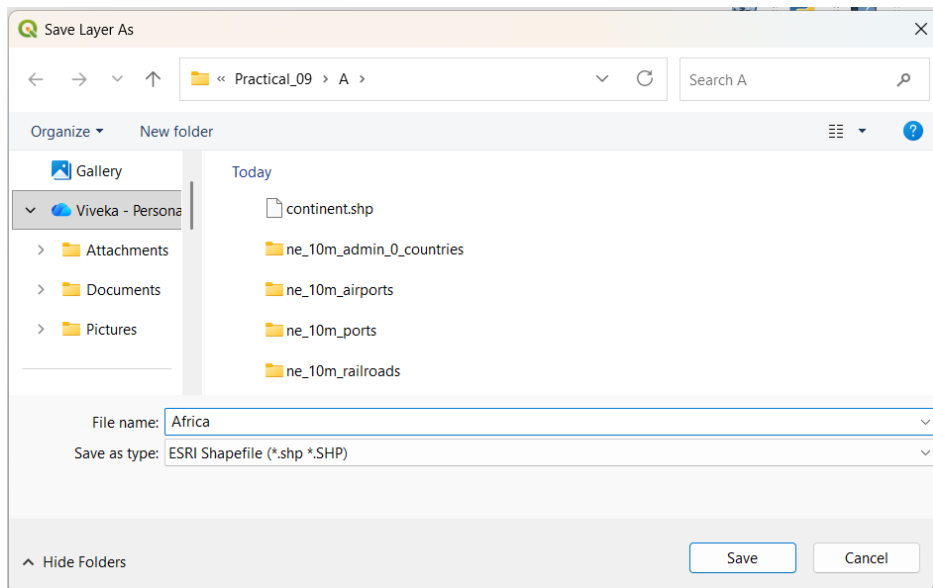


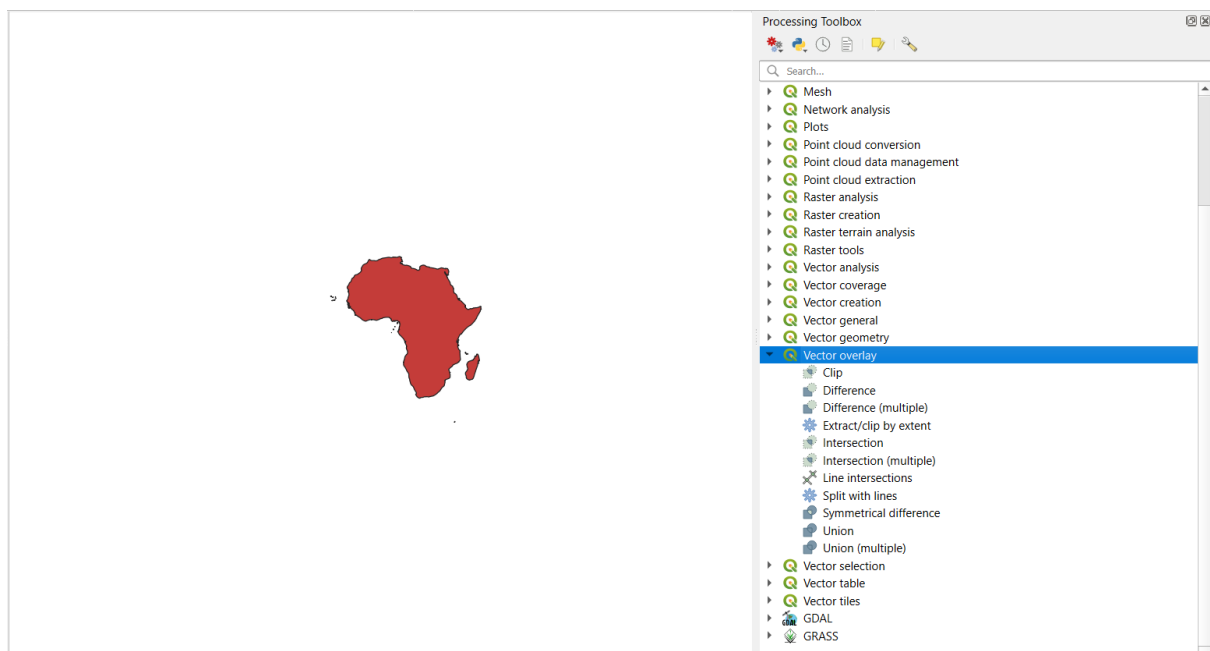
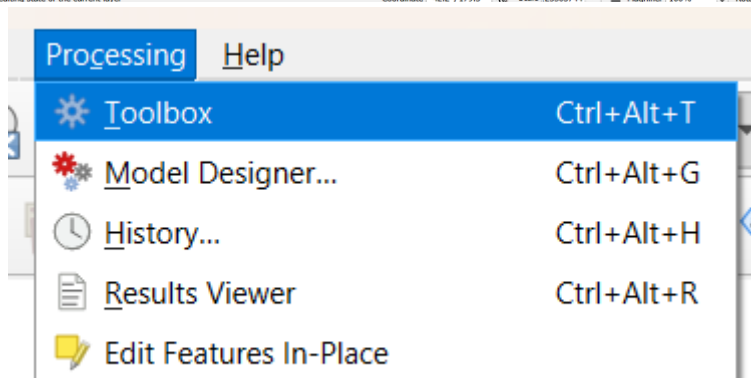
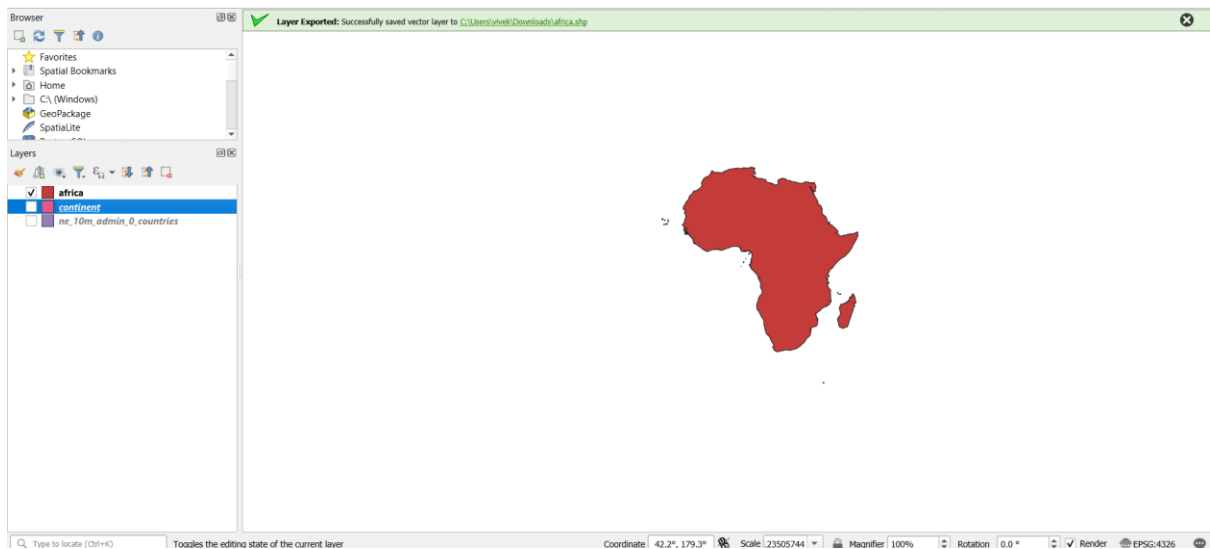


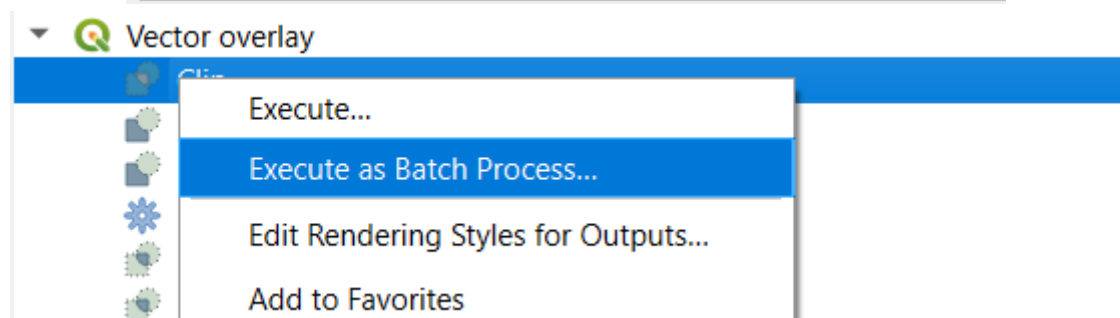
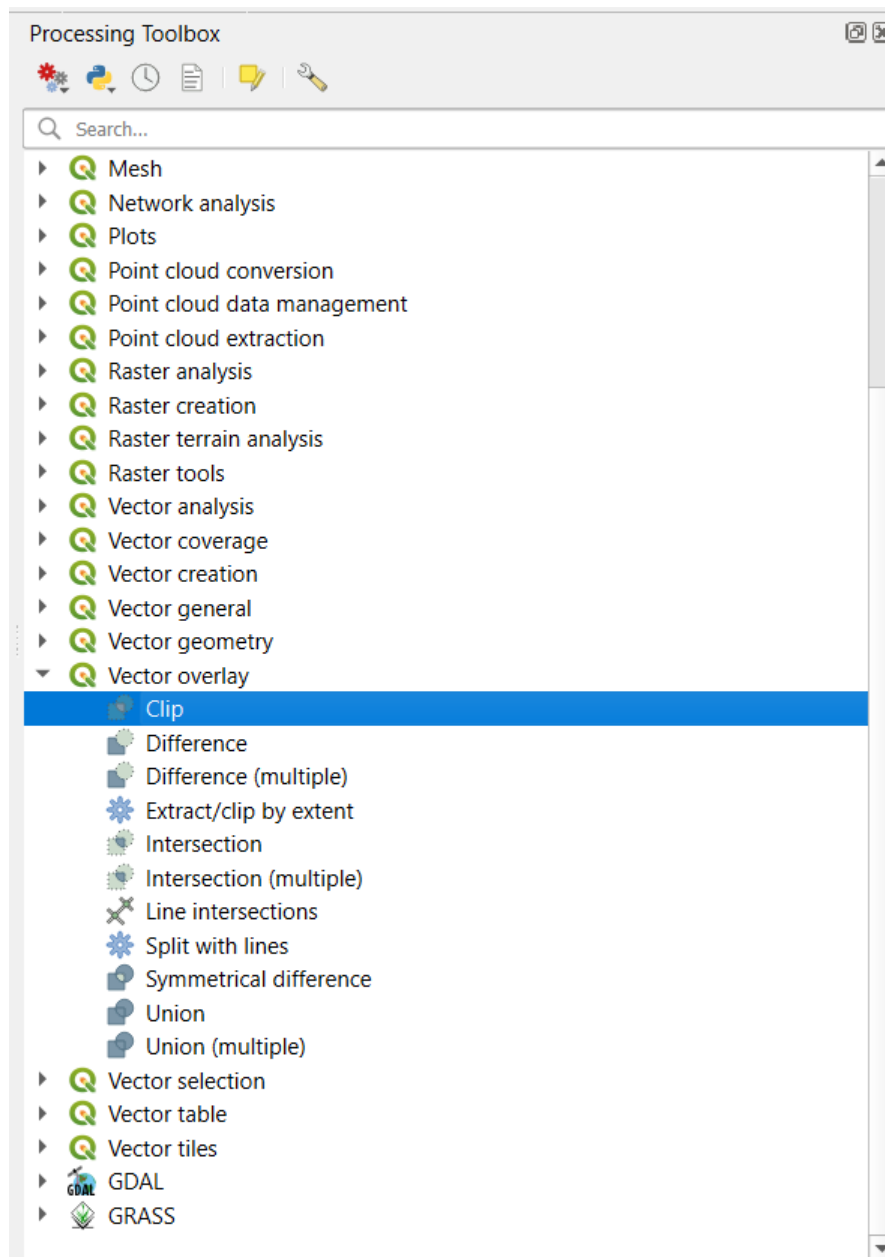


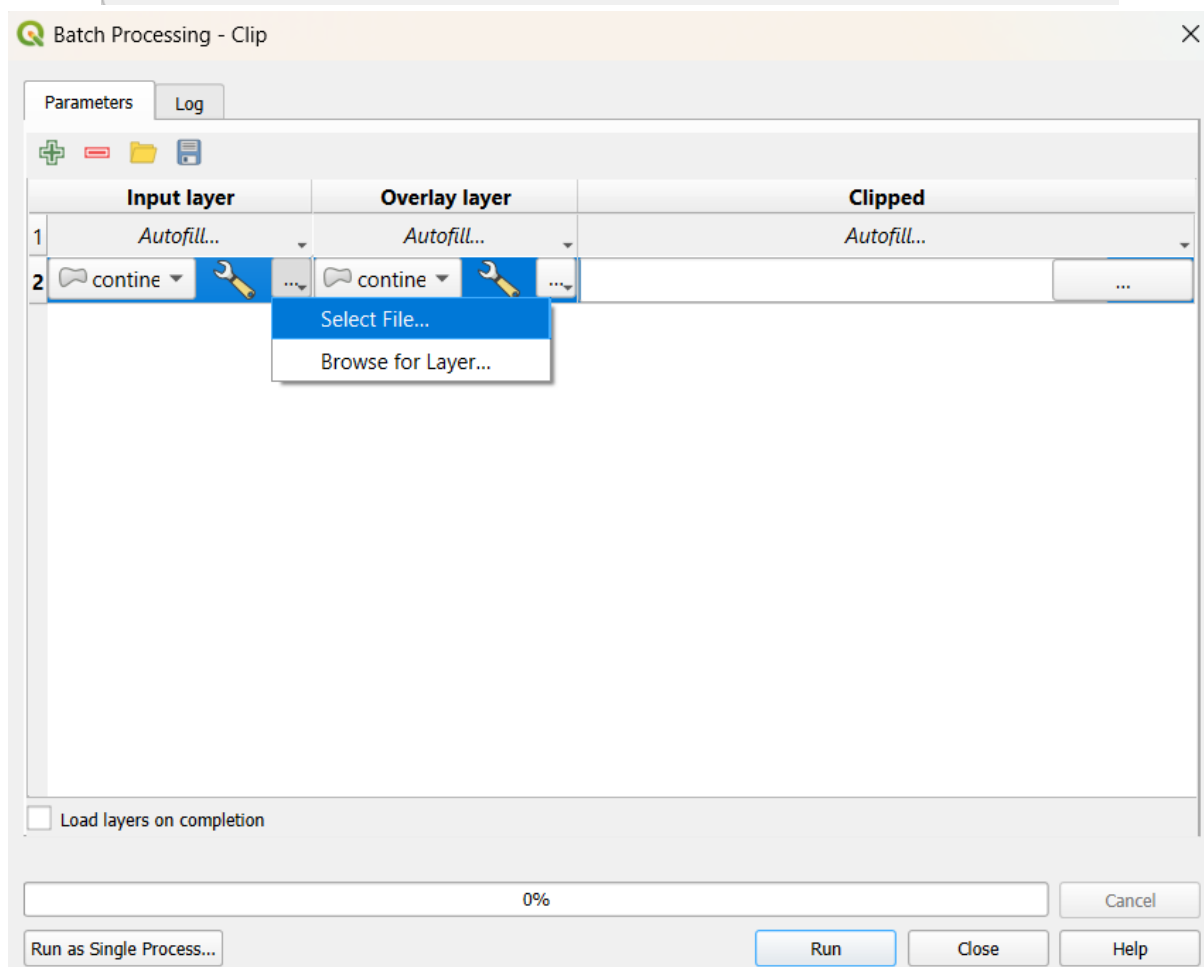
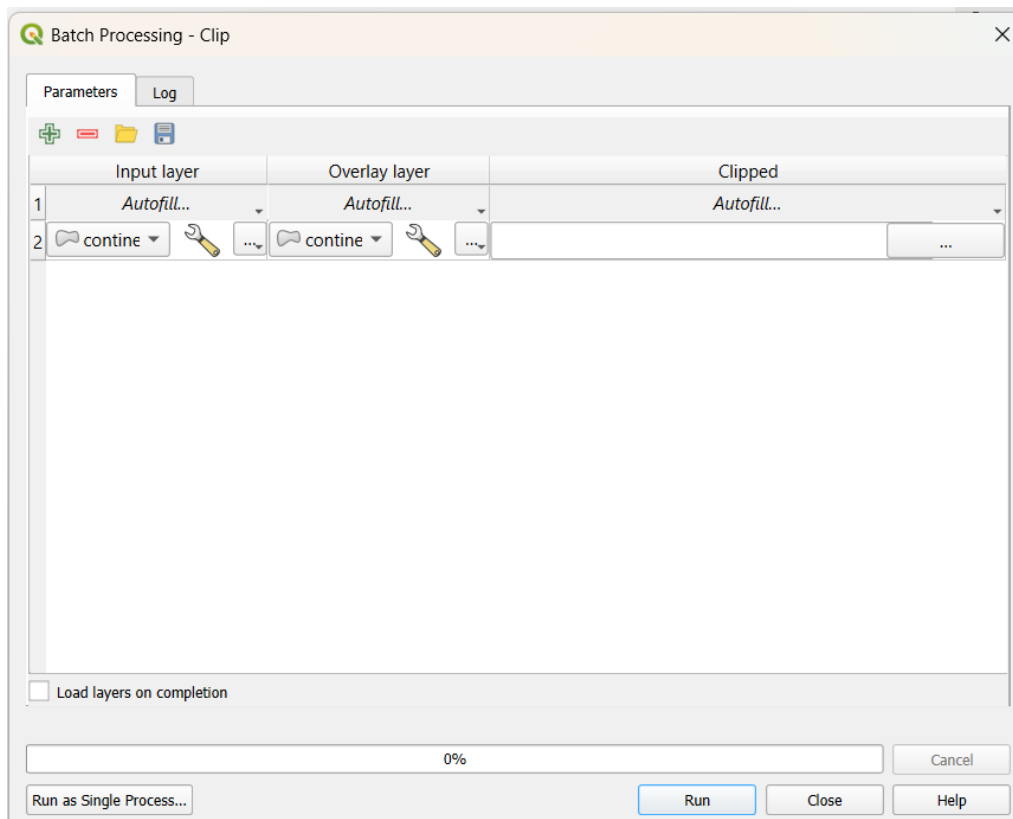


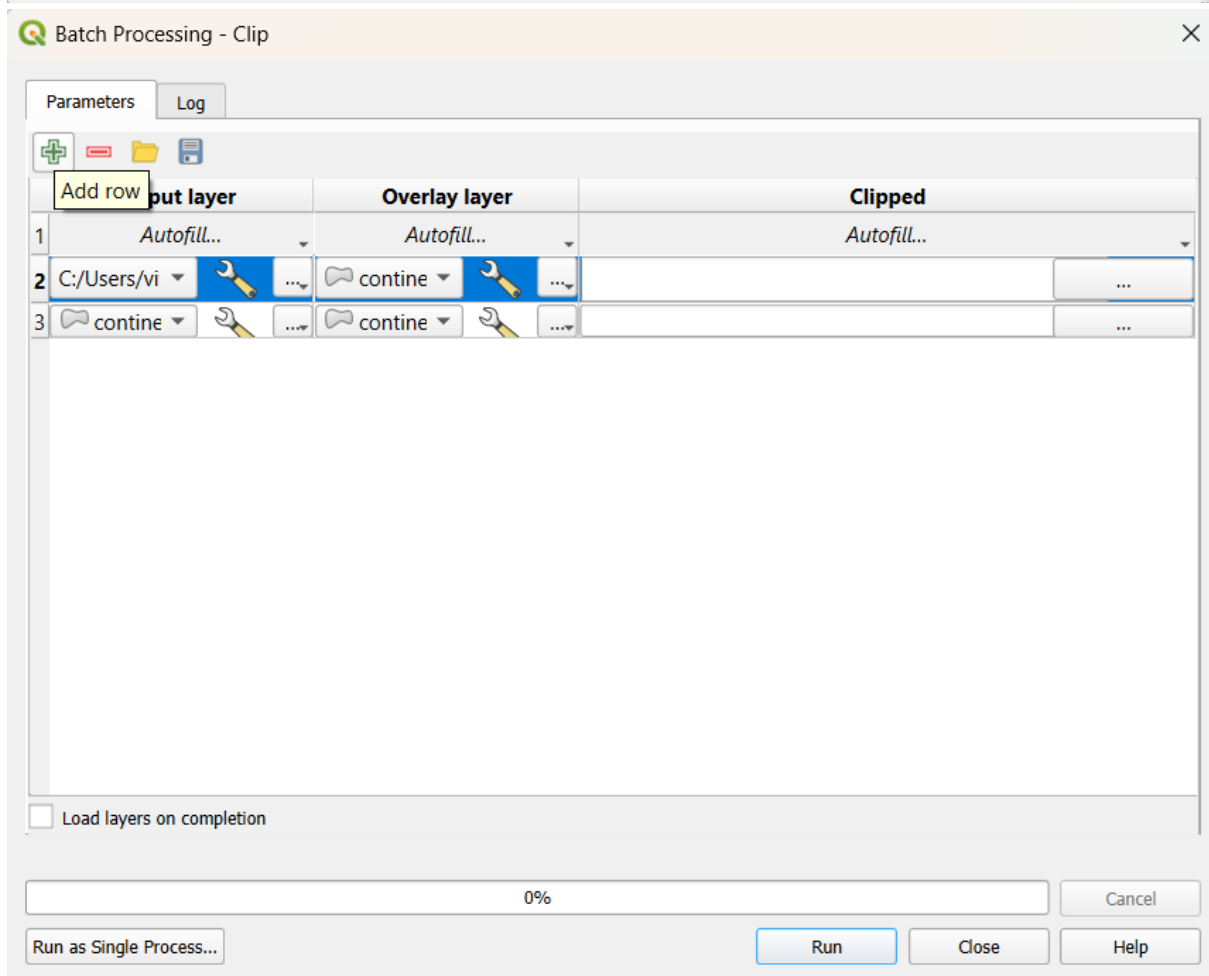
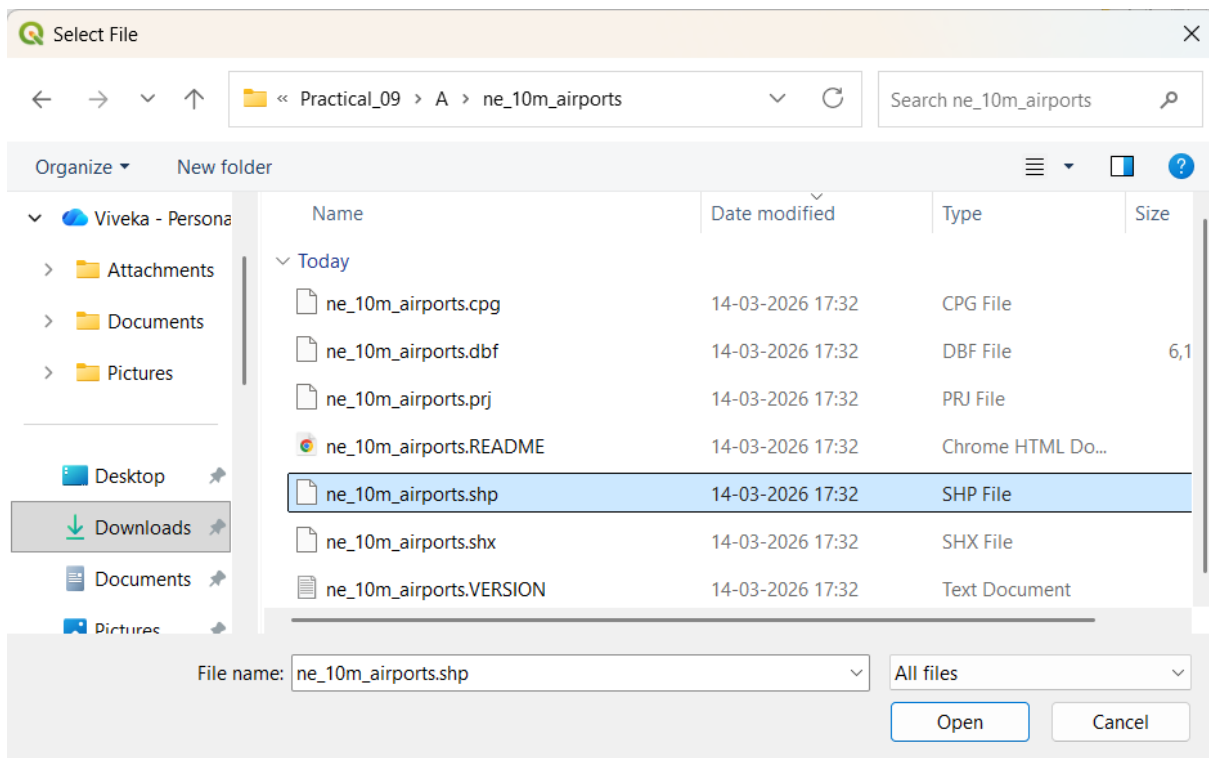


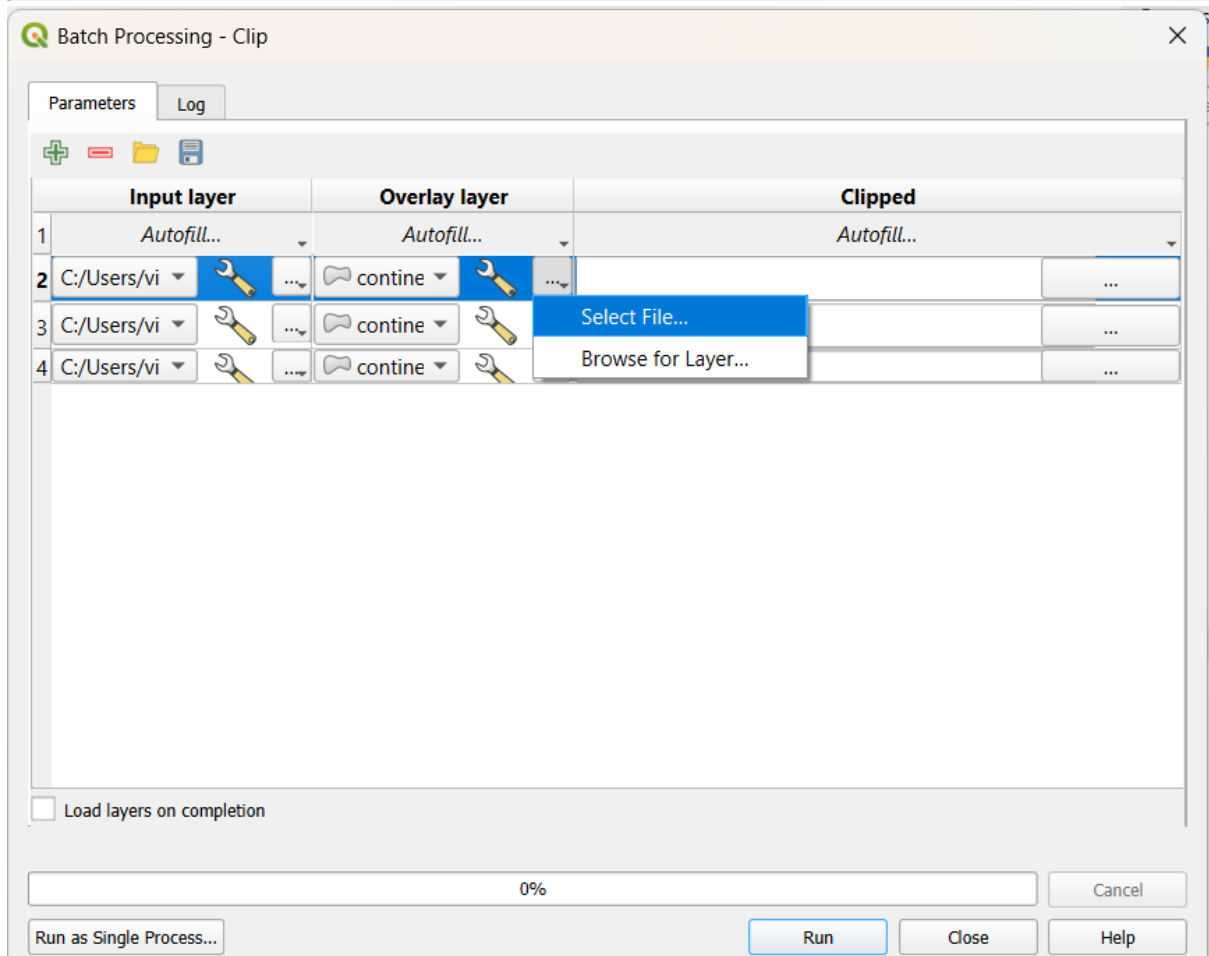
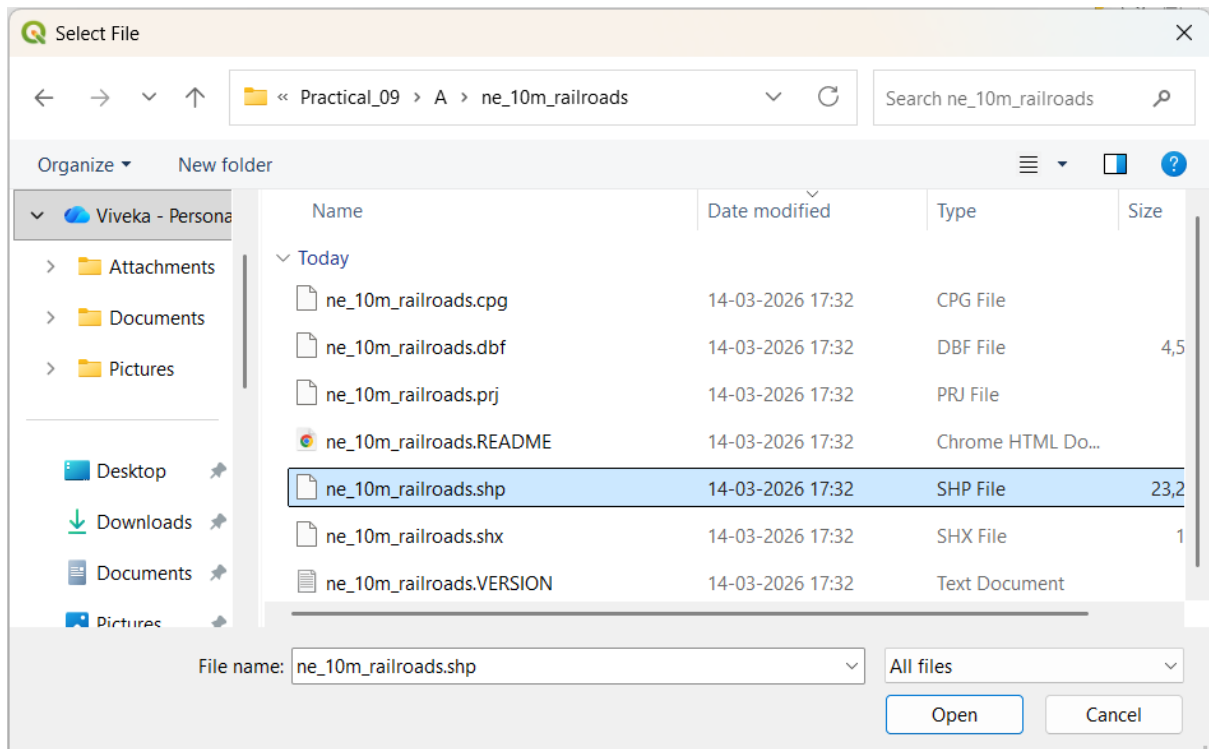


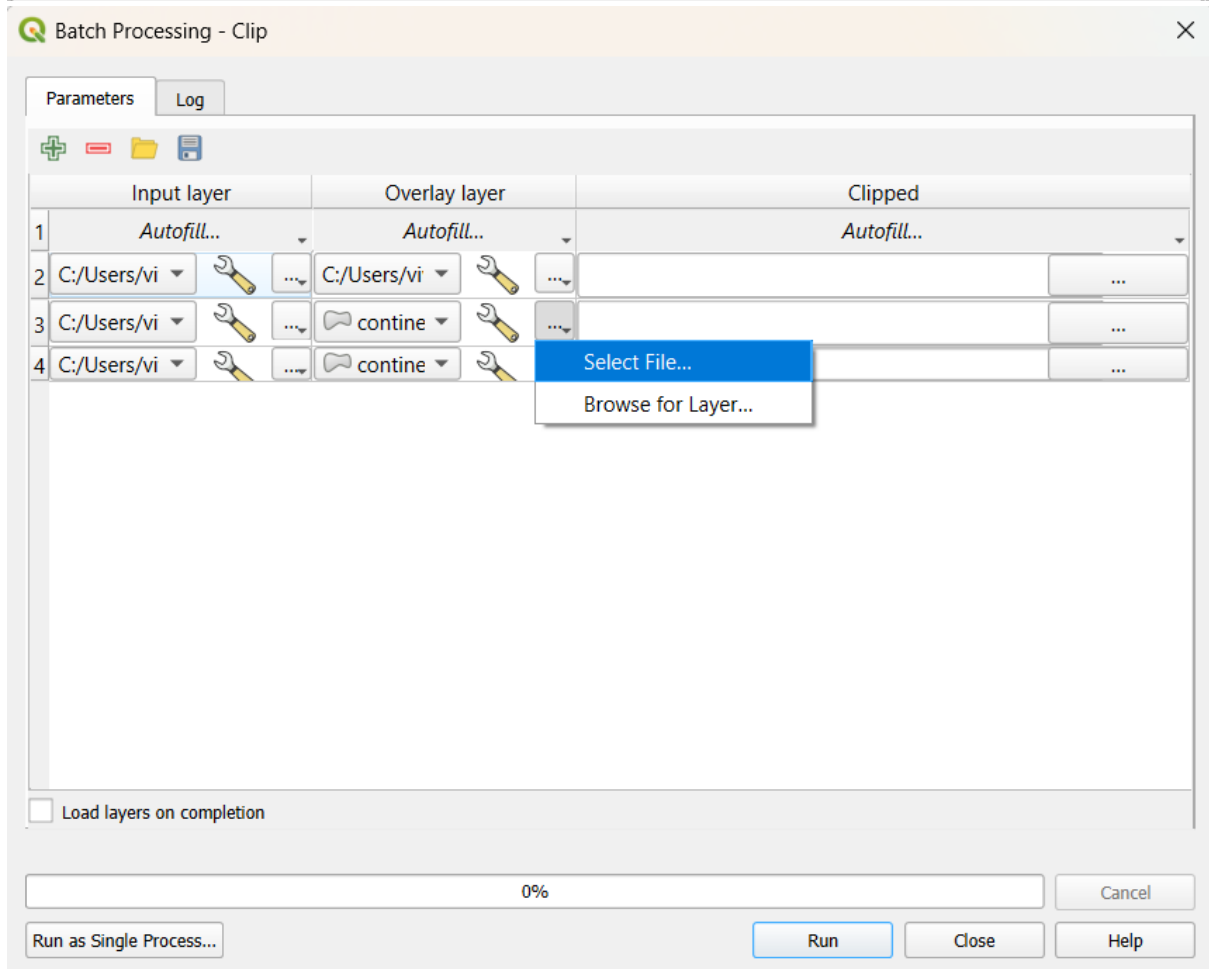
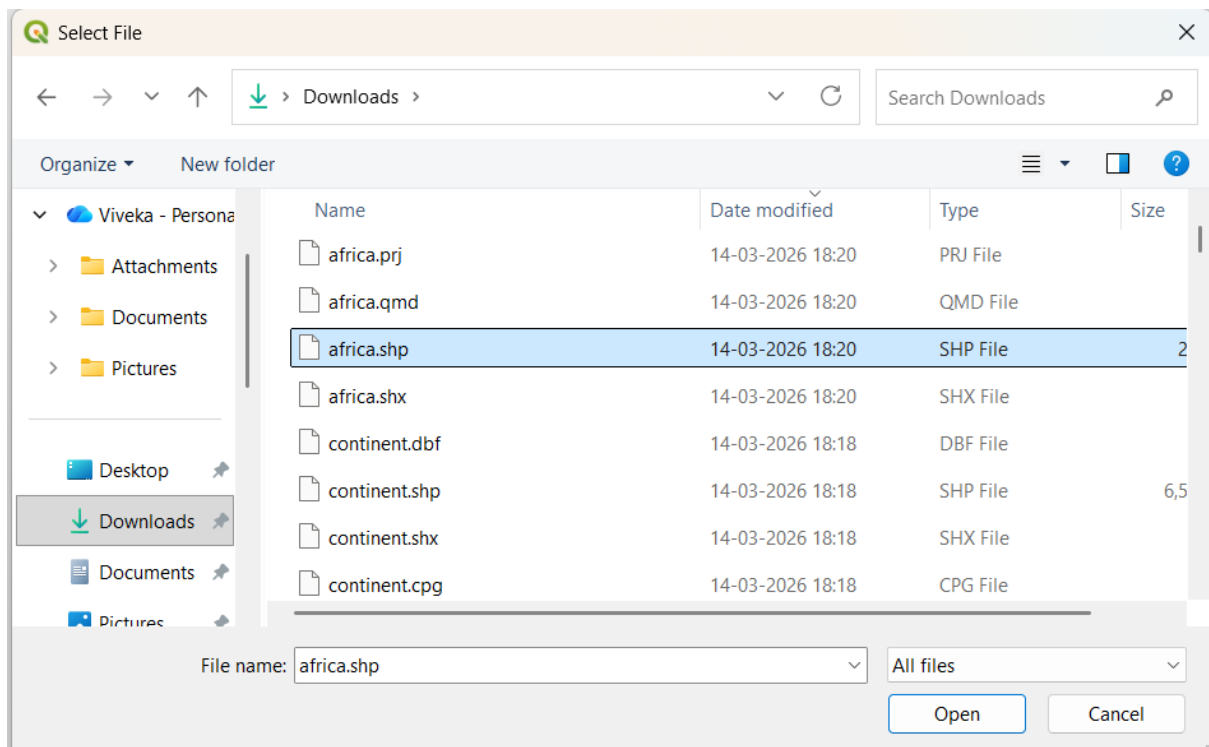


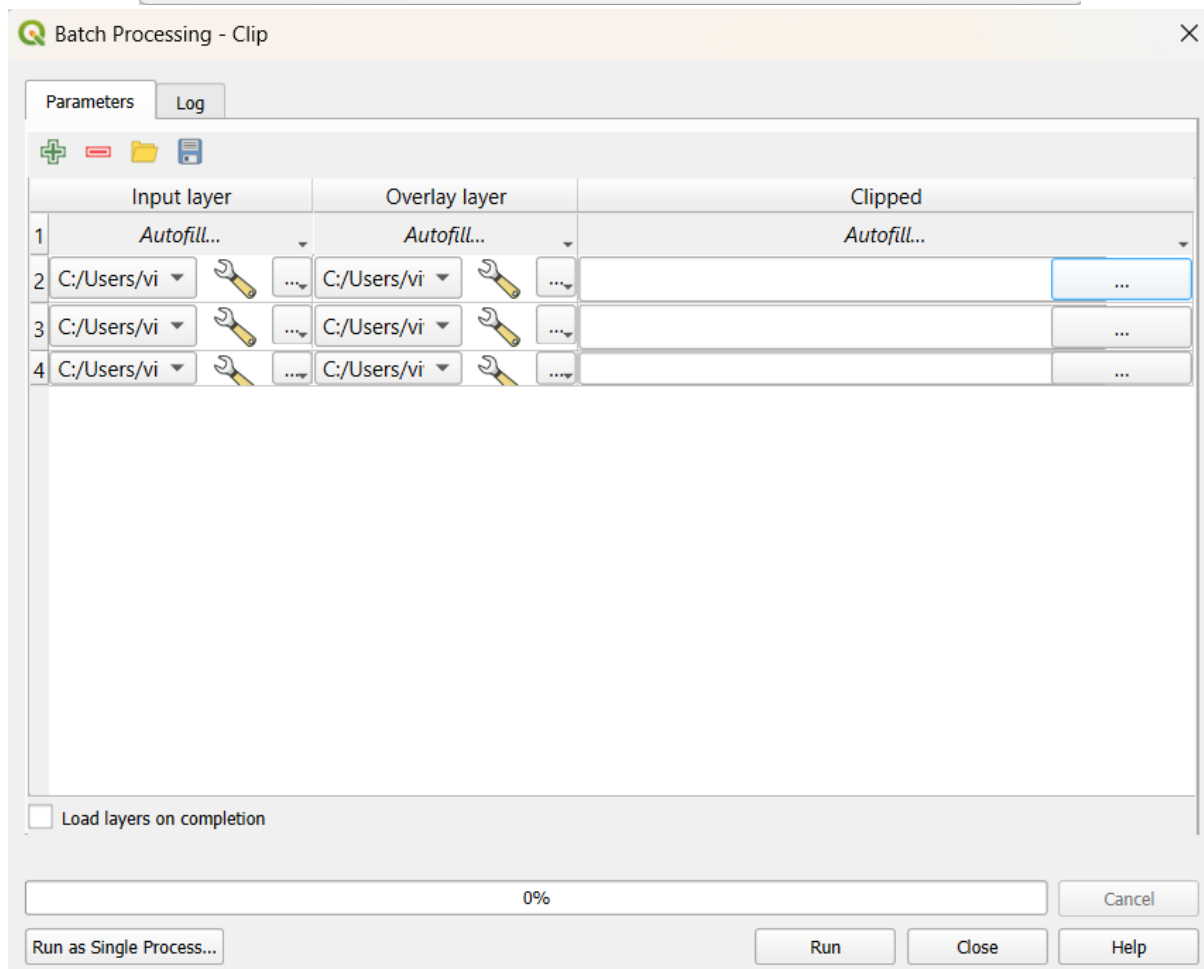
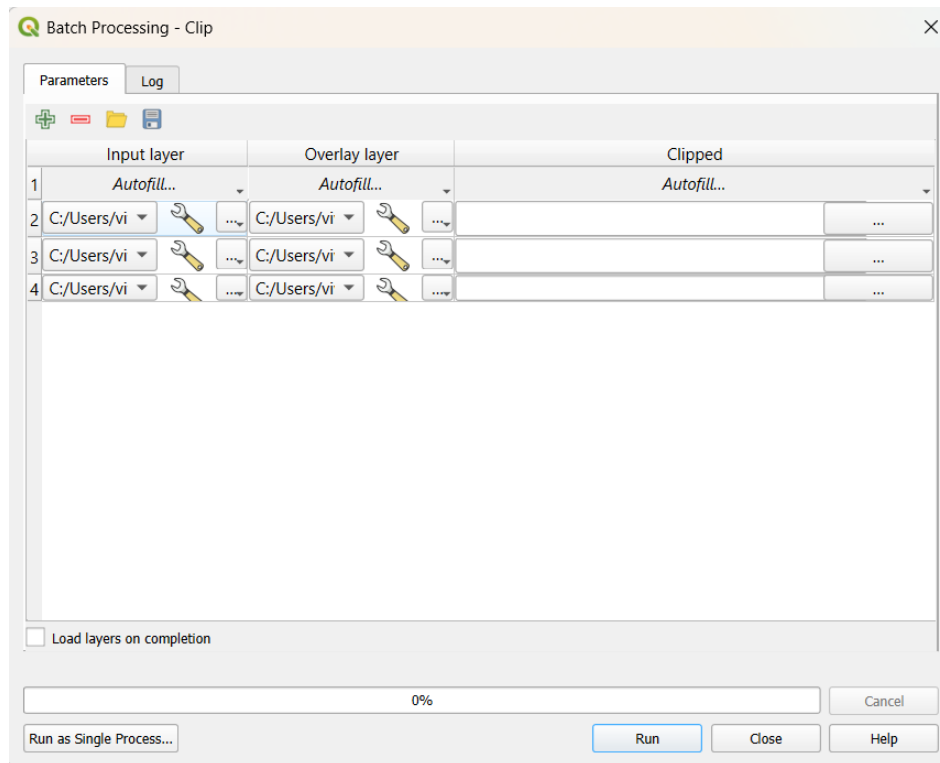


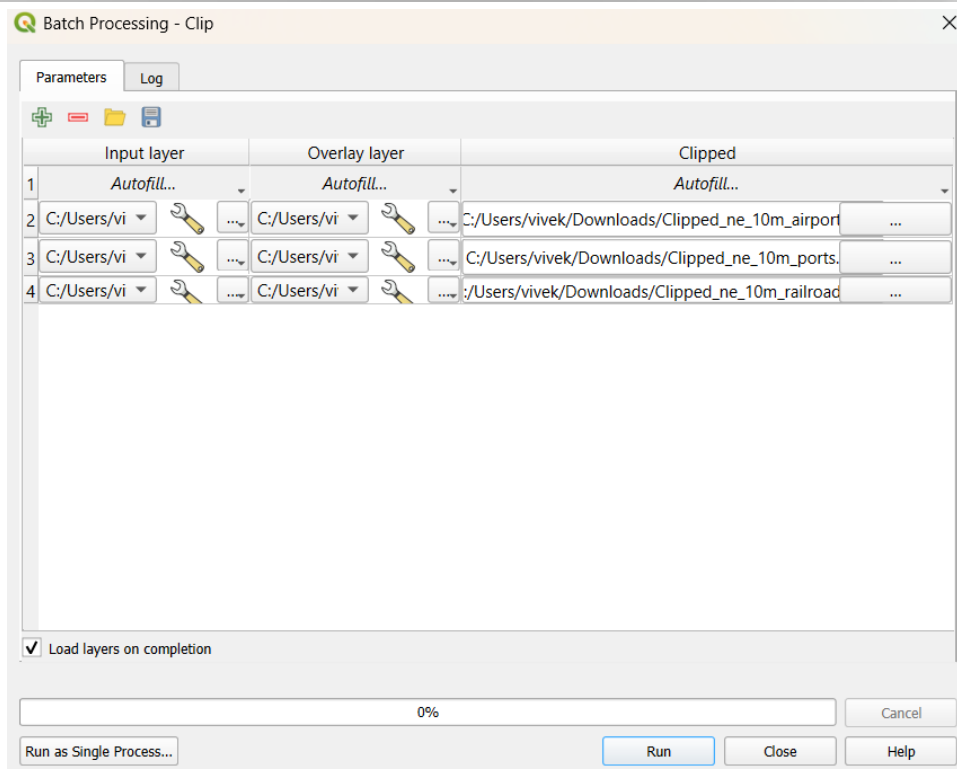
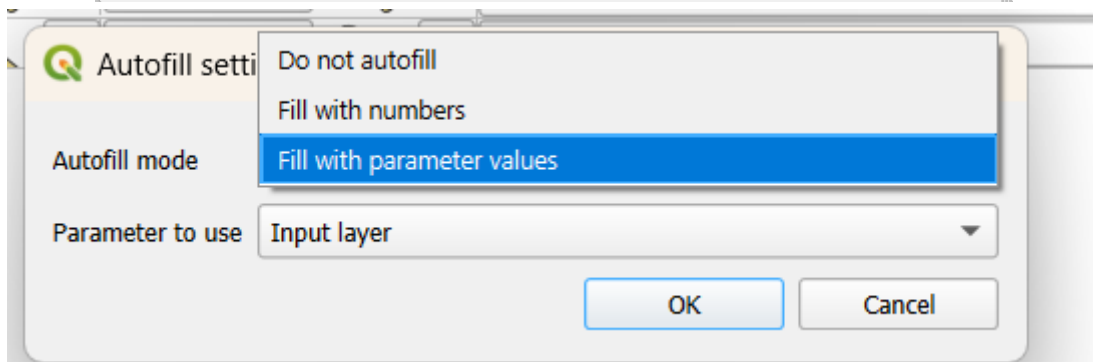
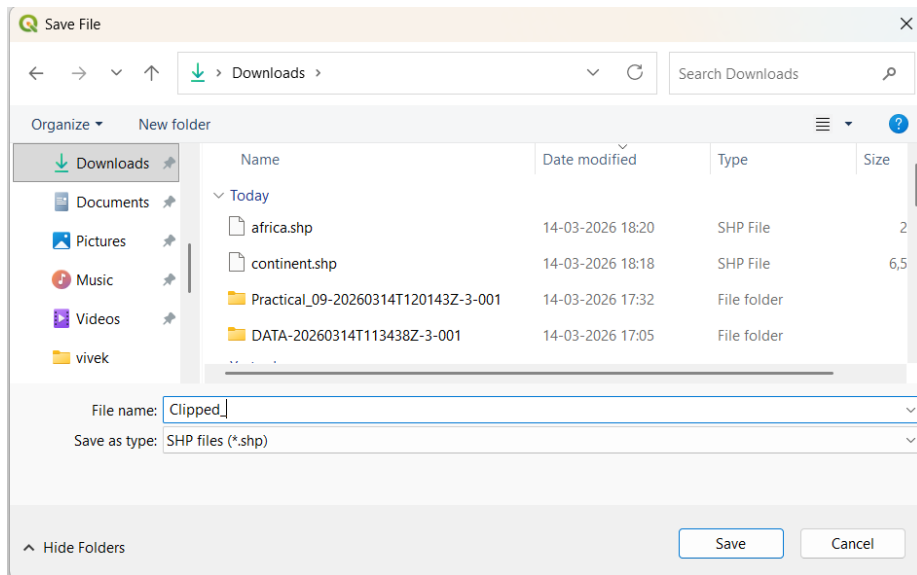


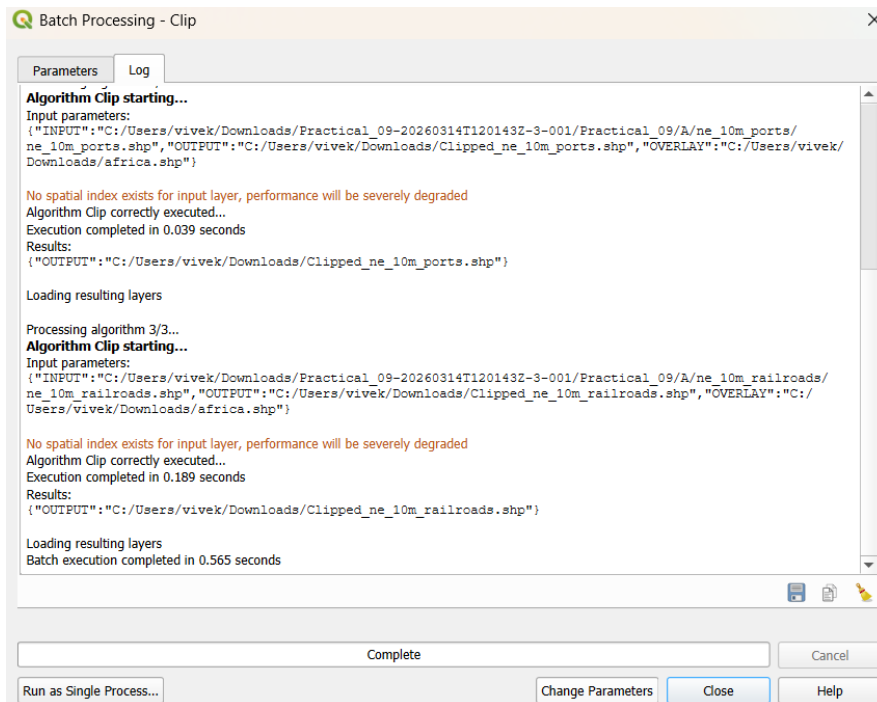


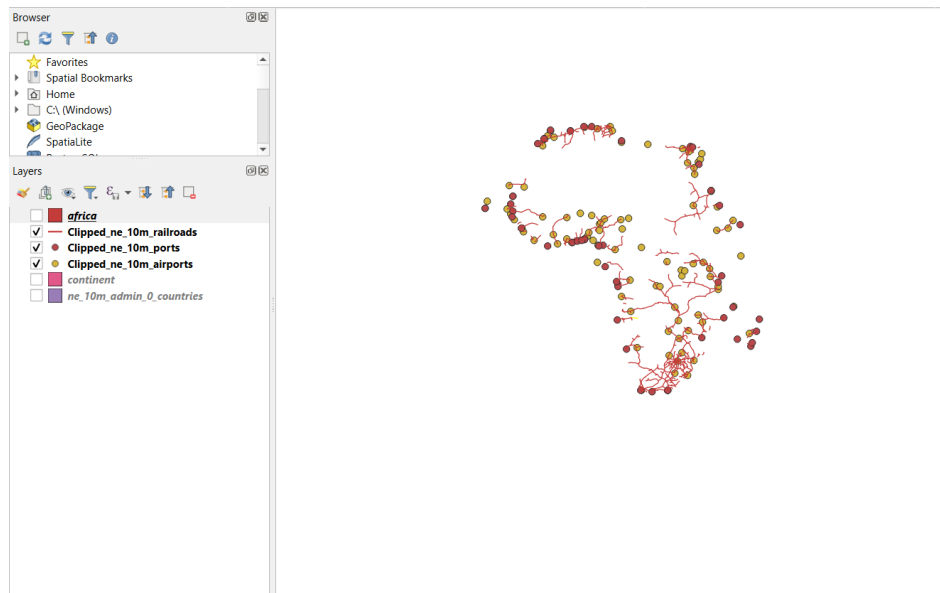




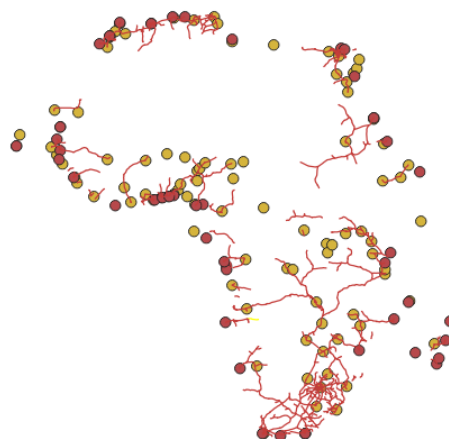








FINAL OUTPUT :



CONCLUSION : Batch processing using processing framework implemented successfully .

